Electronic Supplementary Material for:

Exploring cocoa bean fermentation mechanisms by kinetic modelling

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In this electronic supplementary material, references to equations and tables in the Main Manuscript are labelled with MM. For example, Eq. (1) and Table 1 of the Main Manuscript are referred as Eq. (1.MM) and Table 1.MM, respectively. Particular sections in the Main Manuscript are referred by the number of section followed by MM. For instance, Section 1 in the Main Manuscript, here is referred as Section 1.MM.

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1 Parameter estimates re-scaling to their original units

Consider the kinetics of a given x variable in any of our proposed models, its concentration $[x]$ at any time and its maximum concentration $[x]_{max}$ within the whole interval in which the fermentation took place.

We can then let x' represent a transformation of x as

$$x' = \frac{[x]}{[x]_{max}}, \tag{1}$$

then, the time derivative of x' will be given by

$$\frac{dx'}{dt} = \frac{1}{[x]_{max}} \frac{d\,[x]}{dt}, \tag{2}$$

where $[x]_{max}$ is the maximum concentration of any of the state variables depending on which ODE of a proposed model is expressed in the form of Eq. (2).

Generalizing, one can express all ODEs of a model in the form of Eq. (2). As an example, let's express Eq. (1.MM) for the time derivative of glucose (Glc) as a scaled version in accordance to Eq. (2)

$$\frac{1}{[\mathrm{Glc}]_{max}} \frac{d\,[\mathrm{Glc}]}{dt} = -Y_1 \frac{\mu_1 \frac{[\mathrm{Glc}]}{[\mathrm{Glc}]_{max}}}{\frac{[\mathrm{Glc}]}{[\mathrm{Glc}]_{max}} + Ks_1} \frac{[\mathrm{Y}]}{[\mathrm{Y}]_{max}} - Y_2 \frac{\mu_3 \frac{[\mathrm{Glc}]}{[\mathrm{Glc}]_{max}}}{\frac{[\mathrm{Glc}]}{[\mathrm{Glc}]_{max}} + Ks_3} \frac{[\mathrm{LAB}]}{[\mathrm{LAB}]_{max}}, \tag{3}$$

where μ_1, μ_3, Ks_1, Ks_3, Y_1 and Y_2 are the solutions of this ODE parameters upon the scaled time series of Glc (according to Eq. (1)) for specific maximum growth rates (μ_i), substrate saturation constants (Ks_i) and yield coefficients (Y_i) respectively.

In other words, all aforementioned terms are scaled posterior distributions resulting from our Bayesian framework for which re-scaled versions to their original units can be accomplished by properly working out the scaled ODE.

Hence, Eq. 3 can be simplified to

$$\frac{d\,[\mathrm{Glc}]}{dt} = -Y_1 \frac{[\mathrm{Glc}]_{max}}{[\mathrm{Y}]_{max}} \frac{\mu_1\,[\mathrm{Glc}]}{[\mathrm{Glc}] + Ks_1\,[\mathrm{Glc}]_{max}}\,[\mathrm{Y}] - Y_2 \frac{[\mathrm{Glc}]_{max}}{[\mathrm{LAB}]_{max}} \frac{\mu_3\,[\mathrm{Glc}]}{[\mathrm{Glc}] + Ks_3\,[\mathrm{Glc}]_{max}}\,[\mathrm{LAB}] \tag{4}$$

where the terms μ_1, μ_3, $Ks_1\,[\mathrm{Glc}]_{max}$, $Ks_3\,[\mathrm{Glc}]_{max}$, $Y_1 \frac{[\mathrm{Glc}]_{max}}{[\mathrm{Y}]_{max}}$ and $Y_2 \frac{[\mathrm{Glc}]_{max}}{[\mathrm{LAB}]_{max}}$ are equivalent to $\mu_{\max}^{\mathrm{Y}_{\mathrm{Glc}}}$, $\mu_{\max}^{\mathrm{LAB}_{\mathrm{Glc}}}$, $K_{\mathrm{Glc}}^{\mathrm{Y}}$, $K_{\mathrm{Glc}}^{\mathrm{LAB}}$, $Y_{\mathrm{Glc|Y}}$ and $Y_{\mathrm{Glc|LAB}}$ respectively and the left hand side of the equation corresponds to the state variable with no scaling.

Similarly, transformation factors as those previously described can be then obtained for the rest of parameters in any given MI presented in this study.

2 Fermentation data

From bibliographical review, 15 published research papers met the inclusion criteria described in Section 2.1.MM [1–15]. An additional study lacking time series for LA was also considered due to the limited number of works where heap method was used [16]. For this particular paper, LA's time series were simulated during models calibrations. Overall, data from 32 fermentation trials were gathered. After the calibration of models, 9 out of 32 datasets did not result in successful fits for any MI (see Table S1).

Fermentation trials that resulted in successful fits, according to features mentioned in Section 2.6.MM, were grouped by country in 8 experiments from Brazil [4, 6, 8, 9], 5 from Ghana [1, 16], 5 from Ecuador [5, 13], 2 from Nicaragua [14], 1 from Dominican Republic [2], 1 from Malaysia [10] and 1 from Honduras [12]. Regarding the cocoa cultivar, 4 experiments were performed using Nacional/Trinitario [5], 3 using Criollo/Forastero [1, 4], 1 using Trinitario [2], 1 using PH16 [9], 1 using a mix of IMC-67, UF-29 and UF-668 [12], 1 using Criollo [13], 1 using Nugu [14], 1 using O'payo [14] and 1 using unspecified mixed hybrids [10]. Furthermore, 2 studies with 6 trials in total did not mention any identification about used cultivar [6, 16] and 1 remaining study reported 3 trials with the use of unspecified mixed hybrids [8]. With the aim of not losing information from such studies, cultivars were here categorized as different unknowns on basis of their authorship. Hence, 4 trials with unidentified cultivars performed by Camu *et al.* [16] and 2 conducted by Pereira *et al.* [6] were coded as *un1* and *un2*, respectively. Finally, 3 trials using unspecified mixed hybrids performed by Pereira *et al.* [8] were coded as *un3*.

Regarding the fermentation method, trials were grouped in 11 from wooden boxes [2, 4, 5, 8–10, 12, 14], 5 from heaps [1, 16], 2 from platforms [14], 2 from stainless steel tanks [6, 8] and 2 from plastic boxes [6, 13]. About turning of fermenting mass, 16 experiments reported to have turned cocoa beans during fermentation [2, 4–6, 8, 10, 12, 14, 16], 5 did not conduct turning [1, 5, 16] and 2 did not report whether turning of fermenting cocoa beans was done [9, 13]. In regard to temperature control, only 3 trials reported to have controlled it [6, 13]. Finally, with respect to the use of a starter culture, data gathered from the only study that suited inclusion criteria and reported its use [7], could not be fitted to any MI. A summary of fermentation experiments that could be fitted to at least one MI is shown in Table 3.MM.

3 Mathematical formulation of candidate mechanisms

3.1 M1: Decay of fermentation products

M1 can be formulated by considering a linear term of the form

$$d_i = b_X[X], \tag{5}$$

where d represents the decay i at a rate b of metabolite X, can be added to Eqs. (3.MM), (4.MM) and (5.MM) to account for decrease of EtOH, LA and Ac, respectively:

$$\frac{d\,[\mathrm{EtOH}]}{dt} = Y^{\mathrm{Glc}}_{\mathrm{EtOH|Y}}\,v_1 + Y^{\mathrm{Fru}}_{\mathrm{EtOH|Y}}\,v_2 - Y_{\mathrm{EtOH|AAB}}\,v_4 - d_1, \tag{6}$$

$$\frac{d\,[\mathrm{LA}]}{dt} = Y^{\mathrm{Glc}}_{\mathrm{LA|LAB}}\,v_3 - Y_{\mathrm{LA|AAB}}\,v_5 - d_2 \tag{7}$$

$$\frac{d\,[\mathrm{Ac}]}{dt} = Y^{\mathrm{Glc}}_{\mathrm{Ac|LAB}}\,v_3 + Y^{\mathrm{EtOH}}_{\mathrm{Ac|AAB}}\,v_4 + Y^{\mathrm{LA}}_{\mathrm{Ac|AAB}}\,v_5 - d_3. \tag{8}$$

Therefore, the addition of these terms provides three extra parameters to the baseline model.

3.2 M2: Consumption of fructose (Fru) by lactic acid bacteria (LAB)

M2 can be easily implemented into the baseline model in three steps. First, an extra growth rate for LAB is needed to describe uptake of Fru as carbon source. Such a growth rate can be defined by a Monod type equation of the form

$$v_9 = \frac{\mu_{\max}^{\mathrm{LAB_{Fru}}}\,[\mathrm{Fru}]}{[\mathrm{Fru}] + K_{\mathrm{Fru}}^{\mathrm{LAB}}}\,[\mathrm{LAB}]\,, \tag{9}$$

where, $\mu_{\max}^{\mathrm{LAB_{Fru}}}$ represents the maximum specific growth rate of LAB on Fru and $K_{\mathrm{Fru}}^{\mathrm{LAB}}$ is the substrate saturation constant of LAB growth on Fru. Then, v_9 must be included in Eq. (2.MM) with aid of a LAB-to-Fru yield coefficient ($Y_{\mathrm{Fru|LAB}}$), giving

$$\frac{d\,[\mathrm{Fru}]}{dt} = -Y_{\mathrm{Fru|Y}}\,v_2 - Y_{\mathrm{Fru|LAB}}\,v_9\,. \tag{10}$$

Second, contributions of LAB in producing EtOH, LA and Ac from growth rates v_3 and v_9 are included in Eqs. (3.MM), (4.MM) and (5.MM) with help of four extra parameters, namely LAB-to-EtOH from Glc ($Y_{\mathrm{EtOH|LAB}}^{\mathrm{Glc}}$), LAB-to-EtOH from Fru ($Y_{\mathrm{EtOH|LAB}}^{\mathrm{Fru}}$), LAB-to-LA from Fru ($Y_{\mathrm{LA|LAB}}^{\mathrm{Fru}}$), and LAB-to-Ac from Fru ($Y_{\mathrm{Ac|LAB}}^{\mathrm{Fru}}$) yield coefficients:

$$\frac{d\,[\mathrm{EtOH}]}{dt} = Y_{\mathrm{EtOH|Y}}^{\mathrm{Glc}}\,v_1 + Y_{\mathrm{EtOH|Y}}^{\mathrm{Fru}}\,v_2 + Y_{\mathrm{EtOH|LAB}}^{\mathrm{Glc}}\,v_3 + Y_{\mathrm{EtOH|LAB}}^{\mathrm{Fru}}\,v_9 - Y_{\mathrm{EtOH|AAB}}\,v_4 \tag{11}$$

$$\frac{d\,[\mathrm{LA}]}{dt} = Y_{\mathrm{LA|LAB}}^{\mathrm{Glc}}\,v_3 + Y_{\mathrm{LA|LAB}}^{\mathrm{Fru}}\,v_9 - Y_{\mathrm{LA|AAB}}\,v_5 \tag{12}$$

$$\frac{d\,[\mathrm{Ac}]}{dt} = Y_{\mathrm{Ac|LAB}}^{\mathrm{Glc}}\,v_3 + Y_{\mathrm{Ac|LAB}}^{\mathrm{Fru}}\,v_9 + Y_{\mathrm{Ac|AAB}}^{\mathrm{EtOH}}\,v_4. \tag{13}$$

Finally, increase on microbial population by growth rate v_9 can be added to Eq. (7.MM) as

$$\frac{d\,[\mathrm{LAB}]}{dt} = v_3 + v_9 - v_7. \tag{14}$$

In total, implementation of Eqs. (10) to (14) in the baseline model adds seven extra parameters.

3.3 M3: Production of acetic acid (Ac) by yeasts (Y)

Given that the baseline model already accounts for growth rates of Y on Glc (v_1) and Fru (v_2), deployment of this new mechanism in Eq. (5.MM) needs Y-to-Ac from Glc ($Y_{\mathrm{Ac|Y}}^{\mathrm{Glc}}$) and Y-to-Ac from Fru ($Y_{\mathrm{Ac|Y}}^{\mathrm{Fru}}$) yield coefficients:

$$\frac{d\,[\mathrm{Ac}]}{dt} = Y_{\mathrm{Ac|LAB}}^{\mathrm{Glc}}\,v_3 + Y_{\mathrm{Ac|AAB}}^{\mathrm{EtOH}}\,v_4 + Y_{\mathrm{Ac|AAB}}^{\mathrm{LA}}\,v_5 + Y_{\mathrm{Ac|Y}}^{\mathrm{Glc}}\,v_1 + Y_{\mathrm{Ac|Y}}^{\mathrm{Fru}}\,v_2, \tag{15}$$

adding two more parameters to the baseline model.

3.4 M4: Production of lactic acid (LA) by yeasts (Y)

Similar to M2, M4 can be incorporated to the baseline model in three steps. First, a new growth rate for Y on LA is defined by a Monod type equation of the form

$$v_{10} = \frac{\mu_{\max}^{\mathrm{Y_{LA}}}\,[\mathrm{LA}]}{[\mathrm{LA}] + K_{\mathrm{LA}}^{\mathrm{Y}}}\,[\mathrm{Y}]\,, \tag{16}$$

where, $\mu_{\max}^{\mathrm{Y_{LA}}}$ is the maximum specific growth rate of Y on LA and $K_{\mathrm{LA}}^{\mathrm{Y}}$ represents the substrate saturation constant of Y growth on LA. v_{10} is then added to Eq. (4.MM) with a Y-to-LA yield coefficient ($Y_{\mathrm{LA|Y}}$):

$$\frac{d\,[\mathrm{LA}]}{dt} = Y_{\mathrm{LA|LAB}}^{\mathrm{Glc}}\, v_3 - Y_{\mathrm{LA|AAB}}\, v_5 - Y_{\mathrm{LA|Y}}\, v_{10}. \tag{17}$$

Second, to account the production of EtOH via consumption of LA by Y, v_{10} must be part of Eq. (3.MM) with companion of a Y-to-EtOH from LA yield coefficient ($Y_{\mathrm{EtOH|Y}}^{\mathrm{LA}}$):

$$\frac{d\,[\mathrm{EtOH}]}{dt} = Y_{\mathrm{EtOH|Y}}^{\mathrm{Glc}}\, v_1 + Y_{\mathrm{EtOH|Y}}^{\mathrm{Fru}}\, v_2 + Y_{\mathrm{EtOH|Y}}^{\mathrm{LA}}\, v_{10} - Y_{\mathrm{EtOH|AAB}}\, v_4. \tag{18}$$

Finally, the population size Y (Eq. (6.MM)) is also affected by v_{10}:

$$\frac{d\,[\mathrm{Y}]}{dt} = v_1 + v_2 + v_{10} - v_6. \tag{19}$$

In total, M4 contributes 4 parameters to the baseline model.

3.5 M5: Over-oxidation of acetic acid (Ac) by acetic acid bacteria (AAB)

Similar to M2 and M4, an extra growth rate is required to describe AAB growth on its own main product, Ac:

$$v_{11} = \frac{\mu_{\max}^{\mathrm{AAB_{Ac}}}\,[\mathrm{Ac}]}{[\mathrm{Ac}] + K_{\mathrm{Ac}}^{\mathrm{AAB}}}\,[\mathrm{AAB}]\,, \tag{20}$$

where $\mu_{\max}^{\mathrm{AAB_{Ac}}}$ depicts the maximum specific growth rate of AAB on Ac and $K_{\mathrm{Ac}}^{\mathrm{AAB}}$ is the substrate saturation constant of AAB growth on Ac. Then, v_{11} enters in Eq. (5.MM) with an AAB-to-Ac yield coefficient ($Y_{\mathrm{Ac|AAB}}$):

$$\frac{d\,[\mathrm{Ac}]}{dt} = Y_{\mathrm{Ac|LAB}}^{\mathrm{Glc}}\, v_3 + Y_{\mathrm{Ac|AAB}}^{\mathrm{EtOH}}\, v_4 + Y_{\mathrm{Ac|AAB}}^{\mathrm{LA}}\, v_5 - Y_{\mathrm{Ac|AAB}}\, v_{11}. \tag{21}$$

Lastly, population of AAB described in Eq. (8.MM) is complemented with v_{11}:

$$\frac{d\,[\mathrm{AAB}]}{dt} = v_4 + v_5 + v_{11} - v_8\,. \tag{22}$$

Since we are not interested in simulating carbon dioxide production, no extra equation is needed and M5 ends up adding three extra parameters to the baseline model.

4 Convergence criteria

A deeper understanding of the convergence criteria in this work is explained considering two main points: (1) convergence statistics used, and (2) how these statistics depend on the number of iterations and efficiency of the MCMC sampler.

First point: Our research was conducted using convergence statistics and criteria as proposed by Vehtari *et al.* [17], where the potential scale reduction factor $\widehat{R}$ corresponds to an improved version of the split-$\widehat{R}$ [18]. This procedure relies on a rank-normalized effective sample size (ESS) that ensures obtaining reliable estimates of variances and autocorelations needed for its computation. Additionally, this allows to improve convergence diagnosis by the improved $\widehat{R}$ in presence of heavy tails by introducing two new ESS measures, namely tail-ESS and bulk-ESS. However, for assessing convergence it is not enough to rely on a small value for $\widehat{R}$ as it can occur that reasonable values of this statistic could

be estimated along with poor ESS, specially tail-ESS and bulk-ESS. These last two are important in checking on the stability of the variances of the estimates given they represent specifically the ESS for extreme quantiles (e.g., 5% and 95% quantiles of the posterior).

Second point: As is well known, MCMC methods need to be run long enough to accomplish convergence and obtain reliable uncertainties of the estimates. In this sense, increasing the number of iterations is among the most common practices to asses convergence through the examination of $\widehat{R}$ and ESS. Roughly speaking, larger iterations help by regularizing the variance of the chains and decreasing autocorrelation of the samples. However, the question of how long the chains in any MCMC algorithm should be, does not have a universal answer as it depends highly on the efficiency of the sampler in exploring the parameter space [19]. Usually, classic MCMC algorithms (e.g. Metropolis-Hastings and Gibbs sampler) require quite high number of iterations to obtain reliable uncertainties because of the autocorrelation of their samples which are the result of their inherent random walk nature that governs them. In contrast, newer algorithms based on Hamiltonian dynamics as the MCMC No-U-Turn sampler have satisfactorily shown better efficiency than Metropolis-Hastings and Gibbs sampler in exploring the parameter space [20, 21]. As a result, assessing convergence of a model can require a considerable less amount of iterations, as long as, $\widehat{R}$ and the ESS are within reasonable thresholds. In practical terms of our research, all aforementioned statistics were considered according to their suggested thresholds of 1.05 for $\widehat{R}$ [22], 1000 for ESS [23], and 400 for both, tail-ESS and bulk-ESS [17], which consequently resulted in relative small number of iterations.

5 Supplementary tables and figures

Table S1: Detailed description of the considered data. Author, year of publication, cocoa country of origin, cocoa cultivar, used methodology, code name given in the original trial, re-coded name given in this research, use of starter culture, turning of the fermenting mass and controlled temperature are shown.

Reference	Year	Country	Cultivar	Method	Trial	Code	Starter	Turning	Ctrl. Temp.
Camu *et al.* [1]	2007	Ghana	Criollo/Forastero	heap	heap 5	ghhp1	✗	✗	✗
Lagunes Gálvez *et al.* [2]	2007	Dominican Republic	Trinitario	wooden box	NA	dowb1	✗	✓	✗
Camu *et al.* [16]	2008	Ghana	NA	heap	heap 10	ghhp2	✗	✓	✗
					heap 11	ghhp3	✗	✗	✗
					heap 12	ghhp4	✗	✓	✗
					heap 13	ghhp5	✗	✗	✗
Lefeber *et al.* [3]	2011	Ivory Coast	NA	plastic box	NA	cipb1	✗	✗	✗
Papalexandratou *et al.* [4]	2011	Brazil	Criollo/Forastero	wooden box	box 1	brwb1	✗	✓	✗
					box 2	brwb2	✗	✓	✗
Papalexandratou *et al.* [5]	2011	Ecuador	Nacional/Trinitario	platform	P1	ecpt1	✗	✗	✗
					P2	ecpt2	✗	✗	✗
				wooden box	B1	ecwb1	✗	✓	✗
					B2	ecwb2	✗	✓	✗
Pereira *et al.* [6]	2012	Brazil	NA	plastic box	PC	brpb1	✗	✓	✓
				stainless tank	ST	brst1	✗	✓	✓
Lefeber *et al.* [7]	2012	Malaysia	NA	wooden box	NA	mywb1	✓	✓	NA
					NA	mywb2	✓	✓	NA
		Ghana	NA	heap	NA	ghhp6	✗	NA	NA
					NA	ghhp7	✓	✓	NA
Pereira *et al.* [8]	2013	Brazil	Mixed hybrids	wooden box	WB1	brwb3	✗	✓	✗
					WB2	brwb4	✗	✓	✗
				stainless tank	SST	brst2	✗	✓	✗
Moreira *et al.* [9]	2013	Brazil	PH9, PH15, PH16	wooden box	PH9	brwb5	✗	NA	✗
					PH15	brwb6	✗	NA	✗
					PH16	brwb7	✗	NA	✗
Papalexandratou *et al.* [10]	2013	Malaysia	Mixed hybrids	wooden box	box 2	mywb3	✗	✓	✗
Bastos *et al.* [11]	2018	Brazil	TSH565	wooden box	NA	brwb8	✗	✓	✗
Romanens *et al.* [12]	2018	Honduras	IMC-67, UF-29, UF-668	wooden box	OF-F	hnwb1	✗	✓	✗
†Lee *et al.* [13]	2019	Ecuador	Criollo	plastic box	NA	ecpb1	✗	NA	✓
Papalexandratou *et al.* [14]	2019	Nicaragua	Nugu/O'payo	wooden box	NUGU	niwb1	✗	✓	✗
					O'PAYO	niwb2	✗	✓	✗
†Racine *et al.* [15]	2019	Ecuador	Criollo	plastic box	NA	ecpb2	✗	✓	✓

† Simulated fermentation.

Table S2: Prior distributions in the range of original units for parameters of all model iterations per dataset. All priors are truncated to the positive set of real numbers. Parameters related to lactic acid (LA) for datasets *ghhp2* to *ghhp5* cannot be re-scaled because their LA's time series were simulated (see Section 2).

Parameter	ghhp1	dowb1	ghhp2	ghhp3	ghhp4	ghhp5	cipb1	brwb1
$\mu_{max}^{Y_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{Y_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{Y_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{LAB_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{LAB_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{AAB_{EtOH}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{AAB_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{AAB_{Ac}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
K_{Glc}^{Y}	$\mathcal{N}(25.98, 15.59)$	$\mathcal{N}(28.45, 17.07)$	$\mathcal{N}(23.98, 14.39)$	$\mathcal{N}(20.12, 12.07)$	$\mathcal{N}(20.79, 12.47)$	$\mathcal{N}(24.65, 14.79)$	$\mathcal{N}(47.8, 28.68)$	$\mathcal{N}(27.74, 16.64)$
K_{Fru}^{Y}	$\mathcal{N}(28.87, 17.32)$	$\mathcal{N}(44.4, 26.64)$	$\mathcal{N}(25.69, 15.42)$	$\mathcal{N}(25.43, 15.26)$	$\mathcal{N}(22.11, 13.26)$	$\mathcal{N}(25.7, 15.42)$	$\mathcal{N}(91, 54.6)$	$\mathcal{N}(24.83, 14.9)$
K_{LA}^{Y}	$\mathcal{N}(28.87, 17.32)$	$\mathcal{N}(44.4, 26.64)$	$\mathcal{N}(25.69, 15.42)$	$\mathcal{N}(25.43, 15.26)$	$\mathcal{N}(22.11, 13.26)$	$\mathcal{N}(25.7, 15.42)$	$\mathcal{N}(91, 54.6)$	$\mathcal{N}(24.83, 14.9)$
K_{Glc}^{LAB}	$\mathcal{N}(25.98, 15.59)$	$\mathcal{N}(28.45, 17.07)$	$\mathcal{N}(23.98, 14.39)$	$\mathcal{N}(20.12, 12.07)$	$\mathcal{N}(20.79, 12.47)$	$\mathcal{N}(24.65, 14.79)$	$\mathcal{N}(47.8, 28.68)$	$\mathcal{N}(27.74, 16.64)$
K_{Fru}^{LAB}	$\mathcal{N}(28.87, 17.32)$	$\mathcal{N}(44.4, 26.64)$	$\mathcal{N}(25.69, 15.42)$	$\mathcal{N}(25.43, 15.26)$	$\mathcal{N}(22.11, 13.26)$	$\mathcal{N}(25.7, 15.42)$	$\mathcal{N}(91, 54.6)$	$\mathcal{N}(24.83, 14.9)$
K_{EtOH}^{AAB}	$\mathcal{N}(11.25, 6.75)$	$\mathcal{N}(4.85, 2.91)$	$\mathcal{N}(5.25, 3.15)$	$\mathcal{N}(6.71, 4.03)$	$\mathcal{N}(9.14, 5.48)$	$\mathcal{N}(10.13, 6.08)$	$\mathcal{N}(8.3, 4.98)$	$\mathcal{N}(3.02, 1.81)$
K_{LA}^{AAB}	$\mathcal{N}(2243.66, 2.58)$	$\mathcal{N}(24.95, 0.67)$	-	-	-	-	$\mathcal{N}(55.21, 4.89)$	$\mathcal{N}(286.16, 1.46)$
K_{Ac}^{AAB}	$\mathcal{N}(3.07, 1.84)$	$\mathcal{N}(11.24, 6.74)$	$\mathcal{N}(7.9, 4.74)$	$\mathcal{N}(2.15, 1.29)$	$\mathcal{N}(6.04, 3.62)$	$\mathcal{N}(5.88, 3.53)$	$\mathcal{N}(19.25, 11.55)$	$\mathcal{N}(5.97, 3.58)$
k_Y	$\mathcal{N}(0.0222, 0.0133)$	$\mathcal{N}(0.0515, 0.0309)$	$\mathcal{N}(0.0477, 0.0286)$	$\mathcal{N}(0.0373, 0.0224)$	$\mathcal{N}(0.0274, 0.0164)$	$\mathcal{N}(0.0247, 0.0148)$	$\mathcal{N}(0.0301, 0.0181)$	$\mathcal{N}(0.0829, 0.0497)$
k_{LAB}	$\mathcal{N}(0.0582, 0.0349)$	$\mathcal{N}(0.2222, 0.1333)$	-	-	-	-	$\mathcal{N}(0.0307, 0.0184)$	$\mathcal{N}(0.1026, 0.0616)$
k_{AAB}	$\mathcal{N}(0.0133, 0.008)$	$\mathcal{N}\left(0.001, 6 \times 10^{-4}\right)$	$\mathcal{N}(0.002, 0.0012)$	$\mathcal{N}(0.027, 0.0162)$	$\mathcal{N}(0.0034, 0.0021)$	$\mathcal{N}(0.0036, 0.0022)$	$\mathcal{N}\left(3 \times 10^{-4}, 2 \times 10^{-4}\right)$	$\mathcal{N}(0.0035, 0.0021)$
$Y_{Glc\|Y}$	$\mathcal{N}(27.2, 16.32)$	$\mathcal{N}(29.72, 17.83)$	$\mathcal{N}(47.18, 28.31)$	$\mathcal{N}(56.05, 33.63)$	$\mathcal{N}(32.41, 19.45)$	$\mathcal{N}(73.93, 44.36)$	$\mathcal{N}(316.47, 189.88)$	$\mathcal{N}(130.63, 78.38)$
$Y_{Glc\|LAB}$	$\mathcal{N}(45.89, 27.54)$	$\mathcal{N}(306.37, 183.82)$	$\mathcal{N}(59.55, 35.73)$	$\mathcal{N}(25.28, 15.17)$	$\mathcal{N}(25, 15)$	$\mathcal{N}(14.55, 8.73)$	$\mathcal{N}(57.08, 34.25)$	$\mathcal{N}(74.17, 44.5)$
$Y_{Fru\|Y}$	$\mathcal{N}(30.22, 18.13)$	$\mathcal{N}(46.37, 27.82)$	$\mathcal{N}(50.55, 30.33)$	$\mathcal{N}(70.82, 42.49)$	$\mathcal{N}(34.47, 20.68)$	$\mathcal{N}(77.05, 46.23)$	$\mathcal{N}(602.49, 361.49)$	$\mathcal{N}(116.94, 70.16)$
$Y_{Fru\|LAB}$	$\mathcal{N}(51, 30.6)$	$\mathcal{N}(478.02, 286.81)$	$\mathcal{N}(63.81, 38.29)$	$\mathcal{N}(31.94, 19.17)$	$\mathcal{N}(26.58, 15.95)$	$\mathcal{N}(15.17, 9.1)$	$\mathcal{N}(108.68, 65.21)$	$\mathcal{N}(66.4, 39.84)$
$Y_{EtOH\|Y}^{Glc}$	$\mathcal{N}(11.77, 7.06)$	$\mathcal{N}(5.07, 3.04)$	$\mathcal{N}(10.32, 6.19)$	$\mathcal{N}(18.69, 11.21)$	$\mathcal{N}(14.24, 8.55)$	$\mathcal{N}(30.37, 18.22)$	$\mathcal{N}(54.95, 32.97)$	$\mathcal{N}(14.2, 8.52)$
$Y_{EtOH\|Y}^{Fru}$	$\mathcal{N}(11.77, 7.06)$	$\mathcal{N}(5.07, 3.04)$	$\mathcal{N}(10.32, 6.19)$	$\mathcal{N}(18.69, 11.21)$	$\mathcal{N}(14.24, 8.55)$	$\mathcal{N}(30.37, 18.22)$	$\mathcal{N}(54.95, 32.97)$	$\mathcal{N}(14.2, 8.52)$
$Y_{EtOH\|Y}^{LA}$	$\mathcal{N}(11.77, 7.06)$	$\mathcal{N}(5.07, 3.04)$	$\mathcal{N}(10.32, 6.19)$	$\mathcal{N}(18.69, 11.21)$	$\mathcal{N}(14.24, 8.55)$	$\mathcal{N}(30.37, 18.22)$	$\mathcal{N}(54.95, 32.97)$	$\mathcal{N}(14.2, 8.52)$
$Y_{EtOH\|LAB}^{Glc}$	$\mathcal{N}(19.86, 11.92)$	$\mathcal{N}(52.23, 31.34)$	$\mathcal{N}(13.03, 7.82)$	$\mathcal{N}(8.43, 5.06)$	$\mathcal{N}(10.99, 6.59)$	$\mathcal{N}(5.98, 3.59)$	$\mathcal{N}(9.91, 5.95)$	$\mathcal{N}(8.06, 4.84)$
$Y_{EtOH\|LAB}^{Fru}$	$\mathcal{N}(19.86, 11.92)$	$\mathcal{N}(52.23, 31.34)$	$\mathcal{N}(13.03, 7.82)$	$\mathcal{N}(8.43, 5.06)$	$\mathcal{N}(10.99, 6.59)$	$\mathcal{N}(5.98, 3.59)$	$\mathcal{N}(9.91, 5.95)$	$\mathcal{N}(8.06, 4.84)$
$Y_{EtOH\|AAB}$	$\mathcal{N}(5872.73, 3523.64)$	$\mathcal{N}(107.57, 64.54)$	$\mathcal{N}(712.24, 427.35)$	$\mathcal{N}(481.43, 288.86)$	$\mathcal{N}(795.38, 477.23)$	$\mathcal{N}(1036.07, 621.64)$	$\mathcal{N}(56.22, 33.73)$	$\mathcal{N}(354.28, 212.57)$
$Y_{LA\|LAB}^{Glc}$	$\mathcal{N}(7.59, 4.55)$	$\mathcal{N}(12.11, 7.27)$	-	-	-	-	$\mathcal{N}(9.73, 5.84)$	$\mathcal{N}(6.51, 3.91)$
$Y_{LA\|LAB}^{Fru}$	$\mathcal{N}(7.59, 4.55)$	$\mathcal{N}(12.11, 7.27)$	-	-	-	-	$\mathcal{N}(9.73, 5.84)$	$\mathcal{N}(6.51, 3.91)$
$Y_{LA\|AAB}$	$\mathcal{N}(2243.66, 1346.19)$	$\mathcal{N}(24.95, 14.97)$	-	-	-	-	$\mathcal{N}(55.21, 33.12)$	$\mathcal{N}(286.16, 171.7)$
$Y_{LA\|Y}$	$\mathcal{N}(4.5, 2.7)$	$\mathcal{N}(1.18, 0.71)$	-	-	-	-	$\mathcal{N}(53.96, 32.38)$	$\mathcal{N}(11.47, 6.88)$
$Y_{Ac\|LAB}^{Glc}$	$\mathcal{N}(5.42, 3.25)$	$\mathcal{N}(120.98, 72.59)$	$\mathcal{N}(19.62, 11.77)$	$\mathcal{N}(2.7, 1.62)$	$\mathcal{N}(7.26, 4.36)$	$\mathcal{N}(3.47, 2.08)$	$\mathcal{N}(22.99, 13.79)$	$\mathcal{N}(15.95, 9.57)$
$Y_{Ac\|LAB}^{Fru}$	$\mathcal{N}(5.42, 3.25)$	$\mathcal{N}(120.98, 72.59)$	$\mathcal{N}(19.62, 11.77)$	$\mathcal{N}(2.7, 1.62)$	$\mathcal{N}(7.26, 4.36)$	$\mathcal{N}(3.47, 2.08)$	$\mathcal{N}(22.99, 13.79)$	$\mathcal{N}(15.95, 9.57)$
$Y_{Ac\|AAB}^{EtOH}$	$\mathcal{N}(1602.91, 961.75)$	$\mathcal{N}(249.15, 149.49)$	$\mathcal{N}(1072.81, 643.69)$	$\mathcal{N}(154.45, 92.67)$	$\mathcal{N}(525.57, 315.34)$	$\mathcal{N}(601.38, 360.83)$	$\mathcal{N}(130.4, 78.24)$	$\mathcal{N}(700.87, 420.52)$
$Y_{Ac\|AAB}^{LA}$	$\mathcal{N}(1602.91, 961.75)$	$\mathcal{N}(249.15, 149.49)$	$\mathcal{N}(1072.81, 643.69)$	$\mathcal{N}(154.45, 92.67)$	$\mathcal{N}(525.57, 315.34)$	$\mathcal{N}(601.38, 360.83)$	$\mathcal{N}(130.4, 78.24)$	$\mathcal{N}(700.87, 420.52)$
$Y_{Ac\|Y}^{Glc}$	$\mathcal{N}(3.21, 1.93)$	$\mathcal{N}(11.74, 7.04)$	$\mathcal{N}(15.55, 9.33)$	$\mathcal{N}(6, 3.6)$	$\mathcal{N}(9.41, 5.65)$	$\mathcal{N}(17.63, 10.58)$	$\mathcal{N}(127.45, 76.47)$	$\mathcal{N}(28.1, 16.86)$
$Y_{Ac\|Y}^{Fru}$	$\mathcal{N}(3.21, 1.93)$	$\mathcal{N}(11.74, 7.04)$	$\mathcal{N}(15.55, 9.33)$	$\mathcal{N}(6, 3.6)$	$\mathcal{N}(9.41, 5.65)$	$\mathcal{N}(17.63, 10.58)$	$\mathcal{N}(127.45, 76.47)$	$\mathcal{N}(28.1, 16.86)$
$Y_{Ac\|AAB}$	$\mathcal{N}(1602.91, 961.75)$	$\mathcal{N}(249.15, 149.49)$	$\mathcal{N}(1072.81, 643.69)$	$\mathcal{N}(154.45, 92.67)$	$\mathcal{N}(525.57, 315.34)$	$\mathcal{N}(601.38, 360.83)$	$\mathcal{N}(130.4, 78.24)$	$\mathcal{N}(700.87, 420.52)$
b_{EtOH}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{LA}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{Ac}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$

Table S2 (cont.): Prior distributions in the range of original units for parameters of all model iterations per dataset. All priors are truncated to the positive set of real numbers.

Parameter	brwb2	ecpt1	ecpt2	ecwb1	ecwb2	brpb1	brst1	mywb1
$\mu_{max}^{Y_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{Y_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{Y_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{LAB_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{LAB_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{AAB_{EtOH}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{AAB_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{max}^{AAB_{Ac}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
K_{Glc}^{Y}	$\mathcal{N}(21.47, 12.88)$	$\mathcal{N}(24.56, 14.74)$	$\mathcal{N}(21.33, 12.8)$	$\mathcal{N}(21.04, 12.62)$	$\mathcal{N}(18.34, 11)$	$\mathcal{N}(51.17, 30.7)$	$\mathcal{N}(66.24, 39.74)$	$\mathcal{N}(23.09, 13.85)$
K_{Fru}^{Y}	$\mathcal{N}(33.62, 20.17)$	$\mathcal{N}(26.97, 16.18)$	$\mathcal{N}(19.46, 11.68)$	$\mathcal{N}(26.95, 16.17)$	$\mathcal{N}(16.44, 9.87)$	$\mathcal{N}(47.53, 28.52)$	$\mathcal{N}(61.27, 36.76)$	$\mathcal{N}(27.89, 16.73)$
K_{LA}^{Y}	$\mathcal{N}(33.62, 20.17)$	$\mathcal{N}(26.97, 16.18)$	$\mathcal{N}(19.46, 11.68)$	$\mathcal{N}(26.95, 16.17)$	$\mathcal{N}(16.44, 9.87)$	$\mathcal{N}(47.53, 28.52)$	$\mathcal{N}(61.27, 36.76)$	$\mathcal{N}(27.89, 16.73)$
K_{Glc}^{LAB}	$\mathcal{N}(21.47, 12.88)$	$\mathcal{N}(24.56, 14.74)$	$\mathcal{N}(21.33, 12.8)$	$\mathcal{N}(21.04, 12.62)$	$\mathcal{N}(18.34, 11)$	$\mathcal{N}(51.17, 30.7)$	$\mathcal{N}(66.24, 39.74)$	$\mathcal{N}(23.09, 13.85)$
K_{Fru}^{LAB}	$\mathcal{N}(33.62, 20.17)$	$\mathcal{N}(26.97, 16.18)$	$\mathcal{N}(19.46, 11.68)$	$\mathcal{N}(26.95, 16.17)$	$\mathcal{N}(16.44, 9.87)$	$\mathcal{N}(47.53, 28.52)$	$\mathcal{N}(61.27, 36.76)$	$\mathcal{N}(27.89, 16.73)$
K_{EtOH}^{AAB}	$\mathcal{N}(3.53, 2.12)$	$\mathcal{N}(6.27, 3.76)$	$\mathcal{N}(4.81, 2.89)$	$\mathcal{N}(8.92, 5.35)$	$\mathcal{N}(6.26, 3.76)$	$\mathcal{N}(32.66, 19.6)$	$\mathcal{N}(39.08, 23.45)$	$\mathcal{N}(8.61, 5.16)$
K_{LA}^{AAB}	$\mathcal{N}(76.39, 1.24)$	$\mathcal{N}(173.95, 1.81)$	$\mathcal{N}(41.5, 0.49)$	$\mathcal{N}(929.25, 1.52)$	$\mathcal{N}(471.83, 1.55)$	$\mathcal{N}(26006.84, 9.43)$	$\mathcal{N}(58000.95, 7.04)$	$\mathcal{N}(1844.4, 1.7)$
K_{Ac}^{AAB}	$\mathcal{N}(9.31, 5.59)$	$\mathcal{N}(1.96, 1.18)$	$\mathcal{N}(8.67, 5.2)$	$\mathcal{N}(3.92, 2.35)$	$\mathcal{N}(12.05, 7.23)$	$\mathcal{N}(34.36, 20.62)$	$\mathcal{N}(9.36, 5.61)$	$\mathcal{N}(3.07, 1.84)$
k_Y	$\mathcal{N}(0.0709, 0.0425)$	$\mathcal{N}(0.0399, 0.0239)$	$\mathcal{N}(0.052, 0.0312)$	$\mathcal{N}(0.028, 0.0168)$	$\mathcal{N}(0.0399, 0.024)$	$\mathcal{N}(0.0077, 0.0046)$	$\mathcal{N}(0.0064, 0.0038)$	$\mathcal{N}(0.0291, 0.0174)$
k_{LAB}	$\mathcal{N}(0.1213, 0.0728)$	$\mathcal{N}(0.0829, 0.0497)$	$\mathcal{N}(0.306, 0.1836)$	$\mathcal{N}(0.0988, 0.0593)$	$\mathcal{N}(0.097, 0.0582)$	$\mathcal{N}(0.0159, 0.0095)$	$\mathcal{N}(0.0213, 0.0128)$	$\mathcal{N}(0.0881, 0.0529)$
k_{AAB}	$\mathcal{N}(0.0014, 9 \times 10^{-4})$	$\mathcal{N}(0.0325, 0.0195)$	$\mathcal{N}(0.0017, 0.001)$	$\mathcal{N}(0.0081, 0.0049)$	$\mathcal{N}(9 \times 10^{-4}, 5 \times 10^{-4})$	$\mathcal{N}(1 \times 10^{-4}, 1 \times 10^{-4})$	$\mathcal{N}(0.0014, 9 \times 10^{-4})$	$\mathcal{N}(0.0133, 0.008)$
$Y_{Glc\|Y}$	$\mathcal{N}(88.25, 52.95)$	$\mathcal{N}(22.45, 13.47)$	$\mathcal{N}(24.48, 14.69)$	$\mathcal{N}(25, 15)$	$\mathcal{N}(27.68, 16.61)$	$\mathcal{N}(93.31, 55.99)$	$\mathcal{N}(90.37, 54.22)$	$\mathcal{N}(52.51, 31.51)$
$Y_{Glc\|LAB}$	$\mathcal{N}(30.68, 18.41)$	$\mathcal{N}(44.09, 26.45)$	$\mathcal{N}(71.12, 42.67)$	$\mathcal{N}(32.52, 19.51)$	$\mathcal{N}(28.6, 17.16)$	$\mathcal{N}(152.45, 91.47)$	$\mathcal{N}(594.54, 356.73)$	$\mathcal{N}(28.41, 17.04)$
$Y_{Fru\|Y}$	$\mathcal{N}(138.22, 82.93)$	$\mathcal{N}(24.65, 14.79)$	$\mathcal{N}(22.34, 13.4)$	$\mathcal{N}(32.02, 19.21)$	$\mathcal{N}(24.82, 14.89)$	$\mathcal{N}(86.67, 52)$	$\mathcal{N}(83.59, 50.15)$	$\mathcal{N}(63.44, 38.07)$
$Y_{Fru\|LAB}$	$\mathcal{N}(48.06, 28.83)$	$\mathcal{N}(48.41, 29.04)$	$\mathcal{N}(64.9, 38.94)$	$\mathcal{N}(41.65, 24.99)$	$\mathcal{N}(25.65, 15.39)$	$\mathcal{N}(141.6, 84.96)$	$\mathcal{N}(549.96, 329.98)$	$\mathcal{N}(34.32, 20.59)$
$Y_{EtOH\|Y}^{Glc}$	$\mathcal{N}(14.5, 8.7)$	$\mathcal{N}(5.73, 3.44)$	$\mathcal{N}(5.52, 3.31)$	$\mathcal{N}(10.6, 6.36)$	$\mathcal{N}(9.45, 5.67)$	$\mathcal{N}(59.56, 35.74)$	$\mathcal{N}(53.31, 31.99)$	$\mathcal{N}(19.57, 11.74)$
$Y_{EtOH\|Y}^{Fru}$	$\mathcal{N}(14.5, 8.7)$	$\mathcal{N}(5.73, 3.44)$	$\mathcal{N}(5.52, 3.31)$	$\mathcal{N}(10.6, 6.36)$	$\mathcal{N}(9.45, 5.67)$	$\mathcal{N}(59.56, 35.74)$	$\mathcal{N}(53.31, 31.99)$	$\mathcal{N}(19.57, 11.74)$
$Y_{EtOH\|Y}^{LA}$	$\mathcal{N}(14.5, 8.7)$	$\mathcal{N}(5.73, 3.44)$	$\mathcal{N}(5.52, 3.31)$	$\mathcal{N}(10.6, 6.36)$	$\mathcal{N}(9.45, 5.67)$	$\mathcal{N}(59.56, 35.74)$	$\mathcal{N}(53.31, 31.99)$	$\mathcal{N}(19.57, 11.74)$
$Y_{EtOH\|LAB}^{Glc}$	$\mathcal{N}(5.04, 3.03)$	$\mathcal{N}(11.25, 6.75)$	$\mathcal{N}(16.04, 9.62)$	$\mathcal{N}(13.79, 8.28)$	$\mathcal{N}(9.76, 5.86)$	$\mathcal{N}(97.31, 58.38)$	$\mathcal{N}(350.75, 210.45)$	$\mathcal{N}(10.59, 6.35)$
$Y_{EtOH\|LAB}^{Fru}$	$\mathcal{N}(5.04, 3.03)$	$\mathcal{N}(11.25, 6.75)$	$\mathcal{N}(16.04, 9.62)$	$\mathcal{N}(13.79, 8.28)$	$\mathcal{N}(9.76, 5.86)$	$\mathcal{N}(97.31, 58.38)$	$\mathcal{N}(350.75, 210.45)$	$\mathcal{N}(10.59, 6.35)$
$Y_{EtOH\|AAB}$	$\mathcal{N}(130.75, 78.45)$	$\mathcal{N}(361.41, 216.85)$	$\mathcal{N}(244.26, 146.56)$	$\mathcal{N}(3276.25, 1965.75)$	$\mathcal{N}(1146.34, 687.8)$	$\mathcal{N}(54062.78, 32437.67)$	$\mathcal{N}(193086.14, 115851.68)$	$\mathcal{N}(5592.66, 3355.59)$
$Y_{LA\|LAB}^{Glc}$	$\mathcal{N}(2.95, 1.77)$	$\mathcal{N}(5.42, 3.25)$	$\mathcal{N}(2.72, 1.63)$	$\mathcal{N}(3.91, 2.35)$	$\mathcal{N}(4.02, 2.41)$	$\mathcal{N}(46.81, 28.09)$	$\mathcal{N}(105.36, 63.22)$	$\mathcal{N}(3.49, 2.1)$
$Y_{LA\|LAB}^{Fru}$	$\mathcal{N}(2.95, 1.77)$	$\mathcal{N}(5.42, 3.25)$	$\mathcal{N}(2.72, 1.63)$	$\mathcal{N}(3.91, 2.35)$	$\mathcal{N}(4.02, 2.41)$	$\mathcal{N}(46.81, 28.09)$	$\mathcal{N}(105.36, 63.22)$	$\mathcal{N}(3.49, 2.1)$
$Y_{LA\|AAB}$	$\mathcal{N}(76.39, 45.83)$	$\mathcal{N}(173.95, 104.37)$	$\mathcal{N}(41.5, 24.9)$	$\mathcal{N}(929.25, 557.55)$	$\mathcal{N}(471.83, 283.1)$	$\mathcal{N}(26006.84, 15604.1)$	$\mathcal{N}(58000.95, 34800.57)$	$\mathcal{N}(1844.4, 1106.64)$
$Y_{LA\|Y}$	$\mathcal{N}(8.47, 5.08)$	$\mathcal{N}(2.76, 1.65)$	$\mathcal{N}(0.94, 0.56)$	$\mathcal{N}(3.01, 1.8)$	$\mathcal{N}(3.89, 2.33)$	$\mathcal{N}(28.65, 17.19)$	$\mathcal{N}(16.01, 9.61)$	$\mathcal{N}(6.46, 3.87)$
$Y_{Ac\|LAB}^{Glc}$	$\mathcal{N}(13.31, 7.99)$	$\mathcal{N}(3.52, 2.11)$	$\mathcal{N}(28.91, 17.35)$	$\mathcal{N}(6.06, 3.64)$	$\mathcal{N}(18.79, 11.28)$	$\mathcal{N}(102.37, 61.42)$	$\mathcal{N}(83.97, 50.38)$	$\mathcal{N}(3.77, 2.26)$
$Y_{Ac\|LAB}^{Fru}$	$\mathcal{N}(13.31, 7.99)$	$\mathcal{N}(3.52, 2.11)$	$\mathcal{N}(28.91, 17.35)$	$\mathcal{N}(6.06, 3.64)$	$\mathcal{N}(18.79, 11.28)$	$\mathcal{N}(102.37, 61.42)$	$\mathcal{N}(83.97, 50.38)$	$\mathcal{N}(3.77, 2.26)$
$Y_{Ac\|AAB}^{EtOH}$	$\mathcal{N}(345.07, 207.04)$	$\mathcal{N}(113.06, 67.84)$	$\mathcal{N}(440.34, 264.2)$	$\mathcal{N}(1439.59, 863.75)$	$\mathcal{N}(2206.86, 1324.12)$	$\mathcal{N}(56876.56, 34125.94)$	$\mathcal{N}(46225.84, 27735.5)$	$\mathcal{N}(1993.55, 1196.13)$
$Y_{Ac\|AAB}^{LA}$	$\mathcal{N}(345.07, 207.04)$	$\mathcal{N}(113.06, 67.84)$	$\mathcal{N}(440.34, 264.2)$	$\mathcal{N}(1439.59, 863.75)$	$\mathcal{N}(2206.86, 1324.12)$	$\mathcal{N}(56876.56, 34125.94)$	$\mathcal{N}(46225.84, 27735.5)$	$\mathcal{N}(1993.55, 1196.13)$
$Y_{Ac\|Y}^{Glc}$	$\mathcal{N}(38.28, 22.97)$	$\mathcal{N}(1.79, 1.08)$	$\mathcal{N}(9.95, 5.97)$	$\mathcal{N}(4.66, 2.8)$	$\mathcal{N}(18.19, 10.91)$	$\mathcal{N}(62.66, 37.6)$	$\mathcal{N}(12.76, 7.66)$	$\mathcal{N}(6.98, 4.19)$
$Y_{Ac\|Y}^{Fru}$	$\mathcal{N}(38.28, 22.97)$	$\mathcal{N}(1.79, 1.08)$	$\mathcal{N}(9.95, 5.97)$	$\mathcal{N}(4.66, 2.8)$	$\mathcal{N}(18.19, 10.91)$	$\mathcal{N}(62.66, 37.6)$	$\mathcal{N}(12.76, 7.66)$	$\mathcal{N}(6.98, 4.19)$
$Y_{Ac\|AAB}$	$\mathcal{N}(345.07, 207.04)$	$\mathcal{N}(113.06, 67.84)$	$\mathcal{N}(440.34, 264.2)$	$\mathcal{N}(1439.59, 863.75)$	$\mathcal{N}(2206.86, 1324.12)$	$\mathcal{N}(56876.56, 34125.94)$	$\mathcal{N}(46225.84, 27735.5)$	$\mathcal{N}(1993.55, 1196.13)$
b_{EtOH}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{LA}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{Ac}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$

Table S2 (cont.): Prior distributions in the range of original units for parameters of all model iterations per dataset. All priors are truncated to the positive set of real numbers.

Parameter	mywb2	ghhp6	ghhp7	brwb3	brwb4	brst2	brwb5	brwb6
$\mu_{\max}^{Y_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{Y_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{Y_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{LAB_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{LAB_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{AAB_{EtOH}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{AAB_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{AAB_{Ac}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
K_{Glc}^{Y}	$\mathcal{N}(14.2, 8.52)$	$\mathcal{N}(21.27, 12.76)$	$\mathcal{N}(25.54, 15.32)$	$\mathcal{N}(73.29, 43.98)$	$\mathcal{N}(66.49, 39.89)$	$\mathcal{N}(61.44, 36.87)$	$\mathcal{N}(5.02, 3.01)$	$\mathcal{N}(6.71, 4.03)$
K_{Fru}^{Y}	$\mathcal{N}(14.74, 8.84)$	$\mathcal{N}(39.14, 23.49)$	$\mathcal{N}(29.67, 17.8)$	$\mathcal{N}(81.08, 48.65)$	$\mathcal{N}(74.5, 44.7)$	$\mathcal{N}(68.28, 40.97)$	$\mathcal{N}(6.31, 3.78)$	$\mathcal{N}(8.04, 4.82)$
K_{LA}^{Y}	$\mathcal{N}(14.74, 8.84)$	$\mathcal{N}(39.14, 23.49)$	$\mathcal{N}(29.67, 17.8)$	$\mathcal{N}(81.08, 48.65)$	$\mathcal{N}(74.5, 44.7)$	$\mathcal{N}(68.28, 40.97)$	$\mathcal{N}(6.31, 3.78)$	$\mathcal{N}(8.04, 4.82)$
K_{Glc}^{LAB}	$\mathcal{N}(14.2, 8.52)$	$\mathcal{N}(21.27, 12.76)$	$\mathcal{N}(25.54, 15.32)$	$\mathcal{N}(73.29, 43.98)$	$\mathcal{N}(66.49, 39.89)$	$\mathcal{N}(61.44, 36.87)$	$\mathcal{N}(5.02, 3.01)$	$\mathcal{N}(6.71, 4.03)$
K_{Fru}^{LAB}	$\mathcal{N}(14.74, 8.84)$	$\mathcal{N}(39.14, 23.49)$	$\mathcal{N}(29.67, 17.8)$	$\mathcal{N}(81.08, 48.65)$	$\mathcal{N}(74.5, 44.7)$	$\mathcal{N}(68.28, 40.97)$	$\mathcal{N}(6.31, 3.78)$	$\mathcal{N}(8.04, 4.82)$
K_{EtOH}^{AAB}	$\mathcal{N}(6.69, 4.02)$	$\mathcal{N}(4.59, 2.75)$	$\mathcal{N}(9.06, 5.44)$	$\mathcal{N}(31.81, 19.09)$	$\mathcal{N}(23.92, 14.35)$	$\mathcal{N}(35.5, 21.3)$	$\mathcal{N}(1.54, 0.92)$	$\mathcal{N}(1.99, 1.19)$
K_{LA}^{AAB}	$\mathcal{N}(1126.25, 3.27)$	$\mathcal{N}(33.82, 1.04)$	$\mathcal{N}(108.9, 0.85)$	$\mathcal{N}(9973.04, 5.08)$	$\mathcal{N}(17956.46, 6.62)$	$\mathcal{N}(444524.64, 7.13)$	$\mathcal{N}(2072.3, 0.25)$	$\mathcal{N}(6615.52, 0.27)$
K_{Ac}^{AAB}	$\mathcal{N}(3.84, 2.31)$	$\mathcal{N}(3.4, 2.04)$	$\mathcal{N}(4.14, 2.49)$	$\mathcal{N}(18.09, 10.85)$	$\mathcal{N}(13.46, 8.08)$	$\mathcal{N}(28.71, 17.23)$	$\mathcal{N}(0.8, 0.48)$	$\mathcal{N}(0.85, 0.51)$
k_Y	$\mathcal{N}(0.0374, 0.0224)$	$\mathcal{N}(0.0545, 0.0327)$	$\mathcal{N}(0.0276, 0.0166)$	$\mathcal{N}(0.0079, 0.0047)$	$\mathcal{N}(0.0105, 0.0063)$	$\mathcal{N}(0.007, 0.0042)$	$\mathcal{N}(0.1627, 0.0976)$	$\mathcal{N}(0.1258, 0.0755)$
k_{LAB}	$\mathcal{N}(0.0458, 0.0275)$	$\mathcal{N}(0.1444, 0.0867)$	$\mathcal{N}(0.1757, 0.1054)$	$\mathcal{N}(0.0295, 0.0177)$	$\mathcal{N}(0.0227, 0.0136)$	$\mathcal{N}(0.021, 0.0126)$	$\mathcal{N}(0.5903, 0.3542)$	$\mathcal{N}(0.5482, 0.3289)$
k_{AAB}	$\mathcal{N}(0.0085, 0.0051)$	$\mathcal{N}(0.0108, 0.0065)$	$\mathcal{N}(0.0073, 0.0044)$	$\mathcal{N}\left(4 \times 10^{-4}, 2 \times 10^{-4}\right)$	$\mathcal{N}\left(7 \times 10^{-4}, 4 \times 10^{-4}\right)$	$\mathcal{N}\left(2 \times 10^{-4}, 1 \times 10^{-4}\right)$	$\mathcal{N}(0.197, 0.1182)$	$\mathcal{N}(0.1742, 0.1045)$
$Y_{Glc\|Y}$	$\mathcal{N}(72.33, 43.4)$	$\mathcal{N}(154.4, 92.64)$	$\mathcal{N}(21.14, 12.69)$	$\mathcal{N}(38.72, 23.23)$	$\mathcal{N}(122.92, 73.75)$	$\mathcal{N}(199.25, 119.55)$	$\mathcal{N}(833.73, 500.24)$	$\mathcal{N}(556.73, 334.04)$
$Y_{Glc\|LAB}$	$\mathcal{N}(351.94, 211.16)$	$\mathcal{N}(26.29, 15.78)$	$\mathcal{N}(14.01, 8.4)$	$\mathcal{N}(587.71, 352.62)$	$\mathcal{N}(195.35, 117.21)$	$\mathcal{N}(1070.46, 642.27)$	$\mathcal{N}(2119.36, 1271.62)$	$\mathcal{N}(10588.37, 6353.02)$
$Y_{Fru\|Y}$	$\mathcal{N}(75.05, 45.03)$	$\mathcal{N}(284.17, 170.5)$	$\mathcal{N}(24.56, 14.74)$	$\mathcal{N}(42.84, 25.7)$	$\mathcal{N}(137.74, 82.65)$	$\mathcal{N}(221.42, 132.85)$	$\mathcal{N}(1048.14, 628.89)$	$\mathcal{N}(666.52, 399.91)$
$Y_{Fru\|LAB}$	$\mathcal{N}(365.19, 219.11)$	$\mathcal{N}(48.39, 29.03)$	$\mathcal{N}(16.27, 9.76)$	$\mathcal{N}(650.11, 390.06)$	$\mathcal{N}(218.9, 131.34)$	$\mathcal{N}(1189.6, 713.76)$	$\mathcal{N}(2664.41, 1598.65)$	$\mathcal{N}(12676.39, 7605.84)$
$Y_{EtOH\|Y}^{Glc}$	$\mathcal{N}(34.08, 20.45)$	$\mathcal{N}(33.32, 19.99)$	$\mathcal{N}(7.5, 4.5)$	$\mathcal{N}(16.81, 10.08)$	$\mathcal{N}(44.22, 26.53)$	$\mathcal{N}(115.12, 69.07)$	$\mathcal{N}(255.47, 153.28)$	$\mathcal{N}(164.85, 98.91)$
$Y_{EtOH\|Y}^{Fru}$	$\mathcal{N}(34.08, 20.45)$	$\mathcal{N}(33.32, 19.99)$	$\mathcal{N}(7.5, 4.5)$	$\mathcal{N}(16.81, 10.08)$	$\mathcal{N}(44.22, 26.53)$	$\mathcal{N}(115.12, 69.07)$	$\mathcal{N}(255.47, 153.28)$	$\mathcal{N}(164.85, 98.91)$
$Y_{EtOH\|Y}^{LA}$	$\mathcal{N}(34.08, 20.45)$	$\mathcal{N}(33.32, 19.99)$	$\mathcal{N}(7.5, 4.5)$	$\mathcal{N}(16.81, 10.08)$	$\mathcal{N}(44.22, 26.53)$	$\mathcal{N}(115.12, 69.07)$	$\mathcal{N}(255.47, 153.28)$	$\mathcal{N}(164.85, 98.91)$
$Y_{EtOH\|LAB}^{Glc}$	$\mathcal{N}(165.82, 99.49)$	$\mathcal{N}(5.67, 3.4)$	$\mathcal{N}(4.97, 2.98)$	$\mathcal{N}(255.09, 153.06)$	$\mathcal{N}(70.28, 42.17)$	$\mathcal{N}(618.5, 371.1)$	$\mathcal{N}(649.41, 389.65)$	$\mathcal{N}(3135.19, 1881.12)$
$Y_{EtOH\|LAB}^{Fru}$	$\mathcal{N}(165.82, 99.49)$	$\mathcal{N}(5.67, 3.4)$	$\mathcal{N}(4.97, 2.98)$	$\mathcal{N}(255.09, 153.06)$	$\mathcal{N}(70.28, 42.17)$	$\mathcal{N}(618.5, 371.1)$	$\mathcal{N}(649.41, 389.65)$	$\mathcal{N}(3135.19, 1881.12)$
$Y_{EtOH\|AAB}$	$\mathcal{N}(1381.65, 828.99)$	$\mathcal{N}(89.68, 53.81)$	$\mathcal{N}(693.53, 416.12)$	$\mathcal{N}(37450.12, 22470.07)$	$\mathcal{N}(38957.79, 23374.68)$	$\mathcal{N}(1327683.83, 796610.3)$	$\mathcal{N}(7520.97, 4512.58)$	$\mathcal{N}(28841.36, 17304.81)$
$Y_{LA\|LAB}^{Glc}$	$\mathcal{N}(135.17, 81.1)$	$\mathcal{N}(2.14, 1.28)$	$\mathcal{N}(0.78, 0.47)$	$\mathcal{N}(67.93, 40.76)$	$\mathcal{N}(32.39, 19.44)$	$\mathcal{N}(207.08, 124.25)$	$\mathcal{N}(178.94, 107.36)$	$\mathcal{N}(719.14, 431.48)$
$Y_{LA\|LAB}^{Fru}$	$\mathcal{N}(135.17, 81.1)$	$\mathcal{N}(2.14, 1.28)$	$\mathcal{N}(0.78, 0.47)$	$\mathcal{N}(67.93, 40.76)$	$\mathcal{N}(32.39, 19.44)$	$\mathcal{N}(207.08, 124.25)$	$\mathcal{N}(178.94, 107.36)$	$\mathcal{N}(719.14, 431.48)$
$Y_{LA\|AAB}$	$\mathcal{N}(1126.25, 675.75)$	$\mathcal{N}(33.82, 20.29)$	$\mathcal{N}(108.9, 65.34)$	$\mathcal{N}(9973.04, 5983.83)$	$\mathcal{N}(17956.46, 10773.88)$	$\mathcal{N}(444524.64, 266714.78)$	$\mathcal{N}(2072.3, 1243.38)$	$\mathcal{N}(6615.52, 3969.31)$
$Y_{LA\|Y}$	$\mathcal{N}(27.78, 16.67)$	$\mathcal{N}(12.57, 7.54)$	$\mathcal{N}(1.18, 0.71)$	$\mathcal{N}(4.48, 2.69)$	$\mathcal{N}(20.38, 12.23)$	$\mathcal{N}(38.54, 23.13)$	$\mathcal{N}(70.39, 42.23)$	$\mathcal{N}(37.81, 22.69)$
$Y_{Ac\|LAB}^{Glc}$	$\mathcal{N}(95.23, 57.14)$	$\mathcal{N}(4.2, 2.52)$	$\mathcal{N}(2.27, 1.36)$	$\mathcal{N}(145.04, 87.02)$	$\mathcal{N}(39.55, 23.73)$	$\mathcal{N}(500.19, 300.11)$	$\mathcal{N}(336.54, 201.92)$	$\mathcal{N}(1335.77, 801.46)$
$Y_{Ac\|LAB}^{Fru}$	$\mathcal{N}(95.23, 57.14)$	$\mathcal{N}(4.2, 2.52)$	$\mathcal{N}(2.27, 1.36)$	$\mathcal{N}(145.04, 87.02)$	$\mathcal{N}(39.55, 23.73)$	$\mathcal{N}(500.19, 300.11)$	$\mathcal{N}(336.54, 201.92)$	$\mathcal{N}(1335.77, 801.46)$
$Y_{Ac\|AAB}^{EtOH}$	$\mathcal{N}(793.44, 476.06)$	$\mathcal{N}(66.39, 39.84)$	$\mathcal{N}(317.11, 190.27)$	$\mathcal{N}(21293.36, 12776.02)$	$\mathcal{N}(21923.17, 13153.9)$	$\mathcal{N}(1073699.13, 644219.48)$	$\mathcal{N}(3897.5, 2338.5)$	$\mathcal{N}(12288.04, 7372.83)$
$Y_{Ac\|AAB}^{LA}$	$\mathcal{N}(793.44, 476.06)$	$\mathcal{N}(66.39, 39.84)$	$\mathcal{N}(317.11, 190.27)$	$\mathcal{N}(21293.36, 12776.02)$	$\mathcal{N}(21923.17, 13153.9)$	$\mathcal{N}(1073699.13, 644219.48)$	$\mathcal{N}(3897.5, 2338.5)$	$\mathcal{N}(12288.04, 7372.83)$
$Y_{Ac\|Y}^{Glc}$	$\mathcal{N}(19.57, 11.74)$	$\mathcal{N}(24.67, 14.8)$	$\mathcal{N}(3.43, 2.06)$	$\mathcal{N}(9.56, 5.73)$	$\mathcal{N}(24.89, 14.93)$	$\mathcal{N}(93.1, 55.86)$	$\mathcal{N}(132.39, 79.43)$	$\mathcal{N}(70.23, 42.14)$
$Y_{Ac\|Y}^{Fru}$	$\mathcal{N}(19.57, 11.74)$	$\mathcal{N}(24.67, 14.8)$	$\mathcal{N}(3.43, 2.06)$	$\mathcal{N}(9.56, 5.73)$	$\mathcal{N}(24.89, 14.93)$	$\mathcal{N}(93.1, 55.86)$	$\mathcal{N}(132.39, 79.43)$	$\mathcal{N}(70.23, 42.14)$
$Y_{Ac\|AAB}$	$\mathcal{N}(793.44, 476.06)$	$\mathcal{N}(66.39, 39.84)$	$\mathcal{N}(317.11, 190.27)$	$\mathcal{N}(21293.36, 12776.02)$	$\mathcal{N}(21923.17, 13153.9)$	$\mathcal{N}(1073699.13, 644219.48)$	$\mathcal{N}(3897.5, 2338.5)$	$\mathcal{N}(12288.04, 7372.83)$
b_{EtOH}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{LA}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{Ac}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$

Table S2 (cont.): Prior distributions in the range of original units for parameters of all model iterations per dataset. All priors are truncated to the positive set of real numbers.

Parameter	brwb7	mywb3	brwb8	hnwb1	ecpb1	niwb1	niwb2	ecpb2
$\mu_{\max}^{Y_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{Y_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{Y_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{LAB_{Glc}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{LAB_{Fru}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{AAB_{EtOH}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{AAB_{LA}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
$\mu_{\max}^{AAB_{Ac}}$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
K_{Glc}^{Y}	$\mathcal{N}(7.85, 4.71)$	$\mathcal{N}(19.96, 11.98)$	$\mathcal{N}(40.87, 24.52)$	$\mathcal{N}(21.78, 13.07)$	$\mathcal{N}(14.95, 8.97)$	$\mathcal{N}(12.18, 7.31)$	$\mathcal{N}(10.71, 6.43)$	$\mathcal{N}(14.89, 8.94)$
K_{Fru}^{Y}	$\mathcal{N}(9.29, 5.57)$	$\mathcal{N}(25.92, 15.55)$	$\mathcal{N}(43.24, 25.95)$	$\mathcal{N}(26.8, 16.08)$	$\mathcal{N}(18.32, 10.99)$	$\mathcal{N}(15.62, 9.38)$	$\mathcal{N}(12.77, 7.66)$	$\mathcal{N}(17.64, 10.59)$
K_{LA}^{Y}	$\mathcal{N}(9.29, 5.57)$	$\mathcal{N}(25.92, 15.55)$	$\mathcal{N}(43.24, 25.95)$	$\mathcal{N}(26.8, 16.08)$	$\mathcal{N}(18.32, 10.99)$	$\mathcal{N}(15.62, 9.38)$	$\mathcal{N}(12.77, 7.66)$	$\mathcal{N}(17.64, 10.59)$
K_{Glc}^{LAB}	$\mathcal{N}(7.85, 4.71)$	$\mathcal{N}(19.96, 11.98)$	$\mathcal{N}(40.87, 24.52)$	$\mathcal{N}(21.78, 13.07)$	$\mathcal{N}(14.95, 8.97)$	$\mathcal{N}(12.18, 7.31)$	$\mathcal{N}(10.71, 6.43)$	$\mathcal{N}(14.89, 8.94)$
K_{Fru}^{LAB}	$\mathcal{N}(9.29, 5.57)$	$\mathcal{N}(25.92, 15.55)$	$\mathcal{N}(43.24, 25.95)$	$\mathcal{N}(26.8, 16.08)$	$\mathcal{N}(18.32, 10.99)$	$\mathcal{N}(15.62, 9.38)$	$\mathcal{N}(12.77, 7.66)$	$\mathcal{N}(17.64, 10.59)$
K_{EtOH}^{AAB}	$\mathcal{N}(4.89, 2.93)$	$\mathcal{N}(8.61, 5.17)$	$\mathcal{N}(5.41, 3.25)$	$\mathcal{N}(10.93, 6.56)$	$\mathcal{N}(9.95, 5.97)$	$\mathcal{N}(8.71, 5.22)$	$\mathcal{N}(9.37, 5.62)$	$\mathcal{N}(19.3, 11.58)$
K_{LA}^{AAB}	$\mathcal{N}(1648.18, 0.3)$	$\mathcal{N}(453.45, 3.58)$	$\mathcal{N}(125.38, 3.35)$	$\mathcal{N}(242.02, 1.47)$	$\mathcal{N}(0.98, 3.55)$	$\mathcal{N}(596.41, 1.7)$	$\mathcal{N}(141.89, 2.33)$	$\mathcal{N}(13.6, 1.58)$
K_{Ac}^{AAB}	$\mathcal{N}(1.72, 1.03)$	$\mathcal{N}(4.96, 2.98)$	$\mathcal{N}(0.6, 0.36)$	$\mathcal{N}(7.15, 4.29)$	$\mathcal{N}(12.35, 7.41)$	$\mathcal{N}(7.92, 4.75)$	$\mathcal{N}(4.22, 2.53)$	$\mathcal{N}(4.01, 2.41)$
k_{Y}	$\mathcal{N}(0.0512, 0.0307)$	$\mathcal{N}(0.029, 0.0174)$	$\mathcal{N}(0.0462, 0.0277)$	$\mathcal{N}(0.0229, 0.0137)$	$\mathcal{N}(0.0251, 0.0151)$	$\mathcal{N}(0.0287, 0.0172)$	$\mathcal{N}(0.0267, 0.016)$	$\mathcal{N}(0.013, 0.0078)$
k_{LAB}	$\mathcal{N}(0.4995, 0.2997)$	$\mathcal{N}(0.0419, 0.0251)$	$\mathcal{N}(0.0448, 0.0269)$	$\mathcal{N}(0.1023, 0.0614)$	$\mathcal{N}(0.0423, 0.0254)$	$\mathcal{N}(0.0882, 0.0529)$	$\mathcal{N}(0.0644, 0.0386)$	$\mathcal{N}(0.0951, 0.0571)$
k_{AAB}	$\mathcal{N}(0.0423, 0.0254)$	$\mathcal{N}(0.0051, 0.003)$	$\mathcal{N}(0.3478, 0.2087)$	$\mathcal{N}(0.0024, 0.0015)$	$\mathcal{N}\left(8 \times 10^{-4}, 5 \times 10^{-4}\right)$	$\mathcal{N}(0.002, 0.0012)$	$\mathcal{N}(0.007, 0.0042)$	$\mathcal{N}(0.0078, 0.0047)$
$Y_{Glc\|Y}$	$\mathcal{N}(771.56, 462.94)$	$\mathcal{N}(77.29, 46.38)$	$\mathcal{N}(2.85, 1.71)$	$\mathcal{N}(84.12, 50.47)$	$\mathcal{N}(0.2, 0.12)$	$\mathcal{N}(13.47, 8.08)$	$\mathcal{N}(9.2, 5.52)$	$\mathcal{N}(4.82, 2.89)$
$Y_{Glc\|LAB}$	$\mathcal{N}(5566.1, 3339.66)$	$\mathcal{N}(14.8, 8.88)$	$\mathcal{N}(191.21, 114.72)$	$\mathcal{N}(64.14, 38.48)$	$\mathcal{N}(0.55, 0.33)$	$\mathcal{N}(756.1, 453.66)$	$\mathcal{N}(52.82, 31.69)$	$\mathcal{N}(12.39, 7.43)$
$Y_{Fru\|Y}$	$\mathcal{N}(913.39, 548.04)$	$\mathcal{N}(100.35, 60.21)$	$\mathcal{N}(3.02, 1.81)$	$\mathcal{N}(103.53, 62.12)$	$\mathcal{N}(0.25, 0.15)$	$\mathcal{N}(17.29, 10.37)$	$\mathcal{N}(10.96, 6.58)$	$\mathcal{N}(5.71, 3.43)$
$Y_{Fru\|LAB}$	$\mathcal{N}(6589.31, 3953.58)$	$\mathcal{N}(19.22, 11.53)$	$\mathcal{N}(202.3, 121.38)$	$\mathcal{N}(78.93, 47.36)$	$\mathcal{N}(0.68, 0.41)$	$\mathcal{N}(970.31, 582.19)$	$\mathcal{N}(62.97, 37.78)$	$\mathcal{N}(14.68, 8.81)$
$Y_{EtOH\|Y}^{Glc}$	$\mathcal{N}(480.52, 288.31)$	$\mathcal{N}(33.35, 20.01)$	$\mathcal{N}(0.38, 0.23)$	$\mathcal{N}(42.23, 25.34)$	$\mathcal{N}(0.14, 0.08)$	$\mathcal{N}(9.63, 5.78)$	$\mathcal{N}(8.05, 4.83)$	$\mathcal{N}(6.24, 3.75)$
$Y_{EtOH\|Y}^{Fru}$	$\mathcal{N}(480.52, 288.31)$	$\mathcal{N}(33.35, 20.01)$	$\mathcal{N}(0.38, 0.23)$	$\mathcal{N}(42.23, 25.34)$	$\mathcal{N}(0.14, 0.08)$	$\mathcal{N}(9.63, 5.78)$	$\mathcal{N}(8.05, 4.83)$	$\mathcal{N}(6.24, 3.75)$
$Y_{EtOH\|Y}^{LA}$	$\mathcal{N}(480.52, 288.31)$	$\mathcal{N}(33.35, 20.01)$	$\mathcal{N}(0.38, 0.23)$	$\mathcal{N}(42.23, 25.34)$	$\mathcal{N}(0.14, 0.08)$	$\mathcal{N}(9.63, 5.78)$	$\mathcal{N}(8.05, 4.83)$	$\mathcal{N}(6.24, 3.75)$
$Y_{EtOH\|LAB}^{Glc}$	$\mathcal{N}(3466.49, 2079.89)$	$\mathcal{N}(6.39, 3.83)$	$\mathcal{N}(25.33, 15.2)$	$\mathcal{N}(32.19, 19.32)$	$\mathcal{N}(0.37, 0.22)$	$\mathcal{N}(540.64, 324.38)$	$\mathcal{N}(46.23, 27.74)$	$\mathcal{N}(16.06, 9.63)$
$Y_{EtOH\|LAB}^{Fru}$	$\mathcal{N}(3466.49, 2079.89)$	$\mathcal{N}(6.39, 3.83)$	$\mathcal{N}(25.33, 15.2)$	$\mathcal{N}(32.19, 19.32)$	$\mathcal{N}(0.37, 0.22)$	$\mathcal{N}(540.64, 324.38)$	$\mathcal{N}(46.23, 27.74)$	$\mathcal{N}(16.06, 9.63)$
$Y_{EtOH\|AAB}$	$\mathcal{N}(16093.18, 9655.91)$	$\mathcal{N}(654.59, 392.76)$	$\mathcal{N}(121.73, 73.04)$	$\mathcal{N}(1082.73, 649.64)$	$\mathcal{N}(1.64, 0.99)$	$\mathcal{N}(1830.88, 1098.53)$	$\mathcal{N}(342.53, 205.52)$	$\mathcal{N}(99.86, 59.92)$
$Y_{LA\|LAB}^{Glc}$	$\mathcal{N}(355.02, 213.01)$	$\mathcal{N}(4.42, 2.65)$	$\mathcal{N}(26.09, 15.65)$	$\mathcal{N}(7.2, 4.32)$	$\mathcal{N}(0.22, 0.13)$	$\mathcal{N}(176.11, 105.67)$	$\mathcal{N}(19.15, 11.49)$	$\mathcal{N}(2.19, 1.31)$
$Y_{LA\|LAB}^{Fru}$	$\mathcal{N}(355.02, 213.01)$	$\mathcal{N}(4.42, 2.65)$	$\mathcal{N}(26.09, 15.65)$	$\mathcal{N}(7.2, 4.32)$	$\mathcal{N}(0.22, 0.13)$	$\mathcal{N}(176.11, 105.67)$	$\mathcal{N}(19.15, 11.49)$	$\mathcal{N}(2.19, 1.31)$
$Y_{LA\|AAB}$	$\mathcal{N}(1648.18, 988.91)$	$\mathcal{N}(453.45, 272.07)$	$\mathcal{N}(125.38, 75.23)$	$\mathcal{N}(242.02, 145.21)$	$\mathcal{N}(0.98, 0.59)$	$\mathcal{N}(596.41, 357.85)$	$\mathcal{N}(141.89, 85.13)$	$\mathcal{N}(13.6, 8.16)$
$Y_{LA\|Y}$	$\mathcal{N}(49.21, 29.53)$	$\mathcal{N}(23.1, 13.86)$	$\mathcal{N}(0.39, 0.23)$	$\mathcal{N}(9.44, 5.66)$	$\mathcal{N}(0.08, 0.05)$	$\mathcal{N}(3.14, 1.88)$	$\mathcal{N}(3.33, 2)$	$\mathcal{N}(0.85, 0.51)$
$Y_{Ac\|LAB}^{Glc}$	$\mathcal{N}(1219.34, 731.6)$	$\mathcal{N}(3.68, 2.21)$	$\mathcal{N}(2.8, 1.68)$	$\mathcal{N}(21.06, 12.64)$	$\mathcal{N}(0.46, 0.27)$	$\mathcal{N}(492.02, 295.21)$	$\mathcal{N}(20.82, 12.49)$	$\mathcal{N}(3.34, 2)$
$Y_{Ac\|LAB}^{Fru}$	$\mathcal{N}(1219.34, 731.6)$	$\mathcal{N}(3.68, 2.21)$	$\mathcal{N}(2.8, 1.68)$	$\mathcal{N}(21.06, 12.64)$	$\mathcal{N}(0.46, 0.27)$	$\mathcal{N}(492.02, 295.21)$	$\mathcal{N}(20.82, 12.49)$	$\mathcal{N}(3.34, 2)$
$Y_{Ac\|AAB}^{EtOH}$	$\mathcal{N}(5660.77, 3396.46)$	$\mathcal{N}(377.21, 226.33)$	$\mathcal{N}(13.48, 8.09)$	$\mathcal{N}(708.29, 424.97)$	$\mathcal{N}(2.04, 1.22)$	$\mathcal{N}(1666.22, 999.73)$	$\mathcal{N}(154.28, 92.57)$	$\mathcal{N}(20.76, 12.46)$
$Y_{Ac\|AAB}^{LA}$	$\mathcal{N}(5660.77, 3396.46)$	$\mathcal{N}(377.21, 226.33)$	$\mathcal{N}(13.48, 8.09)$	$\mathcal{N}(708.29, 424.97)$	$\mathcal{N}(2.04, 1.22)$	$\mathcal{N}(1666.22, 999.73)$	$\mathcal{N}(154.28, 92.57)$	$\mathcal{N}(20.76, 12.46)$
$Y_{Ac\|Y}^{Glc}$	$\mathcal{N}(169.02, 101.41)$	$\mathcal{N}(19.22, 11.53)$	$\mathcal{N}(0.04, 0.03)$	$\mathcal{N}(27.62, 16.57)$	$\mathcal{N}(0.17, 0.1)$	$\mathcal{N}(8.77, 5.26)$	$\mathcal{N}(3.63, 2.18)$	$\mathcal{N}(1.3, 0.78)$
$Y_{Ac\|Y}^{Fru}$	$\mathcal{N}(169.02, 101.41)$	$\mathcal{N}(19.22, 11.53)$	$\mathcal{N}(0.04, 0.03)$	$\mathcal{N}(27.62, 16.57)$	$\mathcal{N}(0.17, 0.1)$	$\mathcal{N}(8.77, 5.26)$	$\mathcal{N}(3.63, 2.18)$	$\mathcal{N}(1.3, 0.78)$
$Y_{Ac\|AAB}$	$\mathcal{N}(5660.77, 3396.46)$	$\mathcal{N}(377.21, 226.33)$	$\mathcal{N}(13.48, 8.09)$	$\mathcal{N}(708.29, 424.97)$	$\mathcal{N}(2.04, 1.22)$	$\mathcal{N}(1666.22, 999.73)$	$\mathcal{N}(154.28, 92.57)$	$\mathcal{N}(20.76, 12.46)$
b_{EtOH}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{LA}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$
b_{Ac}	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$	$\mathcal{N}(0.5, 0.3)$

Table S3: Summary of $\widehat{R}$ among successful fits across 31 model iterations (MI). Maximum $\widehat{R}$ in the referred MI is reported. Column "MI( )" indicates combination of mechanisms deployed in MI.

MI( )	ghhp1	dowb1	ghhp2	ghhp3	ghhp4	ghhp5	brwb1	brwb2	ecpt1	ecpt2	ecwb1	ecwb2	brpb1	brst1	brwb3	brwb4	brst2	brwb7	mywb3	hnwb1	ecpb1	niwb1	niwb2
0	1.00	-	-	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	-	1.00
1	1.00	-	-	-	1.00	1.00	1.00	1.00	-	1.00	1.00	1.01	1.01	-	-	1.00	1.00	-	-	-	-	-	1.00
2	1.00	-	-	1.00	1.00	-	1.00	1.01	1.00	1.00	-	1.01	1.00	1.00	-	-	1.00	1.00	1.00	-	-	-	1.00
3	1.00	-	-	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	-	1.00	1.00	1.00	1.00	1.00
4	1.00	-	-	-	1.00	1.00	1.00	1.00	-	1.00	1.00	1.00	1.00	1.00	-	1.00	1.01	-	1.00	1.00	-	-	1.00
5	-	-	-	-	-	-	1.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2	1.00	-	1.00	1.00	1.00	-	1.00	1.00	1.00	-	1.00	-	1.00	-	-	-	-	-	-	-	-	-	-
1,3	1.00	-	-	-	-	1.01	1.00	1.00	1.00	1.00	1.00	1.02	1.00	-	-	1.00	1.00	-	-	-	-	-	-
1,4	-	-	-	-	-	1.00	1.00	1.00	1.00	-	1.01	-	1.00	-	-	-	1.00	-	1.00	-	-	-	1.01
1,5	-	-	-	-	-	-	1.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.00	1.00	-	-	-	-	1.00	-	-	1.00	1.00
2,4	1.00	-	1.00	1.00	1.00	-	-	1.00	-	1.00	1.00	-	1.00	-	-	-	-	-	1.00	-	-	-	1.00
2,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4	1.00	-	-	-	1.00	1.00	1.00	1.00	-	1.00	1.00	-	1.00	1.01	-	-	-	-	1.00	1.00	1.00	-	1.00
3,5	-	-	-	-	-	-	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3	1.00	-	1.00	1.00	-	-	1.00	1.00	-	-	1.00	-	1.00	-	-	-	-	-	-	-	-	-	-
1,2,4	1.00	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	-	1.00	-	1.00	-	-	-	-	-	-	-	-	-	-
1,2,5	-	-	-	-	-	-	1.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4	-	-	-	-	1.01	1.00	1.00	1.00	1.00	-	1.00	-	1.00	-	1.00	-	1.00	-	-	-	-	-	-
1,3,5	-	-	-	-	-	-	1.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,4,5	-	-	-	-	-	-	-	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4	1.00	-	-	1.00	1.00	1.00	1.00	-	1.00	1.00	1.00	-	1.00	-	-	-	1.00	-	1.00	-	-	-	1.00
2,3,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4	1.00	-	1.00	1.00	-	1.00	1.00	1.00	1.00	-	1.00	-	1.00	-	-	-	-	-	-	-	-	-	-
1,2,3,5	-	-	-	-	-	-	1.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table S4: Summary of bulk effective sample size (Bulk-ESS) among successful fits across 31 model iterations (MI). Minimum bulk-ESS in the referred MI is reported. Column "MI( )" indicates combination of mechanisms deployed in MI.

MI( )	ghhp1	dowb1	ghhp2	ghhp3	ghhp4	ghhp5	brwb1	brwb2	ecpt1	ecpt2	ecwb1	ecwb2	brpb1	brst1	brwb3	brwb4	brst2	brwb7	mywb3	hnwb1	ecpb1	niwb1	niwb2
0	3385	-	-	-	2515	1159	3241	2975	3399	2475	3297	775	-	1908	1174	654	596	1415	1501	442	253	-	2548
1	636	-	-	-	407	992	2796	2006	-	848	1160	152	1311	-	-	721	328	-	-	-	-	-	2340
2	2140	-	-	2866	2269	-	1468	509	1297	1247	-	505	1734	1490	-	-	477	1026	1192	-	-	-	1443
3	3009	-	-	-	2429	1025	2594	2878	3087	2859	2794	1639	3040	2078	997	974	504	-	1785	498	1044	726	2260
4	2816	-	-	-	2382	837	3022	3203	-	2791	2966	1012	1263	2119	-	1330	420	-	1399	890	-	-	2437
5	-	-	-	-	-	-	2557	2745	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2	1819	-	2791	728	1370	-	1861	592	940	-	891	-	704	-	-	-	-	-	-	-	-	-	-
1,3	1200	-	-	-	-	112	2569	2296	3396	719	522	245	1062	-	-	1315	352	-	-	-	-	-	-
1,4	-	-	-	-	-	485	1610	2924	3108	-	1235	-	942	-	-	-	3023	-	1774	-	-	-	252
1,5	-	-	-	-	-	-	3349	2294	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3	2723	1654	1748	2638	1940	1176	1306	737	1307	921	509	404	2106	1297	-	-	-	-	1172	-	-	1024	768
2,4	1636	-	2377	2255	3000	-	-	528	-	1275	1015	-	1770	-	-	-	-	-	1416	-	-	-	2273
2,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4	1377	-	-	-	2018	1060	2773	2949	-	2534	3312	-	1747	359	-	-	-	-	1520	1457	1984	-	2377
3,5	-	-	-	-	-	-	2131	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3	1167	-	2427	622	-	-	1395	494	-	-	857	-	571	-	-	-	-	-	-	-	-	-	-
1,2,4	2606	-	2571	1046	1566	1053	1795	775	3073	-	724	-	631	-	-	-	-	-	-	-	-	-	-
1,2,5	-	-	-	-	-	-	1403	624	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4	-	-	-	-	316	197	1674	1945	3131	-	708	-	542	-	824	-	2645	-	-	-	-	-	-
1,3,5	-	-	-	-	-	-	2609	2288	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,4,5	-	-	-	-	-	-	-	2350	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4	2360	-	-	2384	4336	1203	1619	-	1681	800	1158	-	1723	-	-	-	600	-	1114	-	-	-	910
2,3,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4	1624	-	2060	898	-	560	1722	571	2483	-	1097	-	633	-	-	-	-	-	-	-	-	-	-
1,2,3,5	-	-	-	-	-	-	1779	472	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table S5: Summary of tail effective sample size (Tail-ESS) among successful fits across 31 model iterations (MI). Minimum tail-ESS in the referred MI is reported. Column "MI( )" indicates combination of mechanisms deployed in MI.

MI( )	ghhp1	dowb1	ghhp2	ghhp3	ghhp4	ghhp5	brwb1	brwb2	ecpt1	ecpt2	ecwb1	ecwb2	brpb1	brst1	brwb3	brwb4	brst2	brwb7	mywb3	hnwb1	ecpb1	niwb1	niwb2
0	1851	-	-	-	1361	553	2540	2206	1574	2000	2208	1746	-	1185	1746	288	1386	1444	2109	1738	703	-	2290
1	756	-	-	-	103	554	2336	1954	-	313	1017	1130	1423	-	-	199	919	-	-	-	-	-	1536
2	2091	-	-	1334	1695	-	1876	1703	686	1087	-	1154	1312	1217	-	-	1407	1450	1120	-	-	-	1836
3	1687	-	-	-	1532	336	1505	2210	2203	1597	1475	1001	1885	702	1752	710	470	-	1570	1492	1261	828	1944
4	1174	-	-	-	1882	661	1709	1777	-	1529	2013	365	1325	660	-	568	281	-	1775	565	-	-	1930
5	-	-	-	-	-	-	2203	2132	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2	1042	-	1708	810	775	-	1921	1146	1185	-	522	-	1069	-	-	-	-	-	-	-	-	-	-
1,3	853	-	-	-	-	167	1724	2108	1691	214	334	1153	1170	-	-	589	244	-	-	-	-	-	-
1,4	-	-	-	-	-	529	1613	2230	1384	-	1063	-	487	-	-	-	1698	-	1655	-	-	-	234
1,5	-	-	-	-	-	-	2543	1712	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3	2158	1421	1287	1596	800	464	1057	1642	1890	1538	1408	820	1188	865	-	-	-	-	644	-	-	897	1242
2,4	1692	-	1682	1457	1484	-	-	1679	-	1315	1046	-	1353	-	-	-	-	-	1158	-	-	-	2297
2,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4	557	-	-	-	1192	662	1980	1927	-	1257	1709	-	1673	132	-	-	-	-	1501	866	1699	-	1712
3,5	-	-	-	-	-	-	1385	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3	634	-	1706	623	-	-	1613	1394	-	-	1027	-	762	-	-	-	-	-	-	-	-	-	-
1,2,4	1008	-	1295	689	1177	828	1299	1606	1773	-	909	-	777	-	-	-	-	-	-	-	-	-	-
1,2,5	-	-	-	-	-	-	1604	1122	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4	-	-	-	-	1087	414	1776	1182	1836	-	547	-	527	-	687	-	1183	-	-	-	-	-	-
1,3,5	-	-	-	-	-	-	1707	2107	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,4,5	-	-	-	-	-	-	-	1984	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4	2194	-	-	1768	2518	631	1578	-	473	1116	1656	-	1331	-	-	-	1049	-	467	-	-	-	1042
2,3,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4	782	-	1317	1314	-	715	1127	997	1944	-	1084	-	1004	-	-	-	-	-	-	-	-	-	-
1,2,3,5	-	-	-	-	-	-	1703	1198	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table S6: Summary of effective sample size over total number of posterior draws (ESS/N_{draws}) among successful fits across 31 model iterations (MI). Average ESS/N_{draws} in the referred MI is reported. Column "MI( )" indicates combination of mechanisms deployed in MI.

MI( )	ghhp1	dowb1	ghhp2	ghhp3	ghhp4	ghhp5	brwb1	brwb2	ecpt1	ecpt2	ecwb1	ecwb2	brpb1	brst1	brwb3	brwb4	brst2	brwb7	mywb3	hnwb1	ecpb1	niwb1	niwb2
0	0.555	-	-	-	0.591	0.530	0.642	0.607	0.580	0.526	0.619	0.459	-	0.535	0.541	0.416	0.338	0.514	0.501	0.428	0.330	-	0.601
1	0.403	-	-	-	0.267	0.482	0.673	0.592	-	0.462	0.434	0.249	0.467	-	-	0.395	0.440	-	-	-	-	-	0.529
2	0.584	-	-	0.682	0.558	-	0.560	0.340	0.507	0.492	-	0.345	0.551	0.407	-	-	0.351	0.404	0.428	-	-	-	0.533
3	0.536	-	-	-	0.554	0.511	0.602	0.598	0.609	0.615	0.620	0.570	0.626	0.451	0.522	0.366	0.340	-	0.516	0.471	0.448	0.425	0.542
4	0.621	-	-	-	0.586	0.551	0.626	0.604	-	0.578	0.632	0.339	0.522	0.589	-	0.441	0.330	-	0.447	0.462	-	-	0.562
5	-	-	-	-	-	-	0.607	0.596	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2	0.490	-	0.611	0.445	0.471	-	0.557	0.417	0.461	-	0.402	-	0.399	-	-	-	-	-	-	-	-	-	-
1,3	0.481	-	-	-	-	0.257	0.619	0.596	0.695	0.402	0.410	0.274	0.416	-	-	0.490	0.311	-	-	-	-	-	-
1,4	-	-	-	-	-	0.373	0.556	0.677	0.628	-	0.507	-	0.458	-	-	-	0.594	-	0.518	-	-	-	0.272
1,5	-	-	-	-	-	-	0.663	0.547	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3	0.612	0.553	0.521	0.688	0.454	0.512	0.478	0.326	0.583	0.424	0.353	0.330	0.567	0.362	-	-	-	-	0.422	-	-	0.447	0.399
2,4	0.572	-	0.613	0.619	0.694	-	-	0.386	-	0.528	0.510	-	0.532	-	-	-	-	-	0.451	-	-	-	0.438
2,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4	0.556	-	-	-	0.605	0.492	0.607	0.724	-	0.607	0.753	-	0.55	0.487	-	-	-	-	0.509	0.458	0.605	-	0.564
3,5	-	-	-	-	-	-	0.609	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3	0.519	-	0.581	0.393	-	-	0.534	0.401	-	-	0.359	-	0.373	-	-	-	-	-	-	-	-	-	-
1,2,4	0.486	-	0.615	0.493	0.567	0.388	0.525	0.521	0.715	-	0.432	-	0.420	-	-	-	-	-	-	-	-	-	-
1,2,5	-	-	-	-	-	-	0.511	0.388	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4	-	-	-	-	0.369	0.312	0.608	0.587	0.691	-	0.473	-	0.466	-	0.523	-	0.557	-	-	-	-	-	-
1,3,5	-	-	-	-	-	-	0.643	0.578	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,4,5	-	-	-	-	-	-	-	0.611	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4	0.585	-	-	0.599	0.595	0.512	0.569	-	0.583	0.415	0.507	-	0.562	-	-	-	0.440	-	0.459	-	-	-	0.393
2,3,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4	0.536	-	0.512	0.505	-	0.259	0.478	0.482	0.706	-	0.429	-	0.433	-	-	-	-	-	-	-	-	-	-
1,2,3,5	-	-	-	-	-	-	0.536	0.415	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table S7: Summary of Bayesian Model Avering (BMA) among successful fits across 31 model iterations (MI). BMA's weights in the referred MI is reported. Column "MI( )" indicates combination of mechanisms deployed in MI.

MI( )	ghhp1	dowb1	ghhp2	ghhp3	ghhp4	ghhp5	brwb1	brwb2	ecpt1	ecpt2	ecwb1	ecwb2	brpb1	brst1	brwb3	brwb4	brst2	brwb7	mywb3	hnwb1	ecpb1	niwb1	niwb2
0	0.000	-	-	-	0.022	0.238	0.079	0.223	0.000	0.040	0.347	0.000	-	0.747	0.000	0.000	0.000	0.890	0.110	0.659	0.005	-	0.421
1	0.000	-	-	-	0.000	0.002	0.035	0.040	-	0.460	0.004	0.000	0.843	-	-	0.011	0.018	-	-	-	-	-	0.517
2	0.046	-	-	0.086	0.157	-	0.014	0.000	0.038	0.016	-	0.290	0.000	0.007	-	-	0.000	0.110	0.004	-	-	-	0.006
3	0.013	-	-	-	0.012	0.105	0.012	0.088	0.000	0.083	0.118	0.022	0.000	0.009	0.000	0.000	0.000	-	0.778	0.324	0.046	0.057	0.055
4	0.001	-	-	-	0.007	0.175	0.503	0.233	-	0.113	0.308	0.220	0.000	0.233	-	0.000	0.000	-	0.011	0.005	-	-	0.001
5	-	-	-	-	-	-	0.133	0.120	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2	0.000	-	0.481	0.003	0.002	-	0.002	0.001	0.000	-	0.000	-	0.003	-	-	-	-	-	-	-	-	-	-
1,3	0.000	-	-	-	-	0.000	0.001	0.011	0.000	0.103	0.000	0.000	0.007	-	-	0.989	0.000	-	-	-	-	-	-
1,4	-	-	-	-	-	0.000	0.008	0.069	0.001	-	0.004	-	0.146	-	-	-	0.807	-	0.002	-	-	-	0.000
1,5	-	-	-	-	-	-	0.099	0.078	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3	0.398	1.000	0.001	0.328	0.011	0.036	0.005	0.000	0.034	0.008	0.004	0.468	0.000	0.000	-	-	-	-	0.014	-	-	0.943	0.000
2,4	0.018	-	0.001	0.050	0.564	-	-	0.000	-	0.047	0.038	-	0.000	-	-	-	-	-	0.000	-	-	-	0.000
2,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4	0.007	-	-	-	0.007	0.041	0.048	0.087	-	0.121	0.164	-	0.000	0.005	-	-	-	-	0.071	0.013	0.948	-	0.000
3,5	-	-	-	-	-	-	0.044	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3	0.000	-	0.133	0.004	-	-	0.001	0.000	-	-	0.000	-	0.000	-	-	-	-	-	-	-	-	-	-
1,2,4	0.000	-	0.349	0.001	0.016	0.000	0.001	0.002	0.019	-	0.000	-	0.000	-	-	-	-	-	-	-	-	-	-
1,2,5	-	-	-	-	-	-	0.006	0.001	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4	-	-	-	-	0.000	0.000	0.004	0.006	0.007	-	0.000	-	0.001	-	1.000	-	0.175	-	-	-	-	-	-
1,3,5	-	-	-	-	-	-	0.004	0.009	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,4,5	-	-	-	-	-	-	-	0.031	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4	0.518	-	-	0.522	0.201	0.156	0.003	-	0.875	0.009	0.010	-	0.000	-	-	-	0.000	-	0.009	-	-	-	0.000
2,3,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4	0.000	-	0.036	0.006	-	0.000	0.001	0.000	0.027	-	0.000	-	0.000	-	-	-	-	-	-	-	-	-	-
1,2,3,5	-	-	-	-	-	-	0.000	0.000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1,2,3,4,5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table S8: Parameter estimates of model iteration involving mechanisms M1 and M2, MI(2,3) calibrated over different datasets. Means and standard deviations (within parenthesis) of the posterior parameters are shown. All reported values in literature have same units as depicted in table 2.MM with the exception of substrate saturation constants ($\mathrm{mg(substrate)\,mL^{-1}}$). Green coloured cells indicate parameter estimates that lie within reported ranges. Orange coloured cells indicate parameter estimates that are far from reported range by less than the estimate either divided or multiplied by 10. Red coloured cells indicate parameter estimates that are far from reported range by more than the estimate either divided or multiplied by 10. Parameters of MI(2,3) fitted to datasets *ghhp2*, *ghhp3*, *ghhp4* and *ghhp5* are not shown due to their lack of LA time series.

Parameter	ghhp1	dowb1	brwb1	brwb2	ecpt1	ecpt2	ecwb1	Reported values	Reference
$\mu_{max}^{Y_{Glc}}$	0.406 (0.124)	0.133 (0.11)	0.094 (0.052)	0.381 (0.167)	0.162 (0.103)	0.464 (0.245)	0.141 (0.101)	0.0781 - 0.53	[24–31]
$\mu_{max}^{Y_{Fru}}$	0.233 (0.109)	0.145 (0.118)	0.039 (0.027)	0.551 (0.191)	0.167 (0.101)	0.461 (0.224)	0.141 (0.108)	0.01 - 0.83	[29, 31–36]
$\mu_{max}^{LAB_{Glc}}$	0.137 (0.079)	0.382 (0.178)	0.122 (0.109)	0.382 (0.169)	0.279 (0.124)	0.561 (0.287)	0.203 (0.144)	0.0072 - 1.41	[37–42]
$\mu_{max}^{LAB_{Fru}}$	0.193 (0.077)	0.272 (0.177)	0.283 (0.103)	0.328 (0.192)	0.265 (0.122)	0.341 (0.269)	0.216 (0.151)	0.0302 - 0.85	[43, 44]
$\mu_{max}^{AAB_{EtOH}}$	0.466 (0.121)	0.424 (0.154)	0.163 (0.051)	0.181 (0.054)	0.309 (0.115)	0.458 (0.145)	0.223 (0.106)	0.0106 - 0.25	[45–49]
$\mu_{max}^{AAB_{LA}}$	0.016 (0.022)	0.05 (0.071)	0.022 (0.015)	0.02 (0.015)	0.062 (0.03)	0.062 (0.047)	0.038 (0.041)		
K_{Glc}^{Y}	38.569 (12.739)	35.079 (15.842)	35.494 (15.024)	28.897 (11.247)	29.164 (13.642)	24.686 (11.317)	23.998 (11.648)	9.73×10^{-4} - 0.5	[26, 28]
K_{Fru}^{Y}	39.902 (15.329)	54.891 (24.524)	30.313 (13.692)	39.954 (17.818)	31.309 (15.428)	23.367 (10.27)	31.981 (14.899)	5.82 - 161.45	[33, 35, 36]
K_{Glc}^{LAB}	34.266 (13.624)	34.838 (15.049)	35.326 (14.934)	23.098 (11.403)	33.794 (13.191)	23.247 (11.143)	27.604 (11.475)	0.790 - 178.0	[40–42]
K_{Fru}^{LAB}	38.586 (15.226)	56.625 (23.97)	31.16 (13.268)	39.39 (18.242)	37.698 (14.198)	23.568 (10.569)	36.596 (14.592)		
K_{EtOH}^{AAB}	15.995 (5.483)	6.16 (2.507)	3.895 (1.652)	4.193 (1.964)	8.448 (3.276)	5.944 (2.46)	11.8 (4.771)		
K_{LA}^{AAB}	2568.5 (1232.144)	30.1 (13.291)	316.919 (154.179)	81.322 (41.956)	194.145 (92.66)	45.999 (22.447)	1063.022 (498.821)		
k_Y	0.0455 (0.0074)	0.0559 (0.0241)	0.0891 (0.0288)	0.0861 (0.0246)	0.0012 (0.0014)	0.0315 (0.019)	0.0111 (0.0081)		
k_{LAB}	0.007 (0.0016)	0.1558 (0.1003)	0.0553 (0.0151)	0.0452 (0.0212)	0.0572 (0.0174)	0.4004 (0.1669)	0.0561 (0.0219)		
k_{AAB}	0.0096 (0.0023)	0.0011 (4×10^{-4})	4×10^{-4} (2×10^{-4})	4×10^{-4} (2×10^{-4})	0.0158 (0.0064)	0.0014 (7×10^{-4})	0.0067 (0.0036)		
$Y_{Glc\|Y}$	33.305 (10.489)	32.153 (15.894)	266.721 (59.451)	136.564 (38.749)	24.931 (11.133)	31.857 (13.823)	34.711 (13.955)	1.56 - 66.67	[24–26, 30, 31, 50, 51]
$Y_{Glc\|LAB}$	46.411 (22.205)	401.212 (150.668)	75.141 (38.771)	13.355 (15.251)	68.313 (18.895)	108.954 (38.638)	46.171 (17.156)	4.50 - 200	[42, 52]
$Y_{Fru\|Y}$	39.129 (13.619)	52.533 (24.239)	194.391 (62.42)	249.378 (65.934)	27.339 (11.936)	29.489 (11.442)	44.019 (18.671)	43.48 - 200	[31, 32]
$Y_{Fru\|LAB}$	51.41 (20.726)	570.03 (245.658)	53.727 (27.002)	41.31 (28.812)	75.755 (21.129)	88.447 (36.51)	59.887 (22.015)		
$Y_{EtOH\|Y}^{Glc}$	4.236 (2.778)	4.477 (2.656)	7.823 (4.279)	7.798 (4.539)	5.581 (2.913)	4.34 (2.776)	8.798 (5.14)	1.39 - 21.49	[26, 29, 30, 53]
$Y_{EtOH\|Y}^{Fru}$	6.12 (4.207)	4.422 (2.613)	10.794 (6.626)	6.091 (4.129)	5.436 (2.899)	3.434 (2.495)	8.72 (5.209)	5.679 - 5.896	[29]
$Y_{EtOH\|LAB}^{Glc}$	14.762 (9.467)	62.736 (28.211)	6.547 (4.269)	2.258 (2.457)	6.216 (4.03)	10.483 (6.867)	9.482 (6.375)		
$Y_{EtOH\|LAB}^{Fru}$	9.598 (6.119)	58.662 (28.742)	2.446 (2.025)	4.247 (2.964)	6.158 (3.945)	11.696 (7.572)	9.127 (6.483)		
$Y_{EtOH\|AAB}$	1769.652 (915.5)	126.408 (49.954)	450.646 (163.983)	181.209 (57.503)	332.519 (151.239)	371.945 (115.609)	4330.497 (1646.577)	8.06 - 166.67	[46, 48, 54]
$Y_{LA\|LAB}^{Glc}$	9.021 (3.772)	9.765 (5.536)	5.966 (3.558)	3.114 (1.323)	4.167 (2.089)	3.065 (1.291)	3.282 (1.717)		
$Y_{LA\|LAB}^{Fru}$	9.673 (2.733)	10.845 (5.819)	2.994 (1.81)	3.151 (1.527)	4.045 (2.073)	2.967 (1.378)	3.189 (1.666)		
$Y_{LA\|AAB}$	2012.646 (1187.814)	23.54 (13.299)	297.512 (142.921)	63.715 (38.857)	203.33 (81.7)	41.626 (21.469)	913.388 (483.211)	7.51 - 8.34	[48]
$Y_{Ac\|LAB}^{Glc}$	3.277 (2.365)	43.254 (43.771)	9.423 (7.501)	5.236 (6.142)	0.598 (0.527)	9.834 (10.42)	2.382 (2.464)		
$Y_{Ac\|LAB}^{Fru}$	1.809 (1.395)	49.913 (49.765)	2.144 (2.64)	10.774 (7.852)	0.553 (0.499)	13.229 (12.641)	2.356 (2.394)		
$Y_{Ac\|AAB}^{EtOH}$	68.368 (80.311)	29.516 (37.316)	544.767 (252.44)	292.764 (124.828)	19.977 (18.986)	335.779 (179.607)	1271.525 (697.321)		
$Y_{Ac\|AAB}^{LA}$	1170.869 (828.312)	170.617 (128.778)	677.034 (351.821)	376.785 (181.596)	61.421 (45.193)	406.623 (229.638)	1418.934 (760.394)		
$Y_{Ac\|Y}^{Glc}$	1.038 (0.741)	5.706 (5.528)	4.663 (4.576)	3.422 (3.083)	1.106 (0.911)	5.466 (4.86)	2.688 (2.328)		
$Y_{Ac\|Y}^{Fru}$	1.635 (1.13)	5.251 (5.214)	11.526 (11.225)	3.605 (3.209)	0.981 (0.876)	3.594 (4.094)	2.839 (2.355)		
σ	0.134 (0.009)	0.267 (0.039)	0.165 (0.014)	0.176 (0.015)	0.197 (0.015)	0.25 (0.02)	0.21 (0.016)		

Table S8 (cont.): Parameter estimates of model iteration involving mechanisms M1 and M2, MI(2,3) calibrated over different datasets. Means and standard deviations (within parenthesis) of the posterior parameters are shown. All reported values in literature have same units as depicted in table 2.MM with the exception of substrate saturation constants ($\mathrm{mg(substrate)\,mL^{-1}}$). Green coloured cells indicate parameter estimates that lie within reported ranges. Orange coloured cells indicate parameter estimates that are far from reported range by less than the estimate either divided or multiplied by 10. Red coloured cells indicate parameter estimates that are far from reported range by more than the estimate either divided or multiplied by 10. Parameters of MI(2,3) fitted to datasets *ghhp2*, *ghhp3*, *ghhp4* and *ghhp5* are not shown due to their lack of LA time series.

Parameter	ecwb2	brpb1	brst1	mywb3	niwb1	niwb2	Reported values	Reference
$\mu_{max}^{Y_{Glc}}$	0.533 (0.308)	0.419 (0.243)	0.441 (0.244)	0.05 (0.055)	0.068 (0.145)	0.113 (0.063)	0.0781 - 0.53	[24–31]
$\mu_{max}^{Y_{Fru}}$	0.205 (0.224)	0.428 (0.238)	0.433 (0.248)	0.339 (0.126)	0.052 (0.11)	0.06 (0.059)	0.01 - 0.83	[29, 31–36]
$\mu_{max}^{LAB_{Glc}}$	0.403 (0.203)	0.494 (0.225)	0.49 (0.237)	0.33 (0.107)	0.324 (0.176)	0.128 (0.094)	0.0072 - 1.41	[37–42]
$\mu_{max}^{LAB_{Fru}}$	0.249 (0.163)	0.487 (0.226)	0.489 (0.242)	0.129 (0.111)	0.311 (0.165)	0.154 (0.093)	0.0302 - 0.85	[43, 44]
$\mu_{max}^{AAB_{EtOH}}$	0.257 (0.133)	0.313 (0.201)	0.076 (0.113)	0.236 (0.104)	0.233 (0.161)	0.157 (0.055)	0.0106 - 0.25	[45–49]
$\mu_{max}^{AAB_{LA}}$	0.057 (0.042)	0.159 (0.101)	0.393 (0.109)	0.03 (0.035)	0.063 (0.076)	0.022 (0.015)		
K_{Glc}^{Y}	19.578 (9.655)	62.116 (26.394)	80.799 (34.188)	23.619 (11.111)	14.641 (6.638)	13.051 (5.875)	9.73×10^{-4} - 0.5	[26, 28]
K_{Fru}^{Y}	20.005 (8.892)	58.289 (24.438)	74.396 (31.561)	34.473 (14.087)	18.624 (8.353)	15.582 (7.053)	5.82 - 161.45	[33, 35, 36]
K_{Glc}^{LAB}	22.453 (9.579)	60.659 (25.981)	79.018 (34.182)	19.758 (10.912)	15.21 (6.583)	13.484 (5.807)	0.790 - 178.0	[40–42]
K_{Fru}^{LAB}	20.887 (9.011)	56.062 (24.172)	72.225 (32.106)	31.599 (14.311)	19.585 (8.357)	16.055 (7.059)		
K_{EtOH}^{AAB}	7.979 (3.375)	41.536 (17.775)	47.863 (21.327)	11.254 (4.676)	10.808 (4.646)	11.616 (5.046)		
K_{LA}^{AAB}	516.058 (253.489)	29557.579 (14464.123)	88156.259 (30255.39)	509.086 (254.786)	645.104 (323.747)	150.711 (77.189)		
k_{Y}	0.0292 (0.0262)	0.0084 (0.004)	0.0054 (0.0031)	0.0052 (0.0017)	0.0235 (0.0166)	0.0075 (0.0049)		
k_{LAB}	0.019 (0.0083)	0.0083 (0.0075)	0.0019 (0.0011)	0.0095 (0.0035)	0.1128 (0.0314)	0.0075 (0.0061)		
k_{AAB}	7×10^{-4} (3×10^{-4})	2×10^{-4} (1×10^{-4})	0.0028 (5×10^{-4})	0.0028 (8×10^{-4})	0.0014 (0.0013)	0.0018 (0.0011)		
$Y_{Glc\|Y}$	52.08 (19.939)	100.101 (49.917)	83.262 (48.886)	86.243 (41.832)	14.36 (7.361)	10.084 (4.332)	1.56 - 66.67	[24–26, 30, 31, 50, 51]
$Y_{Glc\|LAB}$	46.45 (20.949)	185.227 (70.504)	445.826 (324.583)	23.871 (5.719)	837.761 (367.718)	67.91 (27.827)	4.50 - 200	[42, 52]
$Y_{Fru\|Y}$	34.224 (15.277)	92.125 (45.11)	78.096 (44.916)	184.468 (39.755)	18.344 (9.198)	11.538 (5.614)	43.48 - 200	[31, 32]
$Y_{Fru\|LAB}$	35.514 (13.892)	168.776 (66.785)	420.162 (305.457)	22.443 (10.964)	1090.949 (461.221)	83.376 (29.166)		
$Y_{EtOH\|Y}^{Glc}$	5.967 (4.32)	58.993 (31.485)	54.979 (28.79)	31.918 (17.444)	9.103 (5.12)	5.625 (3.391)	1.39 - 21.49	[26, 29, 30, 53]
$Y_{EtOH\|Y}^{Fru}$	7.332 (4.709)	59.377 (30.913)	55.533 (28.766)	32.111 (11.671)	9.128 (5.101)	6.661 (4.043)	5.679 - 5.896	[29]
$Y_{EtOH\|LAB}^{Glc}$	6.901 (4.98)	84.956 (47.129)	375.841 (182.627)	2.971 (2.127)	471.956 (260.117)	37.133 (22.578)		
$Y_{EtOH\|LAB}^{Fru}$	5.697 (3.694)	85.589 (47.144)	373.244 (183.374)	6.046 (3.483)	470.021 (261.764)	30.268 (20.045)		
$Y_{EtOH\|AAB}$	1359.87 (563.799)	65259.148 (29128.894)	185454.191 (102105.223)	300.007 (230.429)	2253.568 (956.26)	402.877 (145.085)	8.06 - 166.67	[46, 48, 54]
$Y_{LA\|LAB}^{Glc}$	4.156 (1.908)	53.487 (23.831)	132.892 (52.066)	5.605 (1.335)	212.055 (92.244)	22.434 (9.313)		
$Y_{LA\|LAB}^{Fru}$	4.683 (1.598)	53.988 (23.427)	132.034 (52.079)	4.972 (2.354)	214.081 (92.846)	22.042 (8.087)		
$Y_{LA\|AAB}$	490.432 (245.904)	29932.608 (13944.798)	13414.789 (19209.268)	294.357 (239.17)	614.49 (314.846)	148.154 (71.871)	7.51 - 8.34	[48]
$Y_{Ac\|LAB}^{Glc}$	10.368 (9.221)	76.167 (46.329)	36.943 (32.826)	1.108 (0.926)	284.313 (221.157)	10.935 (9.642)		
$Y_{Ac\|LAB}^{Fru}$	5.178 (6.049)	76.538 (47.826)	39.531 (34.132)	3.291 (1.975)	279.148 (211.528)	7.524 (7.958)		
$Y_{Ac\|AAB}^{EtOH}$	1722.008 (1032.276)	53537.995 (30498.644)	40471.046 (24408.066)	126.499 (101.311)	1377.261 (843.113)	78.41 (47.761)		
$Y_{Ac\|AAB}^{LA}$	2000.661 (1164.352)	51184.995 (30136.054)	10189.867 (16933.627)	325.67 (203.901)	1605.76 (881.987)	149.246 (79.831)		
$Y_{Ac\|Y}^{Glc}$	7.947 (8.111)	54.115 (32.154)	8.596 (6.597)	15.564 (10.007)	6.982 (4.479)	1.111 (1.28)		
$Y_{Ac\|Y}^{Fru}$	9.826 (8.814)	54.611 (31.819)	8.981 (6.679)	7.59 (4.948)	7.082 (4.488)	1.934 (1.767)		
σ	0.253 (0.02)	0.278 (0.023)	0.234 (0.024)	0.161 (0.013)	0.316 (0.07)	0.182 (0.031)		

Table S9: Explained variance (EV) by the first two principal components of PCAs performed with medians and samples contained within 95% credible intervals (CI) of the posterior distributions. Columns EV_1 and EV_2 show EV for PCAs performed over MIs reported in Section 3.4MM using medians and 95% CI, respectively. EV is expressed as percentage.

Feature	EV_1	EV_2
Cultivar	48.36	23.76
Temperature	41.11	20.63
Method	51.23	26.58
Country	89.82	71.69

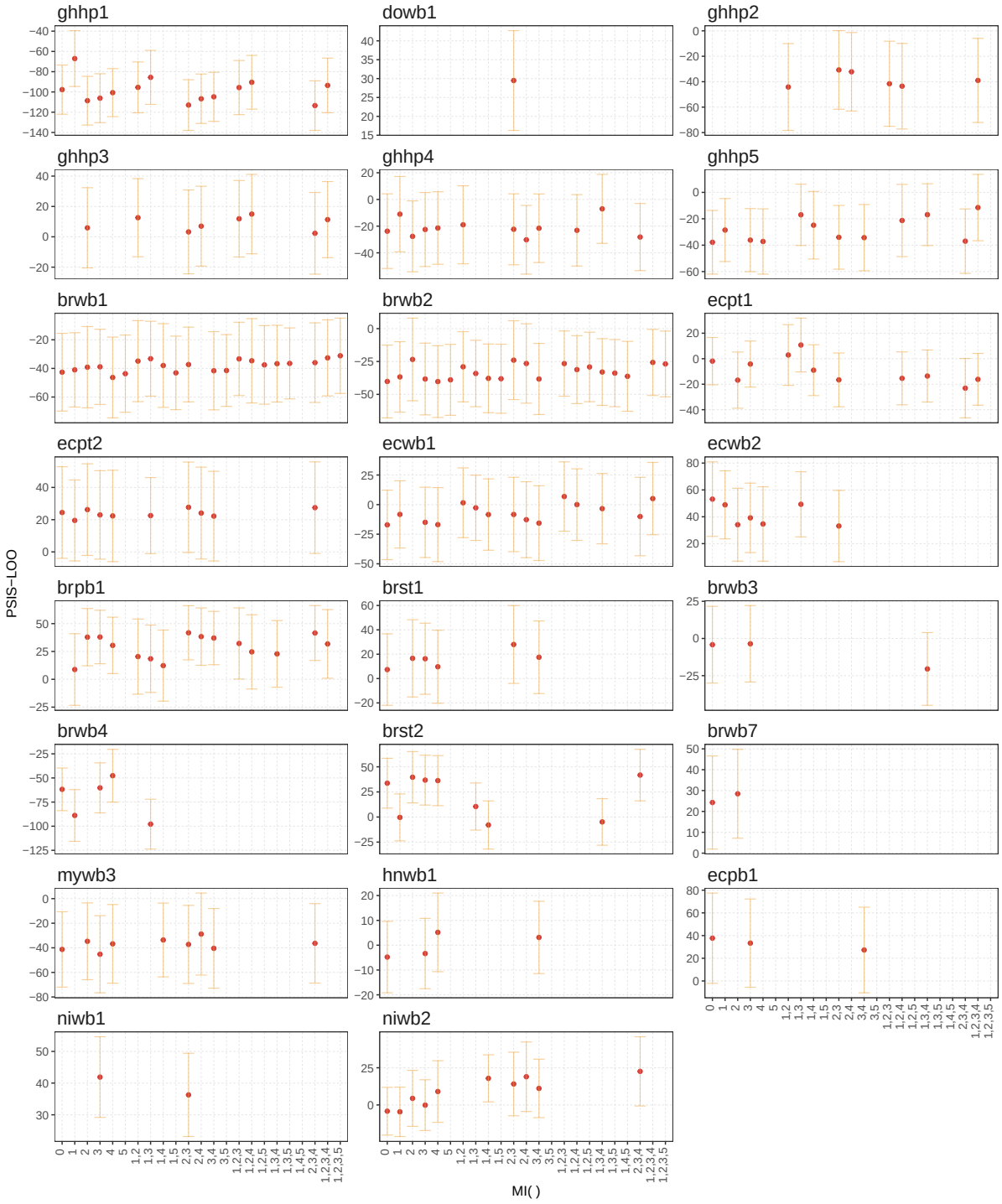


Figure S1: Pareto-smoothed importance sampling leave-one out cross validation (PSIS-LOO) values for model iterations (MIs) over considered datasets. Error bars correspond to standard error of PSIS-LOO. "MI( )" indicates combination of mechanisms deployed in MI.

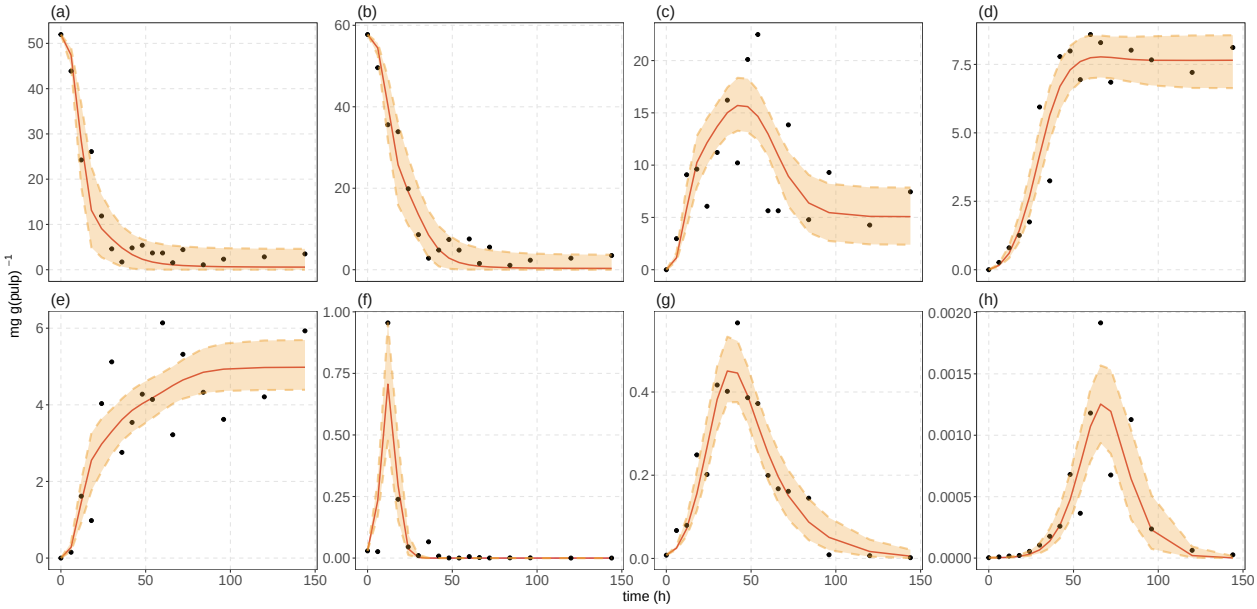


Figure S2: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ghhp1* reported by Camu *et al.* [1]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

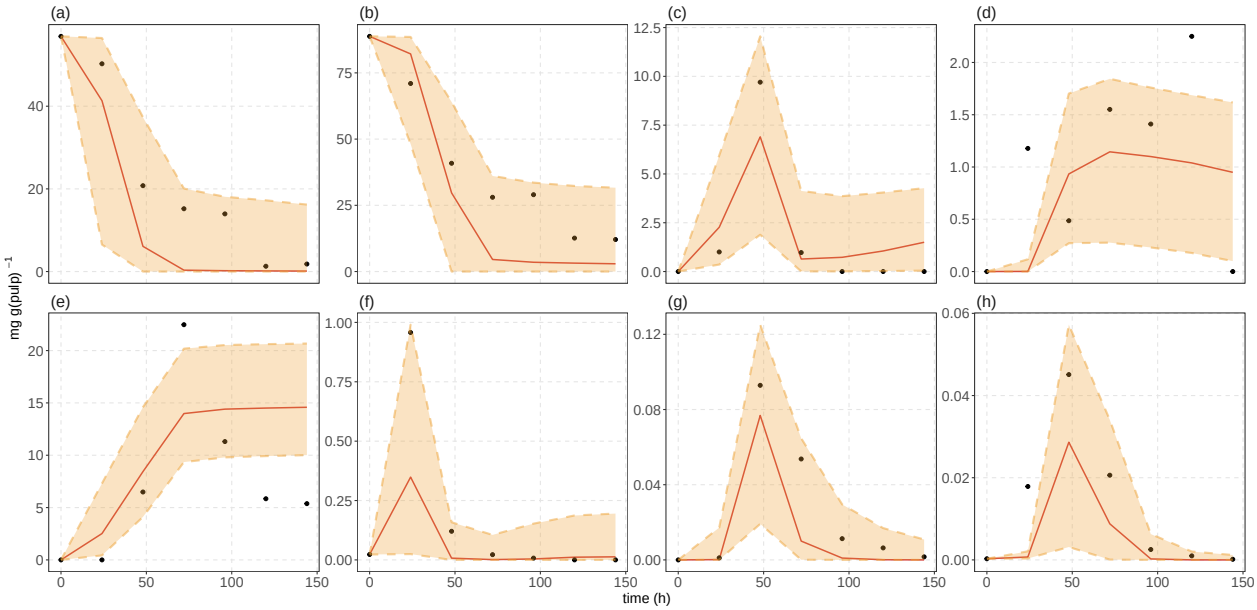


Figure S3: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *dowb1* reported by Lagunes Gálvez *et al.* [2]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

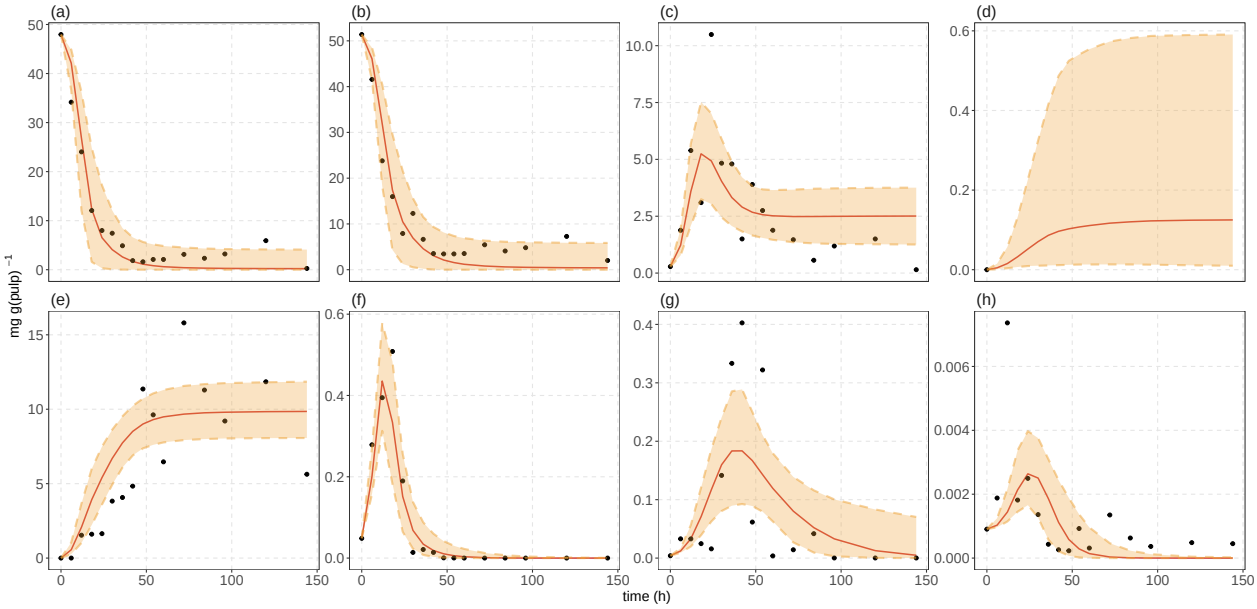


Figure S4: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ghhp2* reported by Camu *et al.* [16]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid (simulated time series scaled to 1) and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

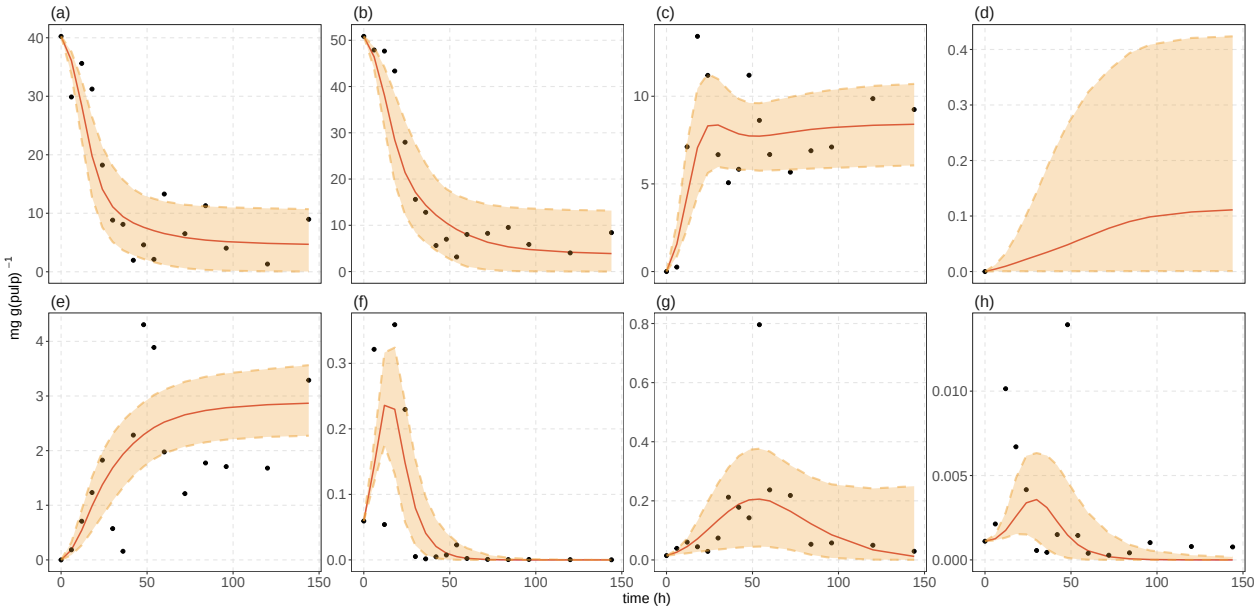


Figure S5: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ghhp3* reported by Camu *et al.* [16]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid (simulated time series scaled to 1) and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

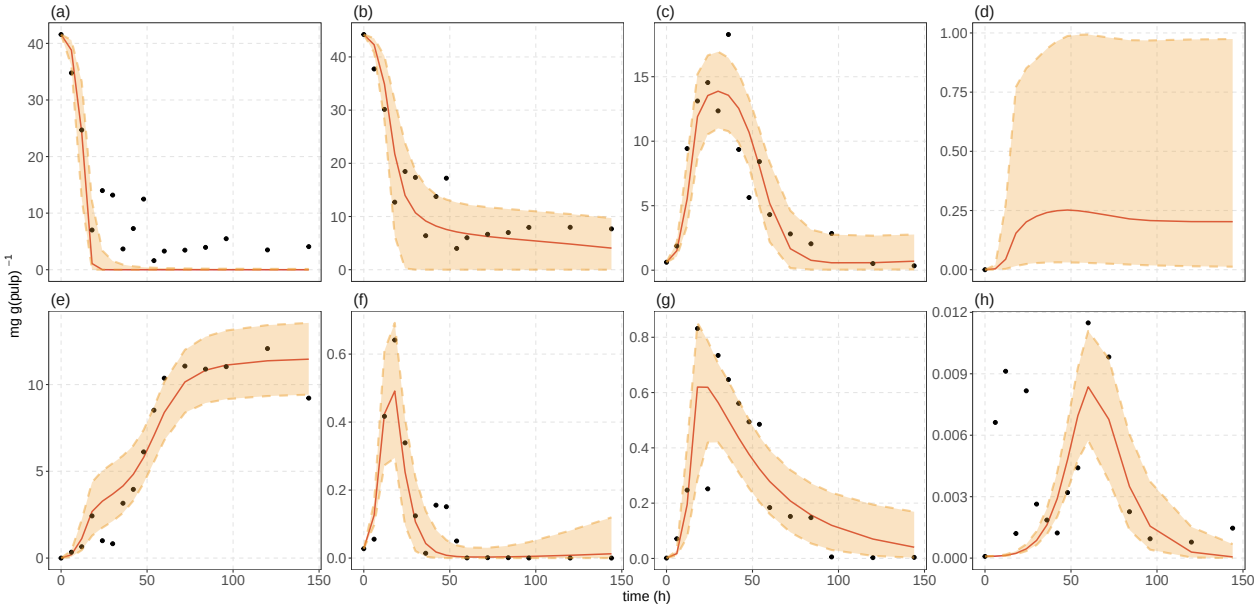


Figure S6: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ghhp4* reported by Camu *et al.* [16]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid (simulated time series scaled to 1) and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

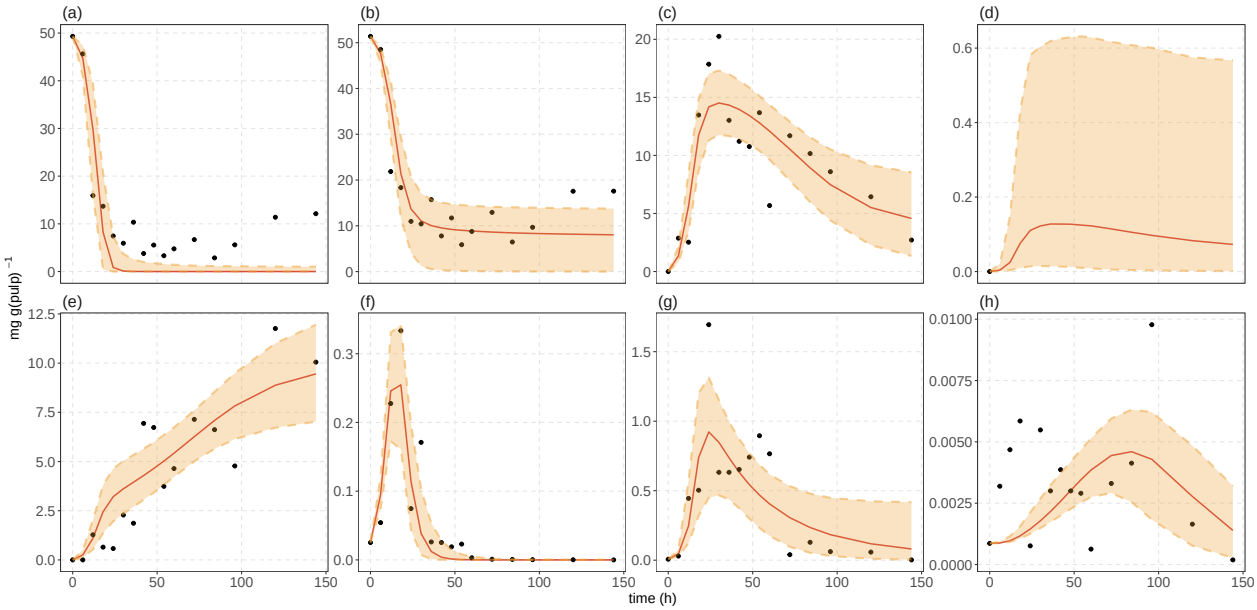


Figure S7: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ghhp5* reported by Camu *et al.* [16]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid (simulated time series scaled to 1) and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

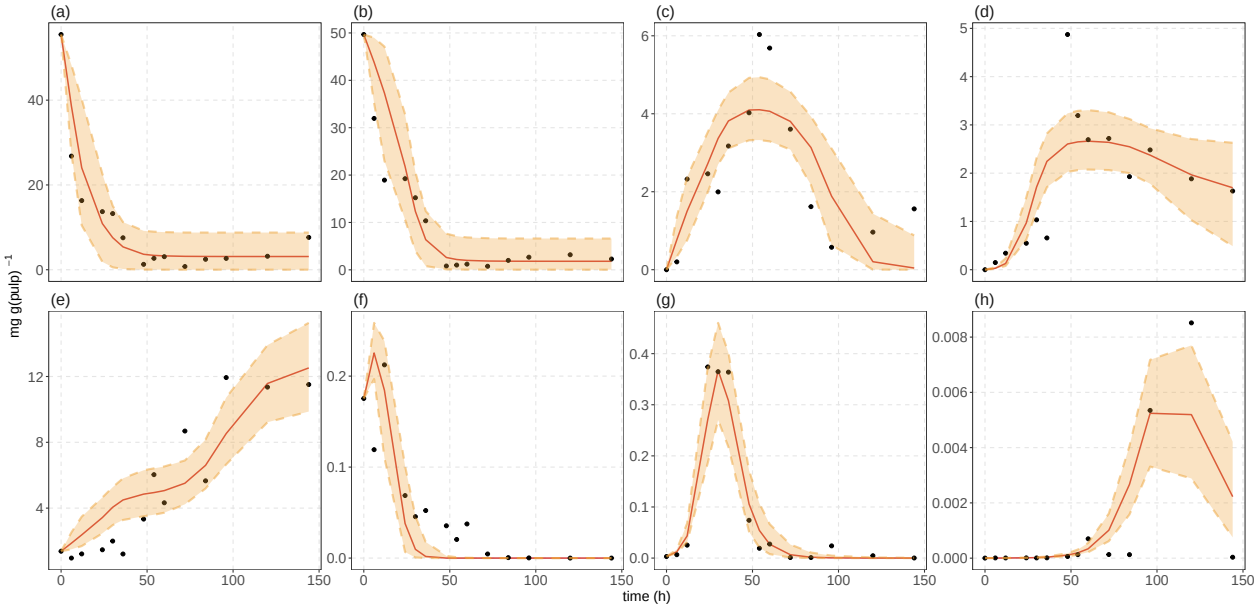


Figure S8: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *brwb1* reported by Papalexandratou *et al.* [4]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

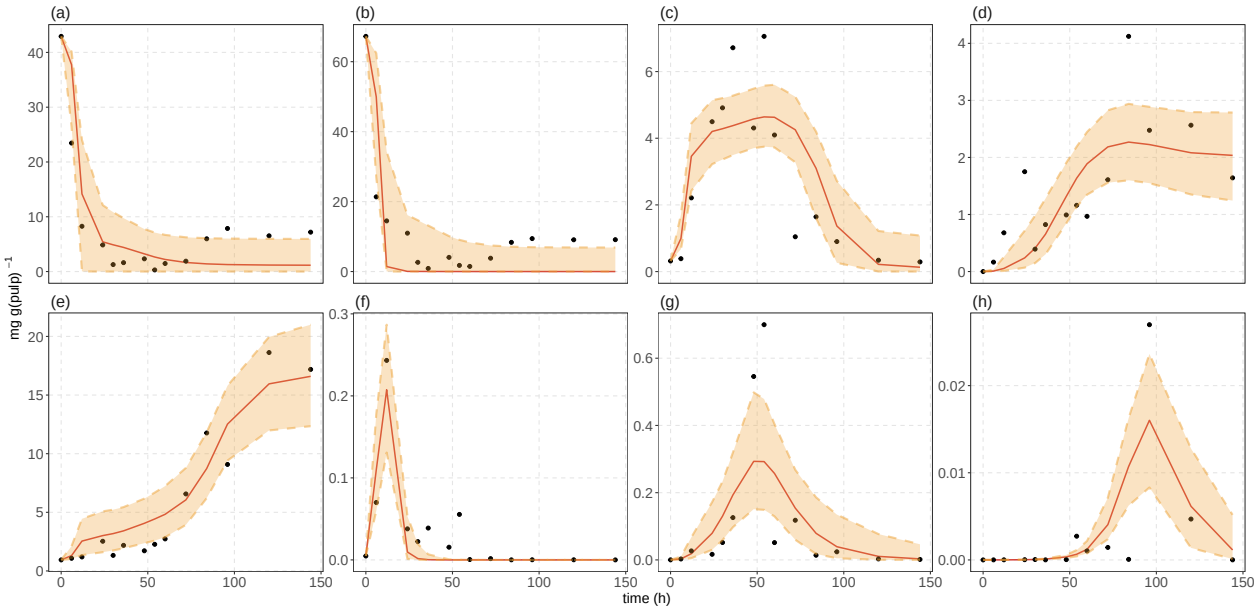


Figure S9: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *brwb2* reported by Papalexandratou *et al.* [4]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

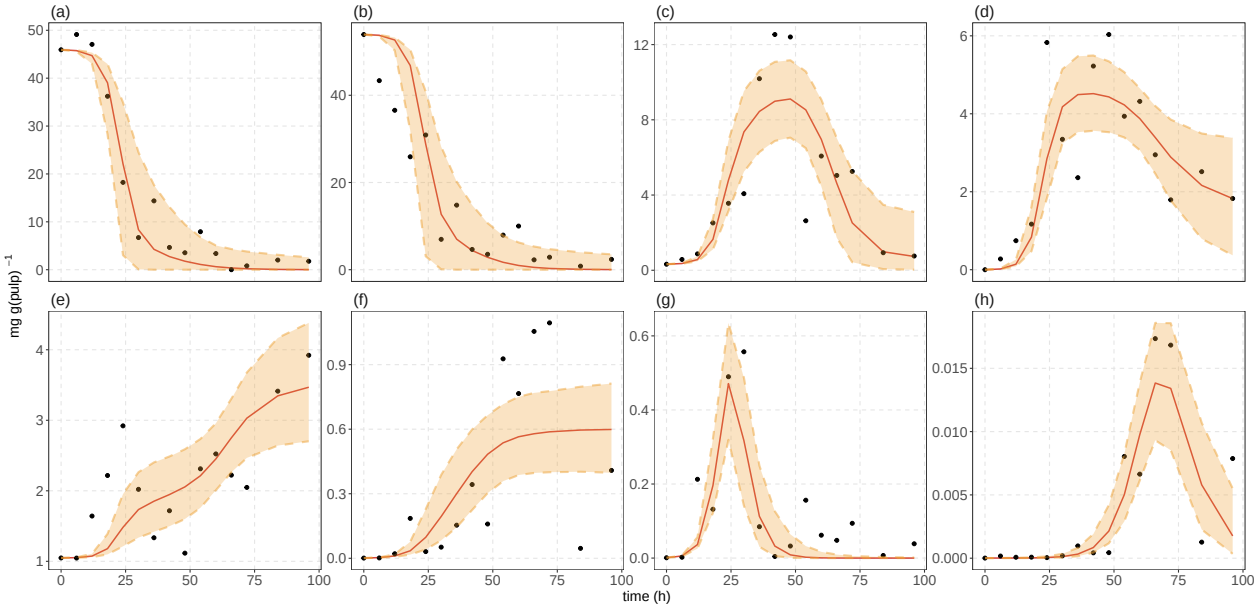


Figure S10: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ecpt1* reported by Papalexandratou *et al.* [5]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

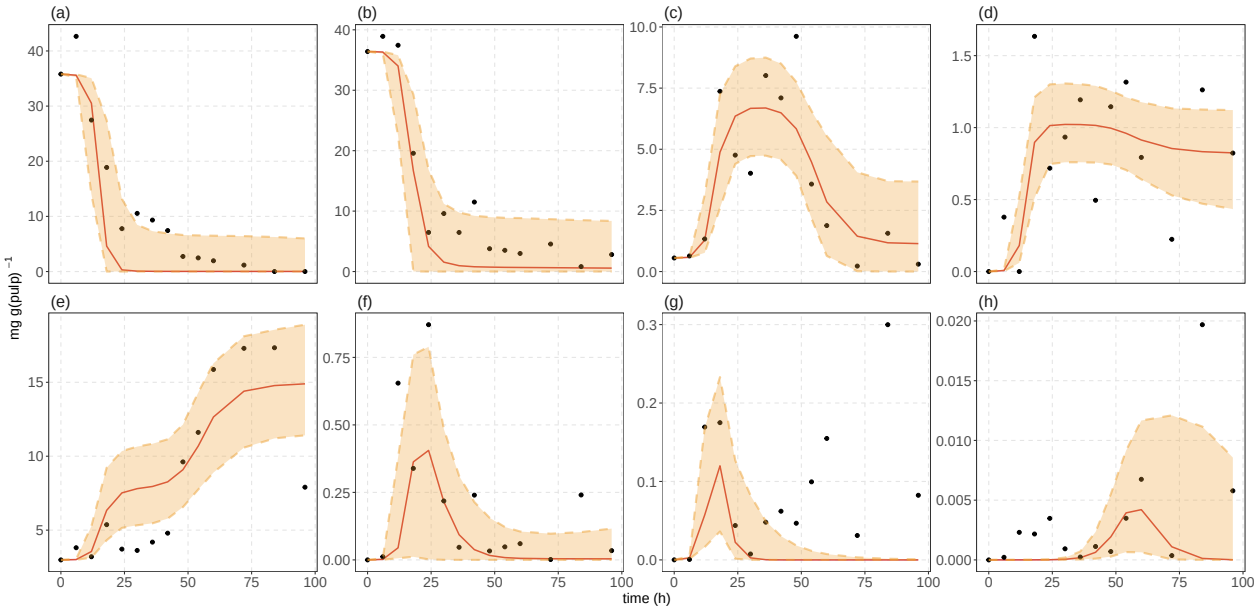


Figure S11: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ecpt2* reported by Papalexandratou *et al.* [5]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

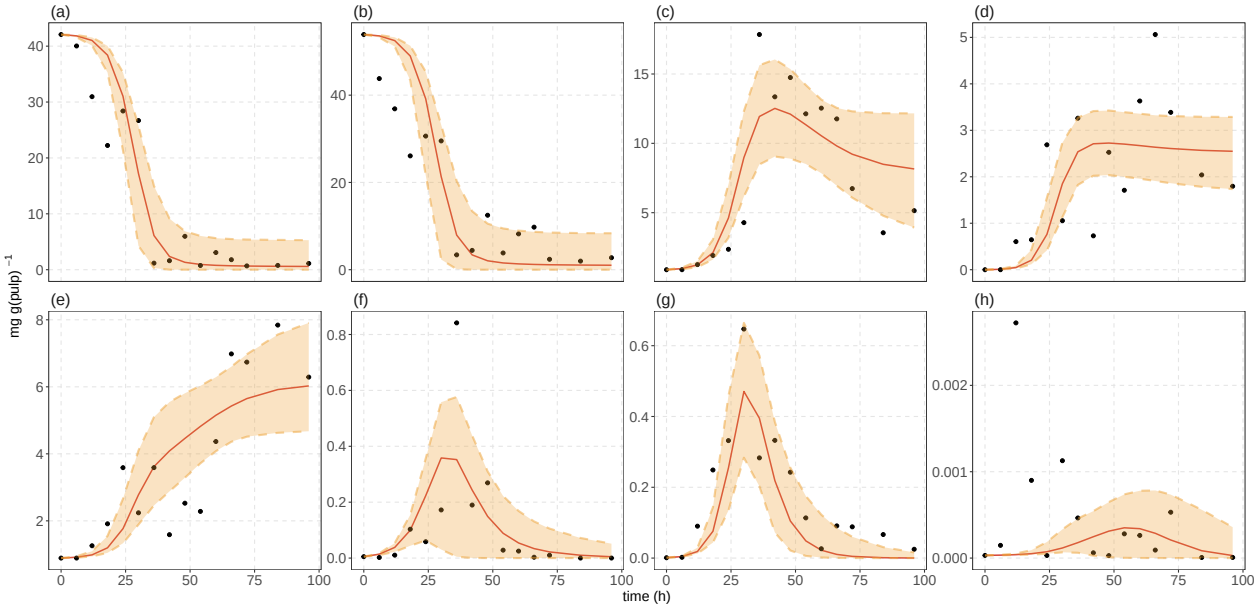


Figure S12: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ecwb1* reported by Papalexandratou *et al.* [5]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

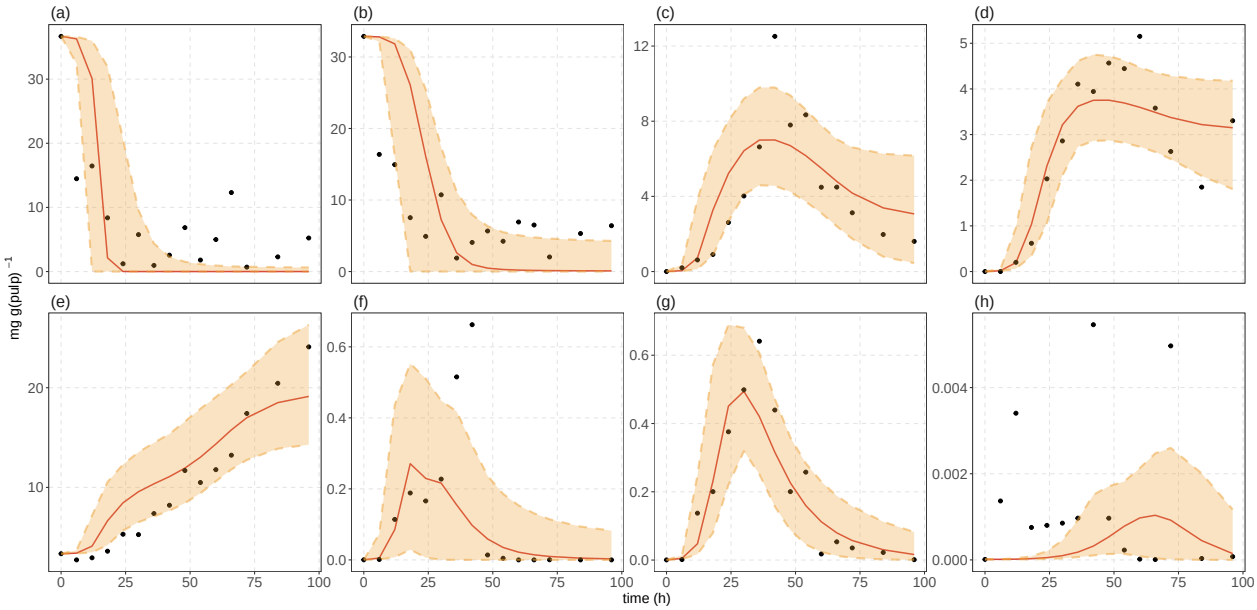


Figure S13: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *ecwb2* reported by Papalexandratou *et al.* [5]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

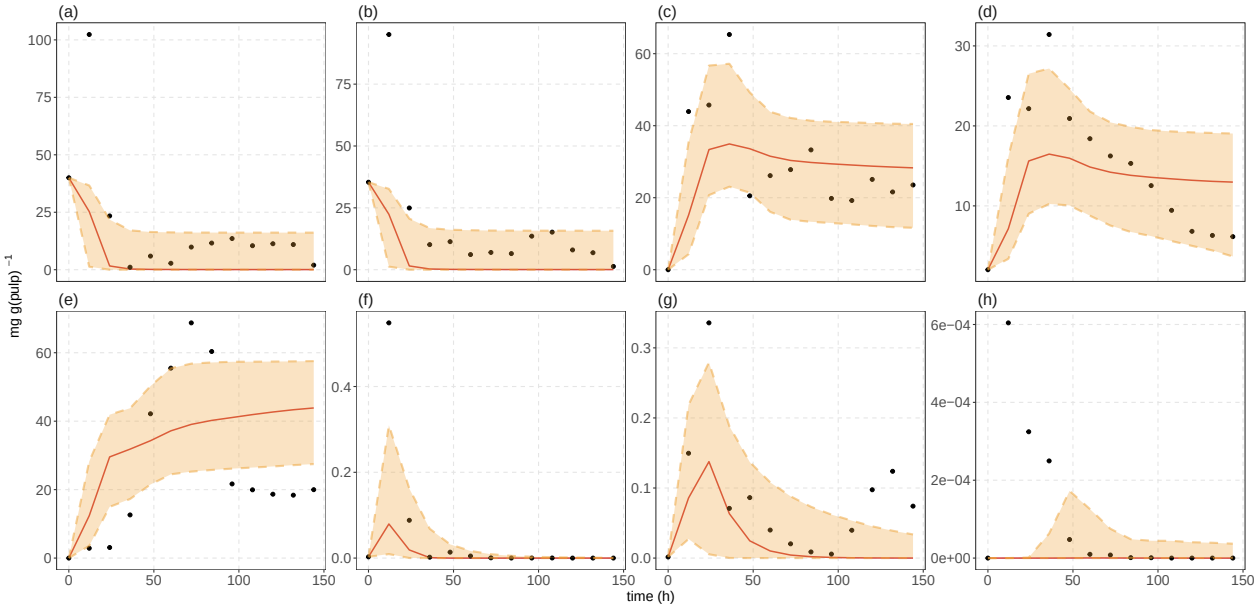


Figure S14: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *brpb1* reported by Pereira *et al.* [6]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

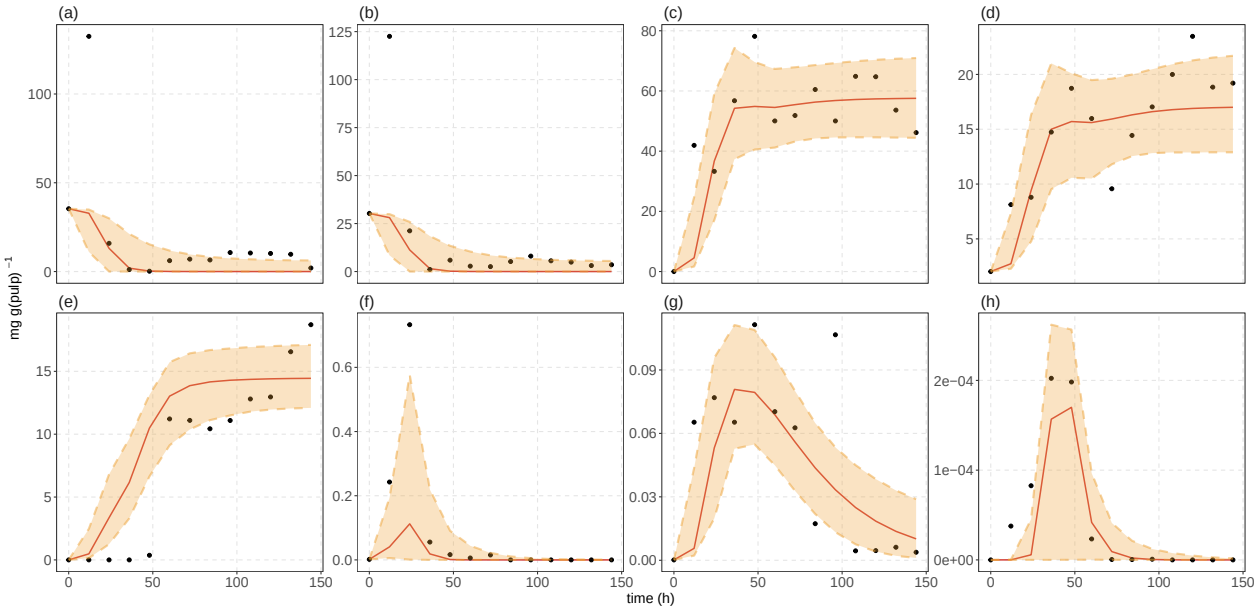


Figure S15: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *brst1* reported by Pereira *et al.* [6]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

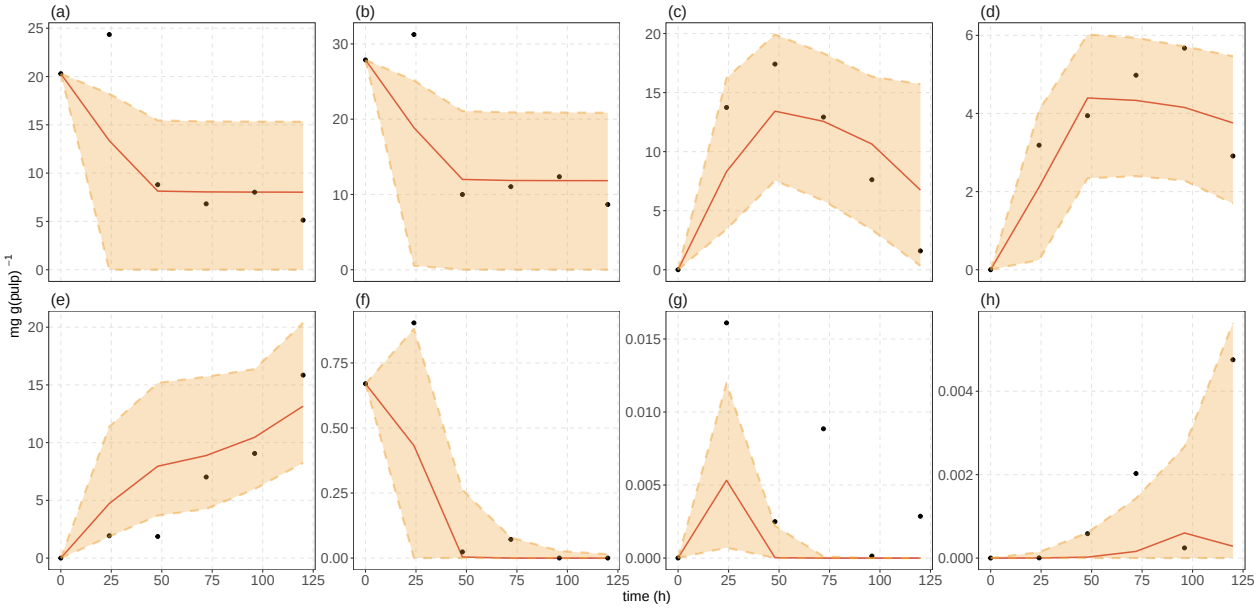


Figure S16: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *niwb1* reported by Papalexandratou *et al.* [14]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

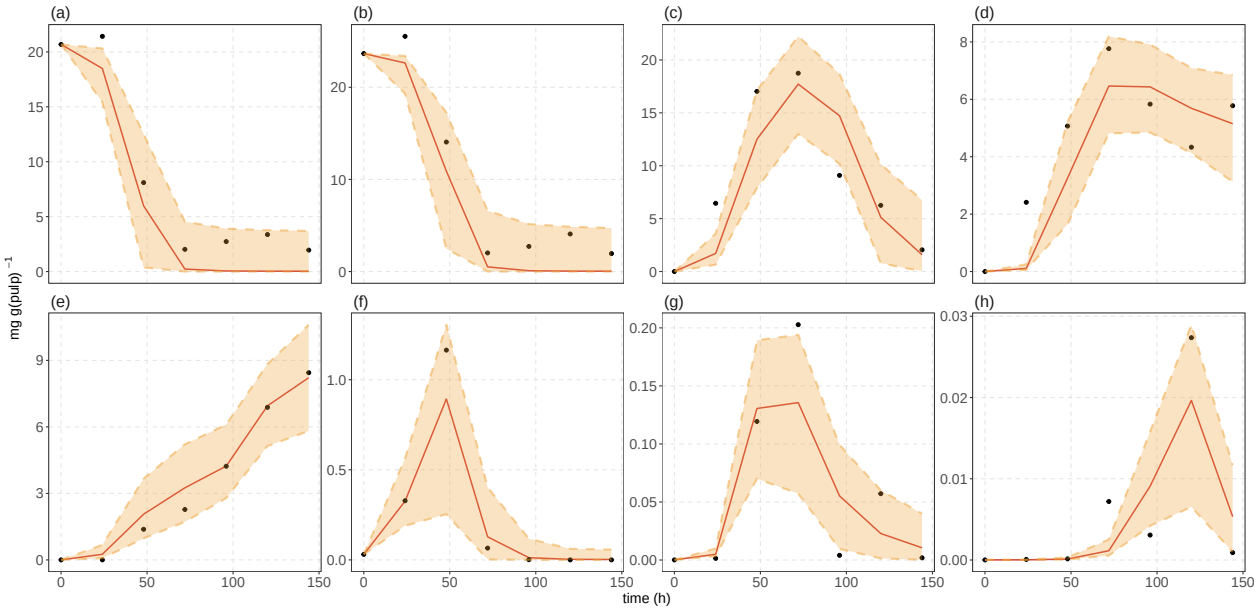


Figure S17: Posterior predictions of model iteration corresponding to mechanisms M2 and M3, MI(2,3), fitted to dataset *niwb2* reported by Papalexandratou *et al.* [14]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

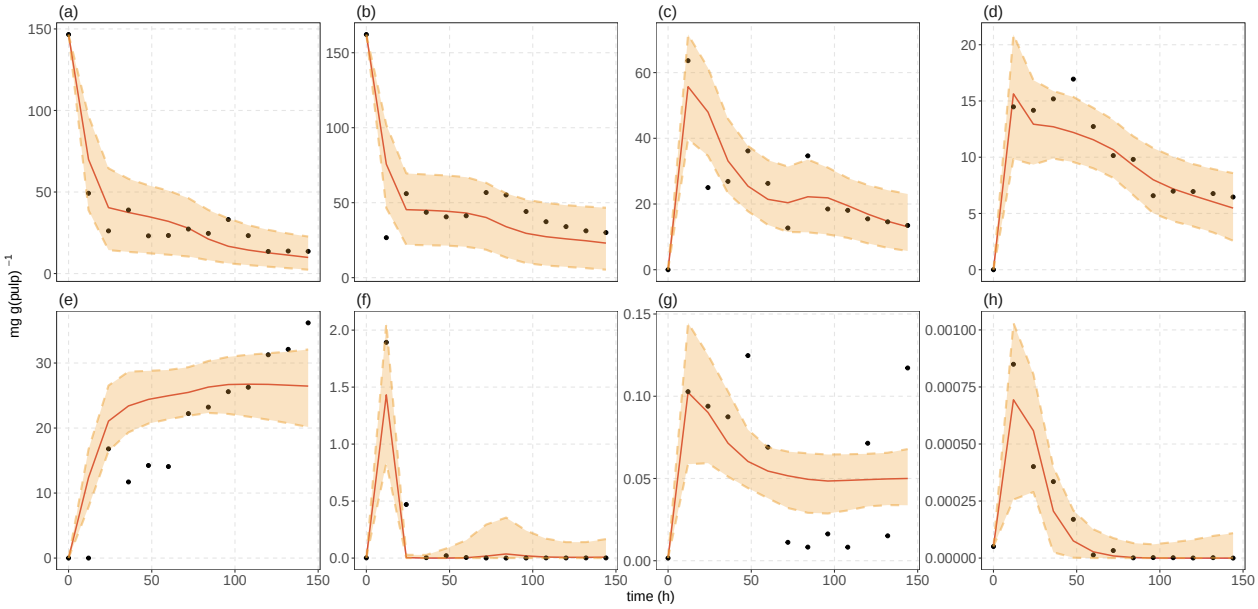


Figure S18: Posterior predictions of model iteration corresponding to mechanisms M1, M3 and M4, MI(1,3,4), fitted to dataset *brwb3* reported by Pereira *et al.* [8]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

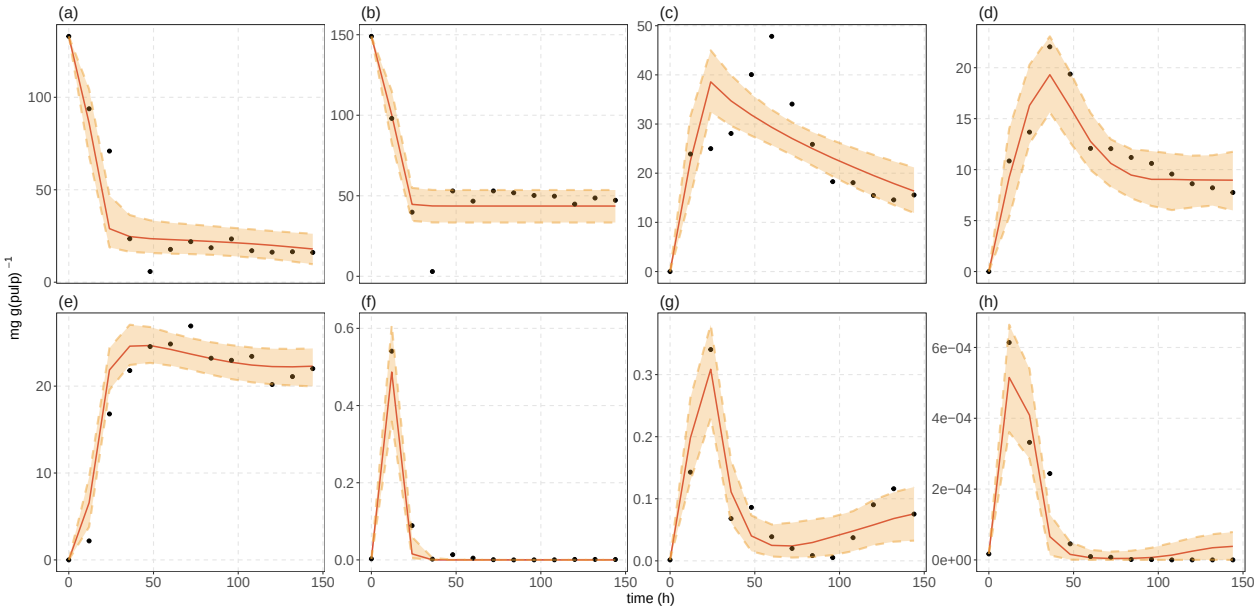


Figure S19: Posterior predictions of model iteration corresponding to mechanisms M1 and M3, MI(1,3), fitted to dataset *brwb4* reported by Pereira *et al.* [8]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

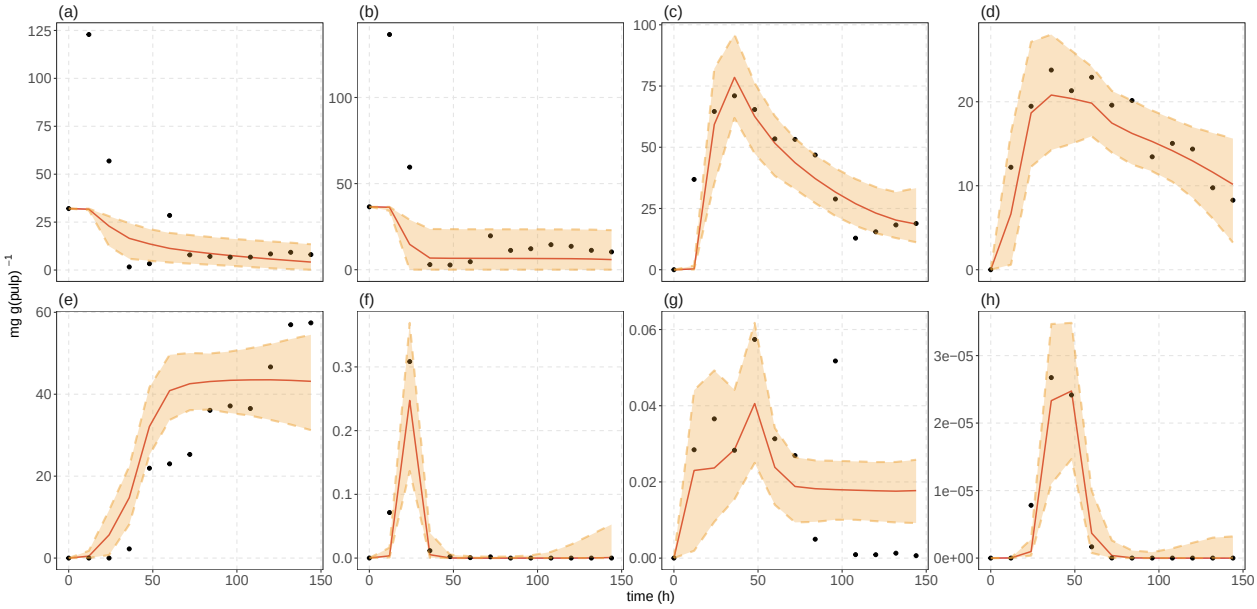


Figure S20: Posterior predictions of model iteration corresponding to mechanisms M1 and M4, MI(1,4), fitted to dataset *brst2* reported by Pereira *et al.* [8]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

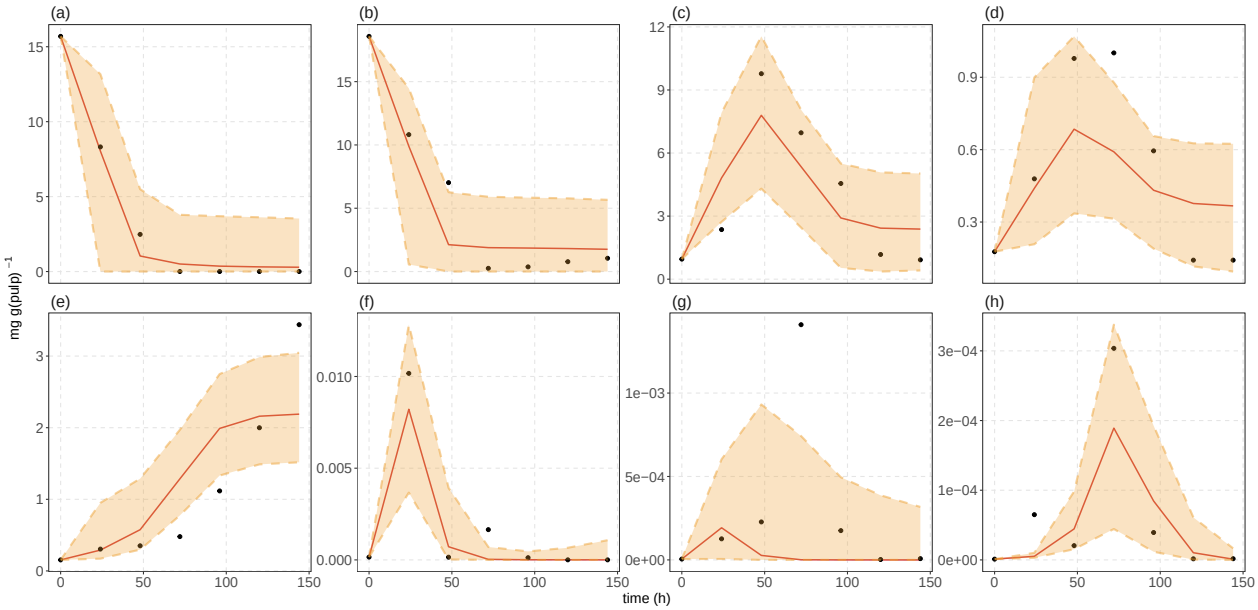


Figure S21: Posterior predictions of model iteration corresponding to baseline model, MI(0), fitted to dataset *brwb7* reported by Moreira *et al.* [9]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

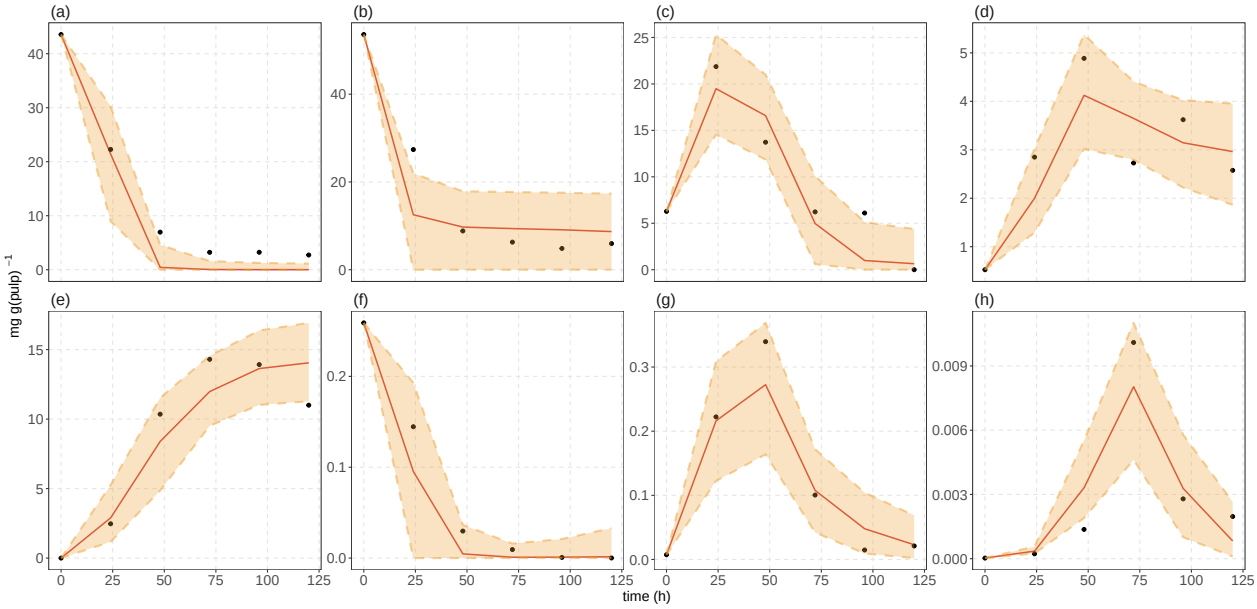


Figure S22: Posterior predictions of model iteration corresponding to baseline model, MI(0), fitted to dataset *hnwb1* reported by Romanens *et al.* [12]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

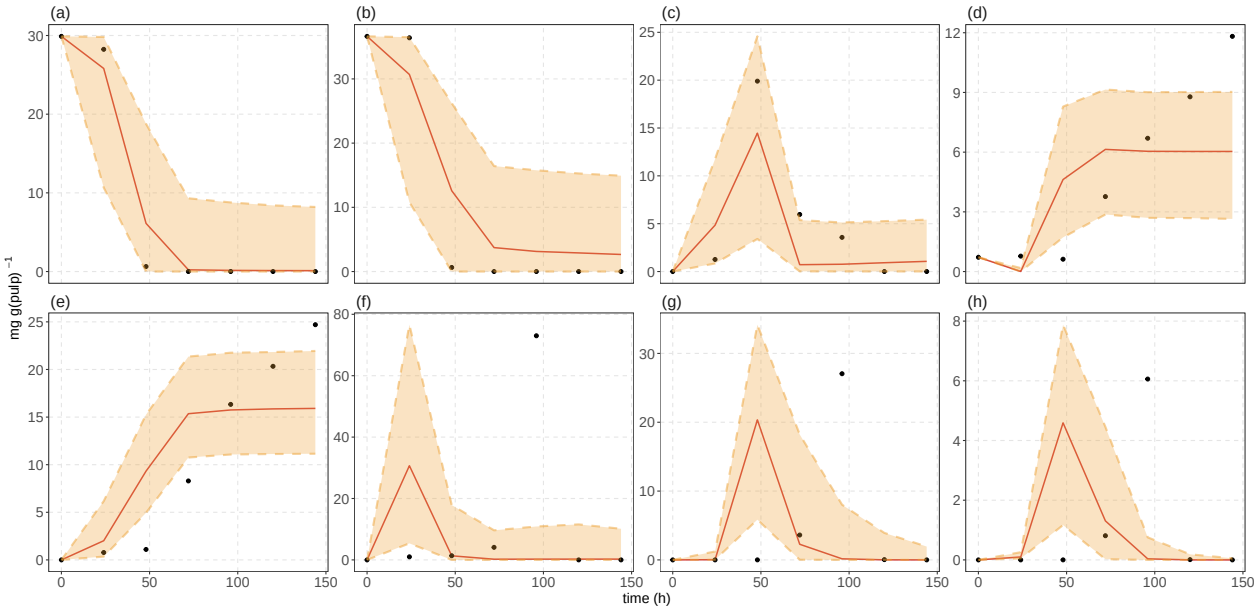


Figure S23: Posterior predictions of model iteration corresponding to mechanisms M3 and M4, MI(3,4), fitted to dataset *ecpb1* reported by Lee *et al.* [13]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

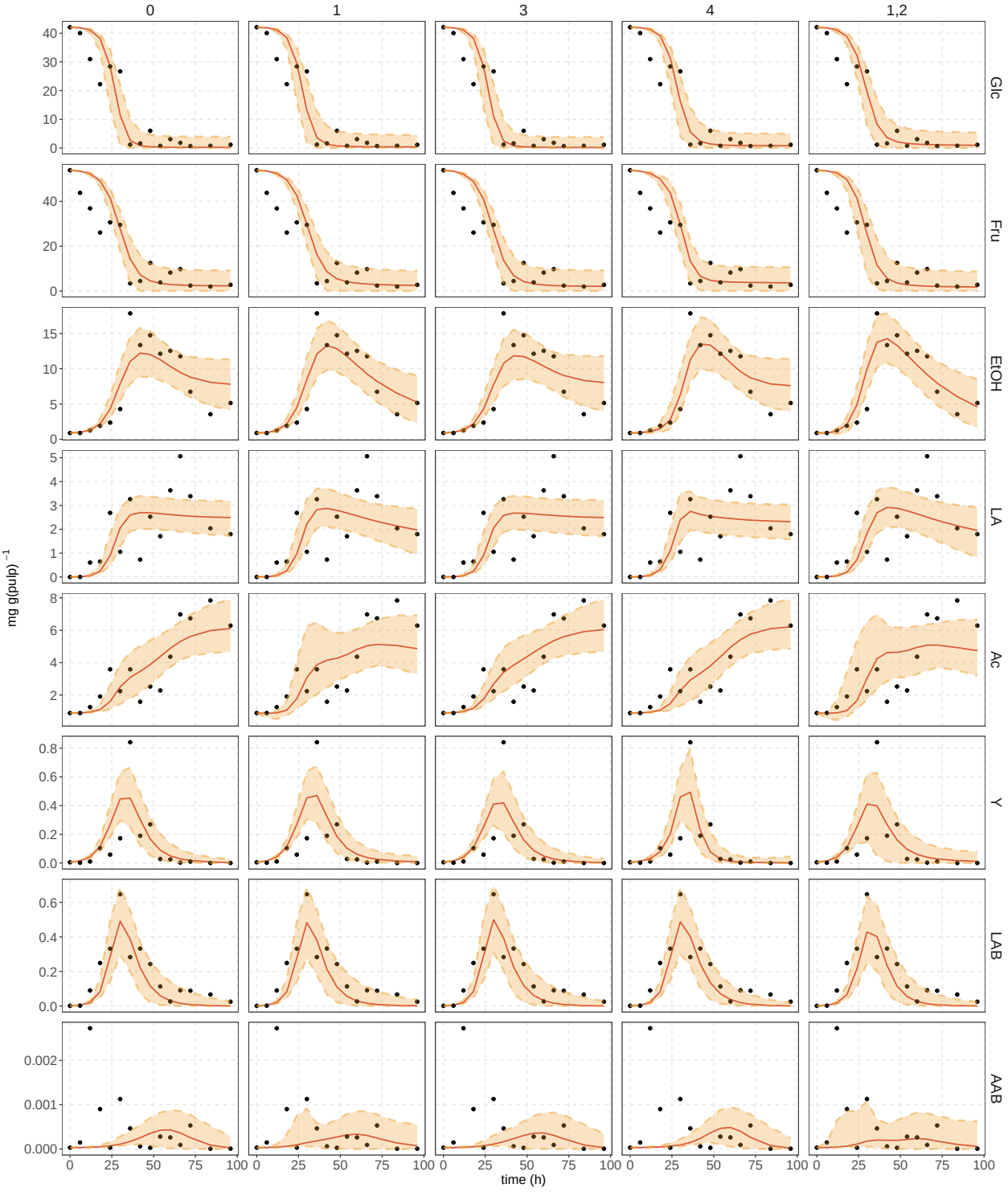


Figure S24: Posterior predictions across model iterations fitted on dataset *ecwb1* reported by Papalexandratou *et al.* [5]. Columns names refer to mechanisms involved in each model iteration. Row names refer to metabolites and microbial groups: glucose (Glc), fructose (Fru), ethanol (EtOH), lactic acid (LA), acetic acid (Ac), yeasts (Y), lactic acid bacteria (LAB), and acetic acid bacteria (AAB). Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

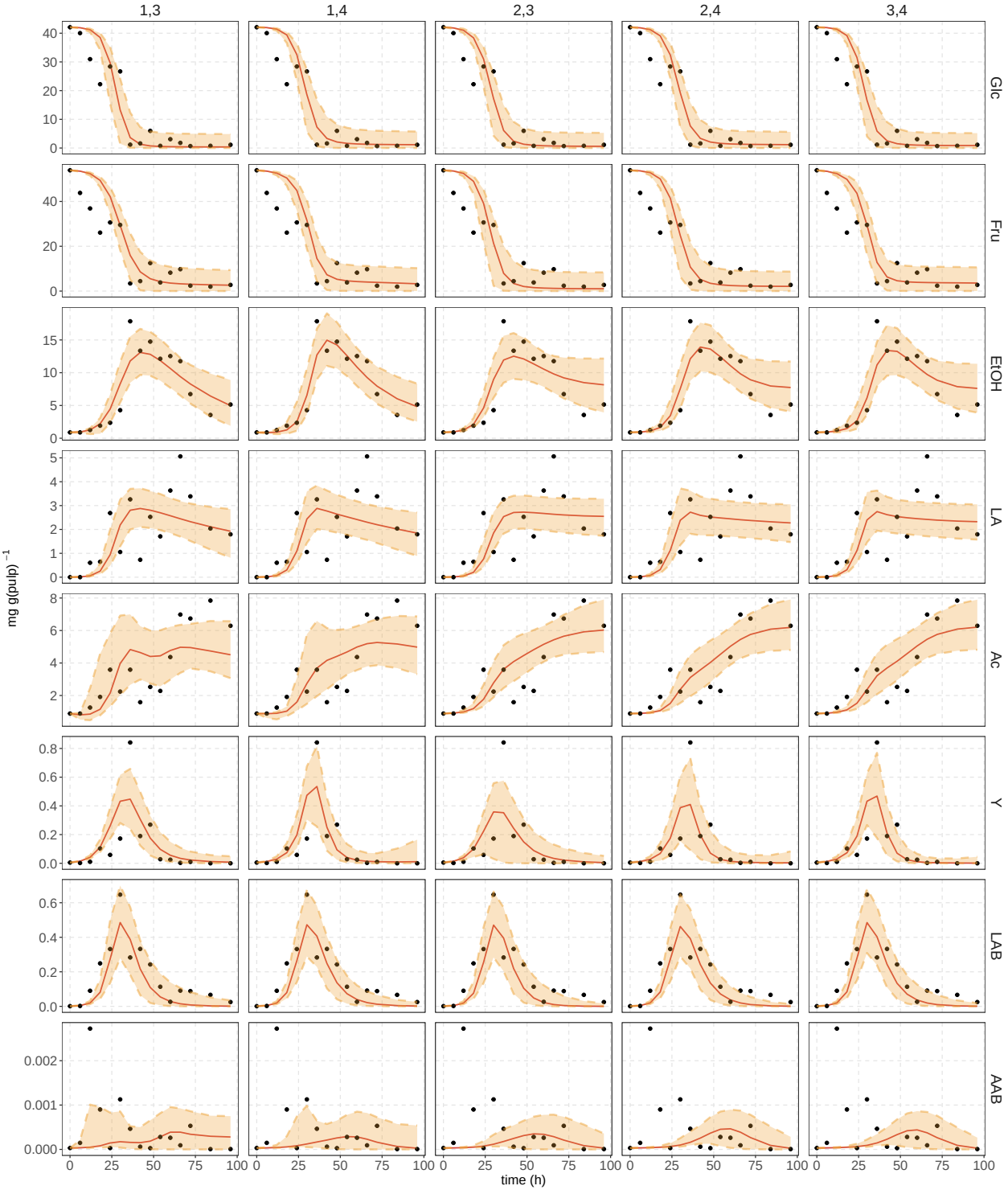


Figure S24 (cont.): Posterior predictions across model iterations fitted on dataset *ecwb1* reported by Papalexandratou *et al.* [5]. Columns names refer to mechanisms involved in each model iteration. Row names refer to metabolites and microbial groups: glucose (Glc), fructose (Fru), ethanol (EtOH), lactic acid (LA), acetic acid (Ac), yeasts (Y), lactic acid bacteria (LAB), and acetic acid bacteria (AAB). Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

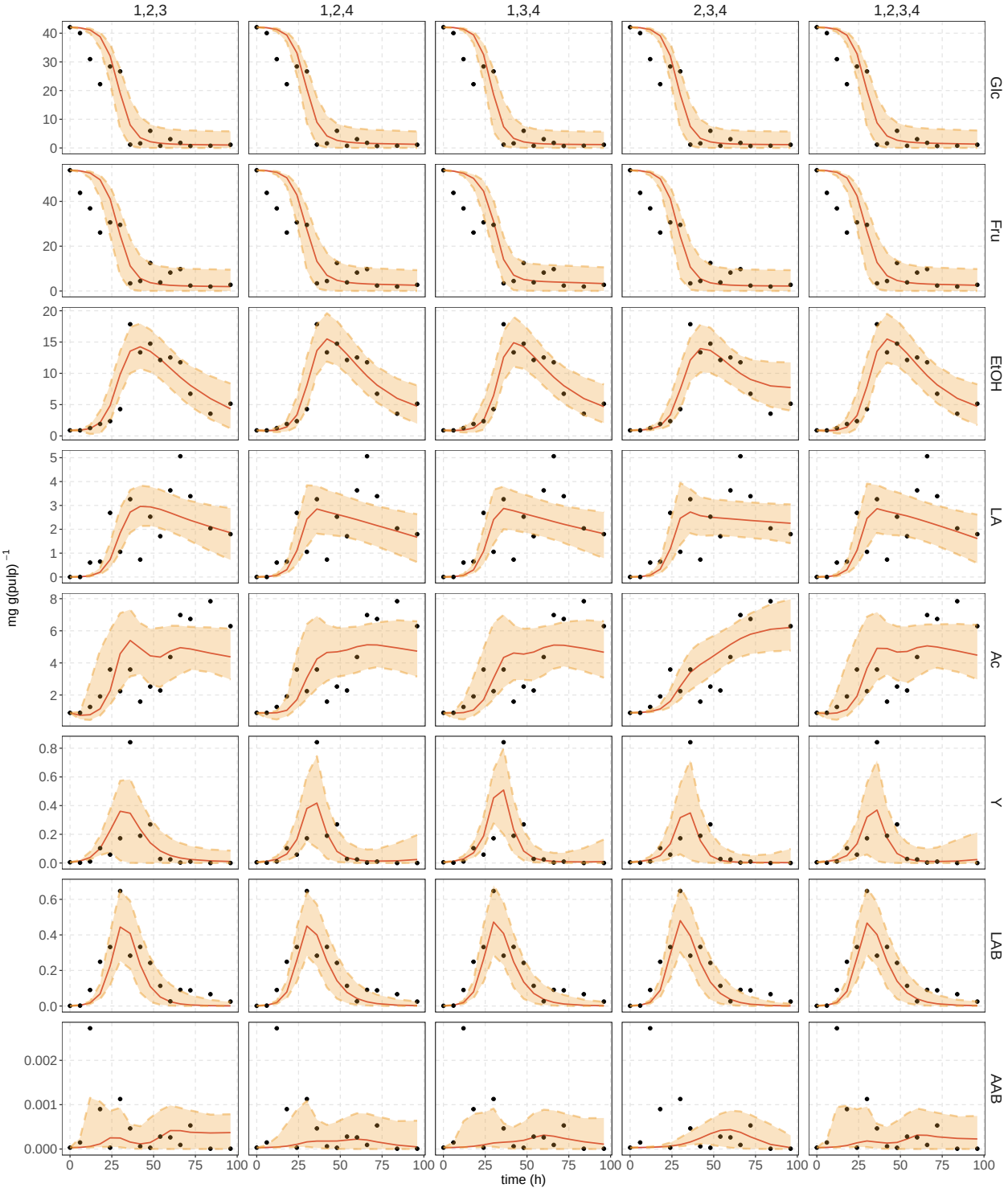


Figure S24 (cont.): Posterior predictions across model iterations fitted on dataset *ecwb1* reported by Papalexandratou *et al.* [5]. Columns names refer to mechanisms involved in each model iteration. Row names refer to metabolites and microbial groups: glucose (Glc), fructose (Fru), ethanol (EtOH), lactic acid (LA), acetic acid (Ac), yeasts (Y), lactic acid bacteria (LAB), and acetic acid bacteria (AAB). Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

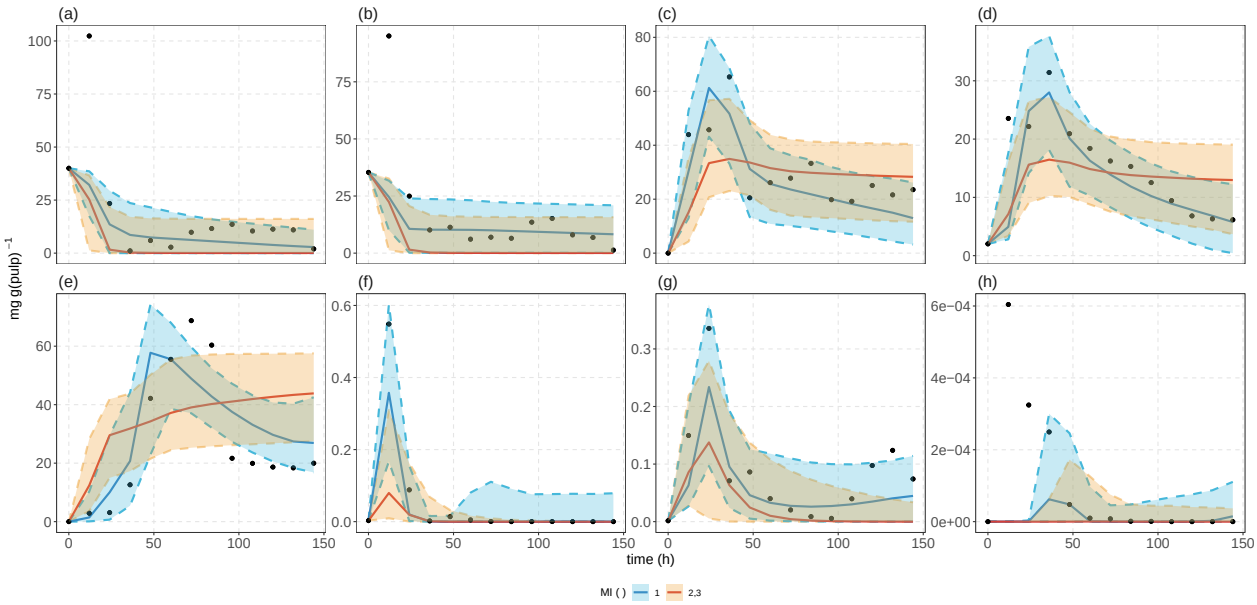


Figure S25: Comparison of posterior predictions for model iterations MI(1) and MI(2,3), fitted to dataset *brpb1* reported by Pereira *et al.* [6]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. For MI(1), solid blue lines represent medians of the posterior predictions and sky-blue ribbons describe their 95% credible intervals. For MI(2,3), solid red lines represent medians of the posterior predictions and orange ribbons describe their 95% credible intervals. Solid black points denote experimental data.

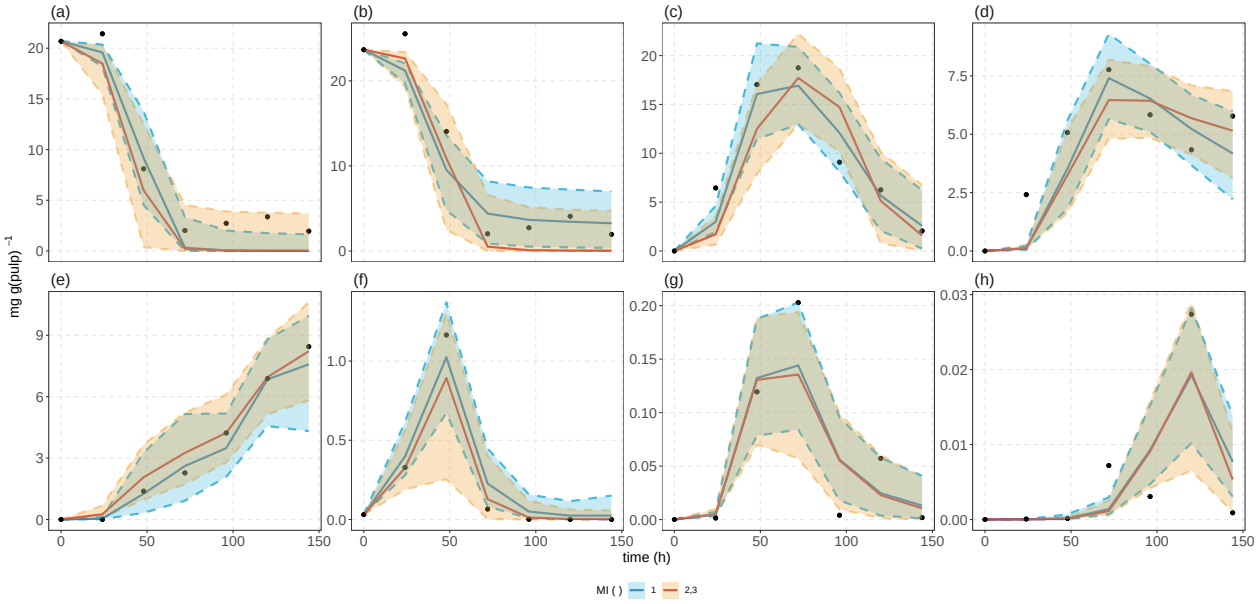


Figure S26: Comparison of posterior predictions for model iterations MI(1) and MI(2,3), fitted to dataset *niwb2* reported by Papalexandratou *et al.* [14]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. For MI(1), solid blue lines represent medians of the posterior predictions and sky-blue ribbons describe their 95% credible intervals. For MI(2,3), solid red lines represent medians of the posterior predictions and orange ribbons describe their 95% credible intervals. Solid black points denote experimental data.

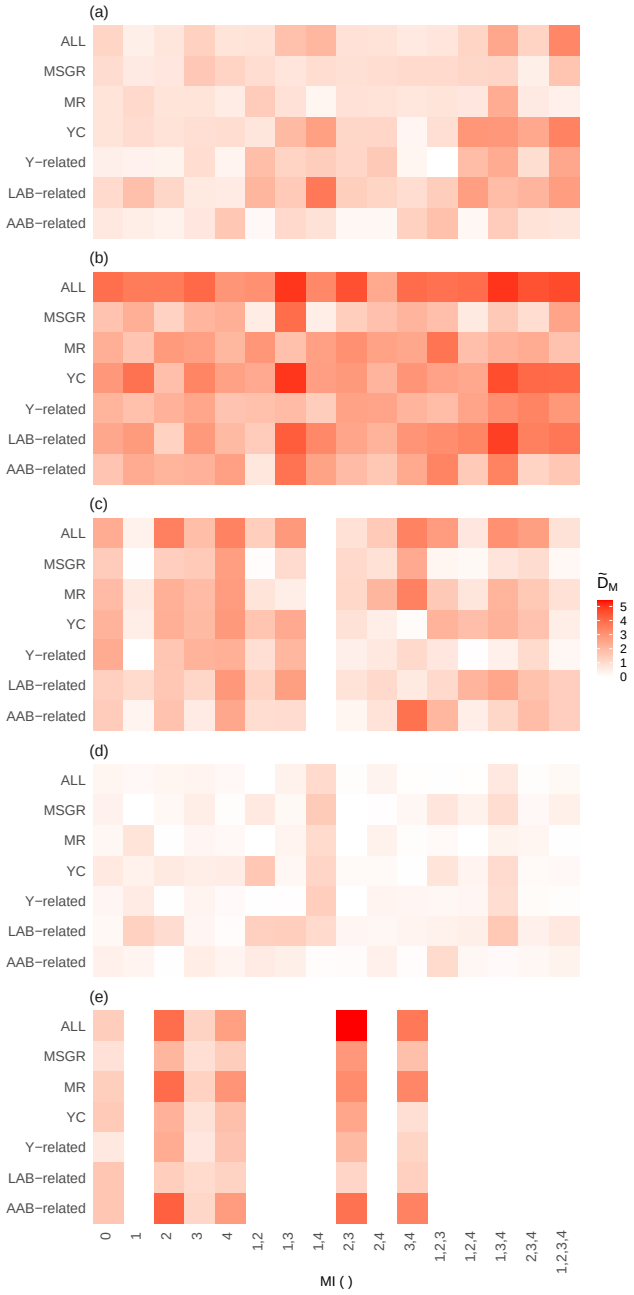


Figure S27: Heat map of medians of pairwise squared Malahanobis distances ($\tilde{D}_M$) between centroids of grouping classes scores from Principal Component Analysis (PCA) per model iteration and subgroups of parameter estimates. (a) Country, (b) cocoa cultivar, (c) fermentation method, (d) turning of fermenting mass, and (e) use of controlled temperature. White rows in (b), (c) and (e) correspond to model iterations where only one group class was available and PCA groupings could not be performed. Subgroups ALL, MSGR, MR, YC, Y-related, LAB-related and AAB-related correspond to all common MIs parameter estimates, maximum specific growth rates, yield coefficients, yeast-related parameters, lactic acid bacteria-related parameters, and acetic acid bacteria-related paremeters, respectively.

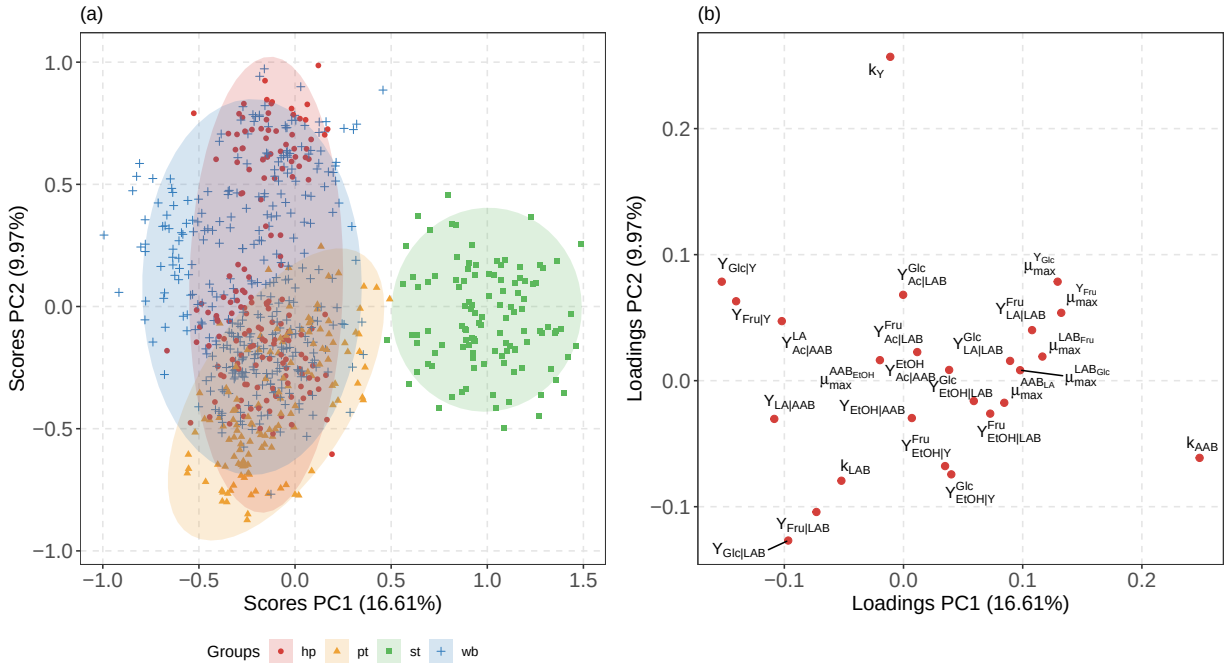


Figure S28: PCA score (a) and loading plot (b) from all parameters of model iteration MI(2). For visualization purposes, scores of only 10% of posterior draws are shown. Heap (hp), platform (pt), stainless-steel tank (st) and wooden box (wb) fermentation methods are shown. Parameters located on the left and right with respect to 0 in the horizontal axis in the loading plot determine differentiation between stainless-steel tank and rest of fermentation methods, respectively.

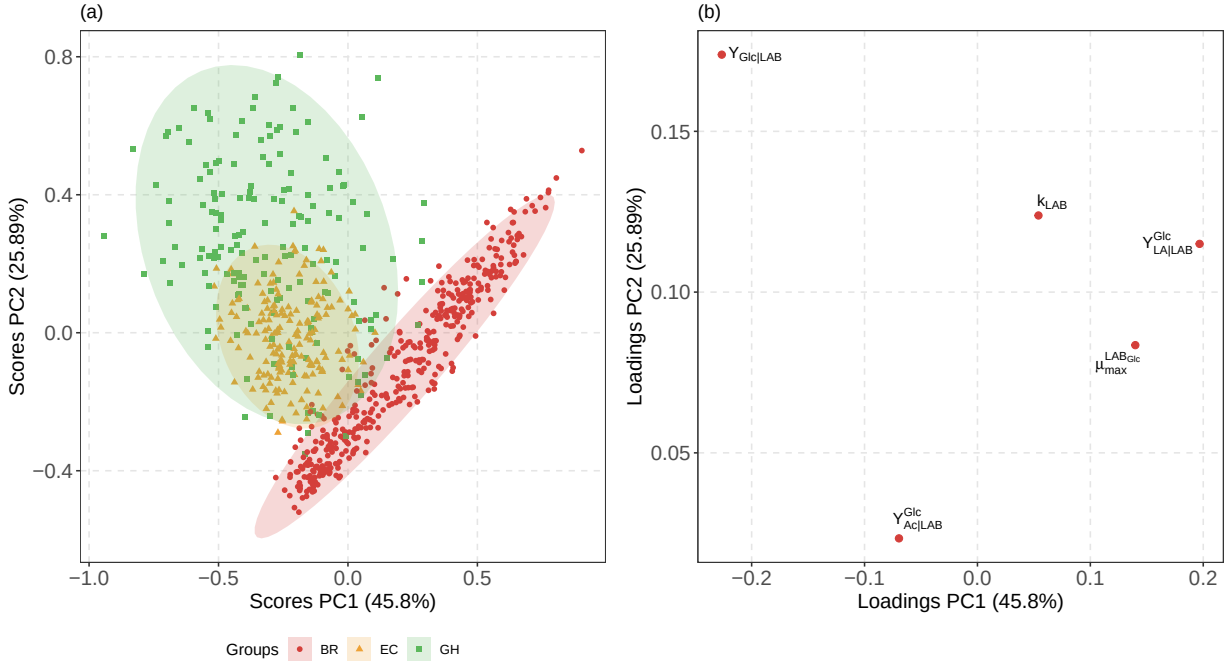


Figure S29: PCA score (a) and loading plot (b) from lactic acid bacteria-related parameters of model iteration MI(1,3,4). For visualization purposes, scores of only 10% of posterior draws are shown. Brazil (BR) is clearly separated from Ecuador (EC) and Ghana (GH). Parameters located on the left and right with respect to 0 in the horizontal axis in the loading plot determine differentiation between all classes.

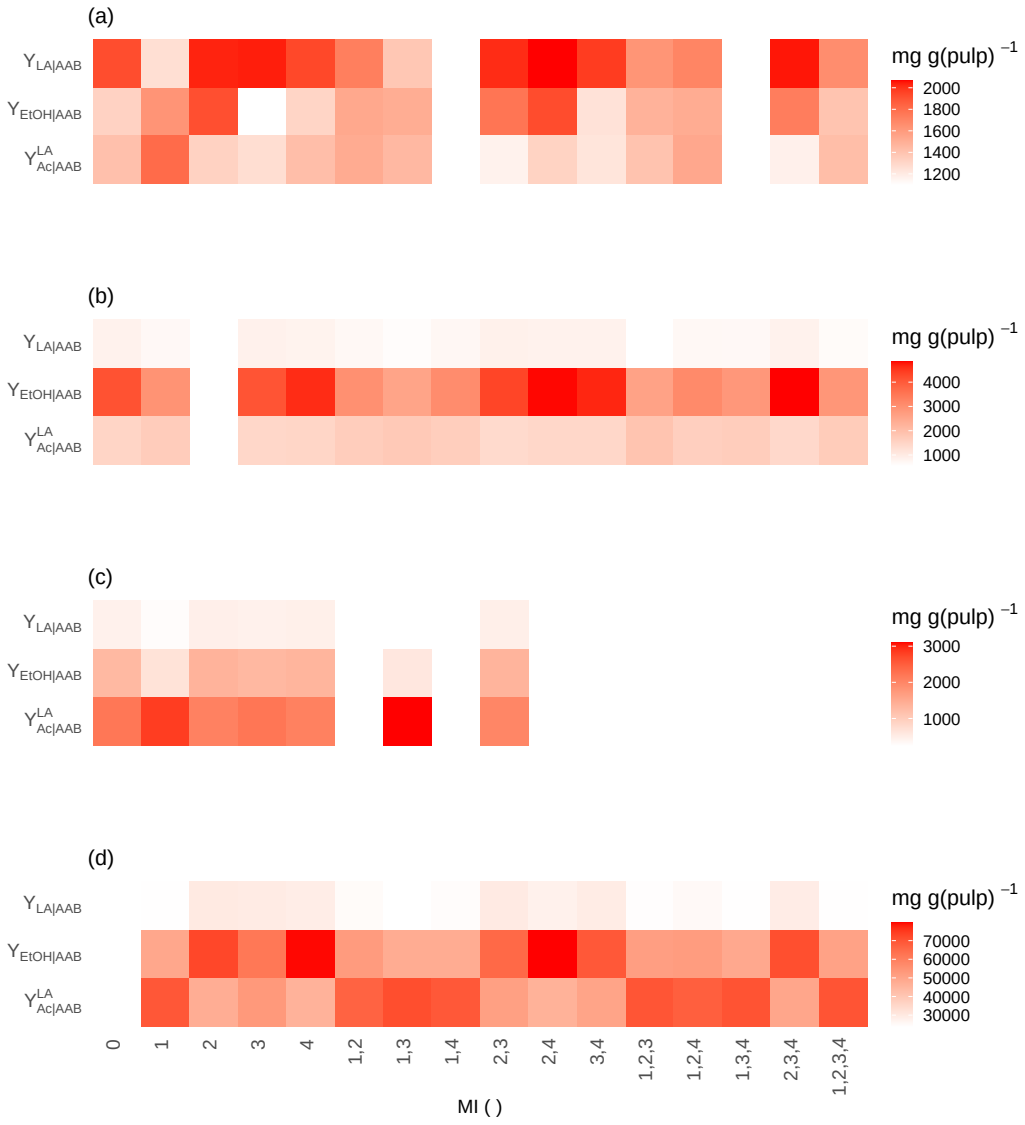


Figure S30: Heat map of posterior distributions means of $Y_{\mathrm{LA|AAB}}$, $Y_{\mathrm{EtOH|AAB}}$, and $Y^{\mathrm{LA}}_{\mathrm{Ac|AAB}}$. Dataset : (a) *ghhp1*, (b) *ecwb1*, (c) *ecwb2* and (d) *brpb1*.

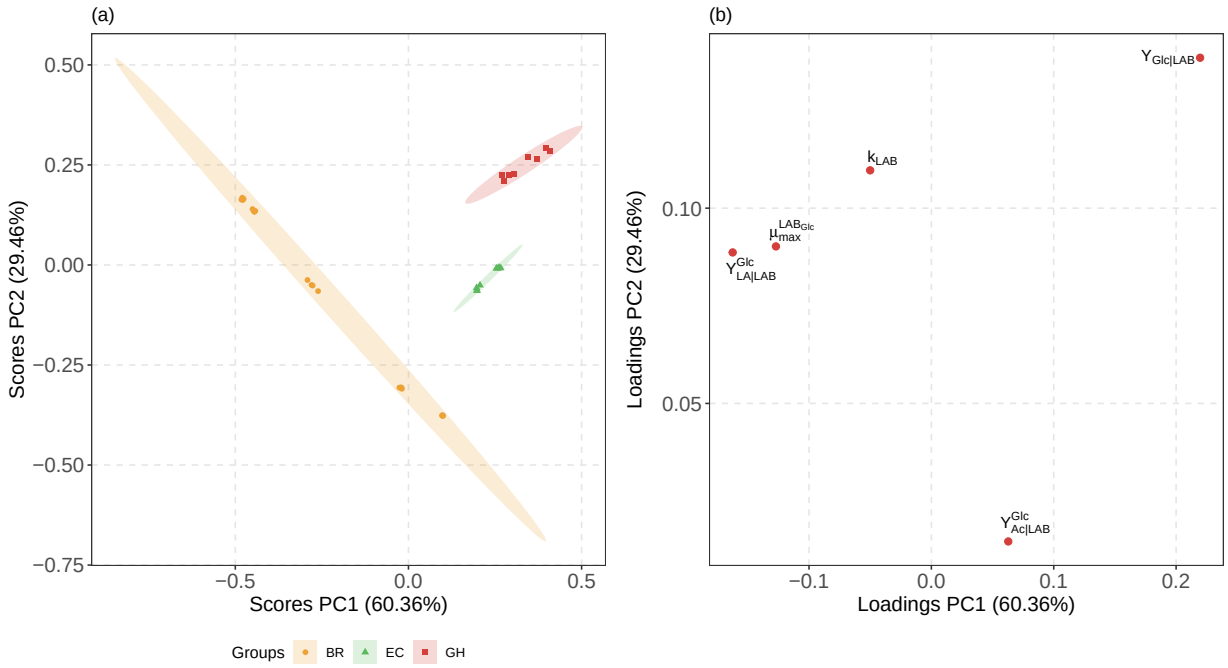


Figure S31: PCA score (a) and loading plot (b) from lactic acid bacteria-related parameters of model iteration MI(1,3,4) using medians of posterior distributions. All countries, Brazil (BR), Ecuador (EC) and Ghana (GH) are clearly separated. Parameters located on the left and right with respect to 0 in the horizontal axis in the loading plot determine differentiation between all classes.

6 Temperature in cocoa bean fermentation kinetic modelling

6.1 Introduction

From a bibliographical point of view, several authors have widely discussed the effect of temperature upon fermentation of cocoa beans. It is well known that increase of temperature within the fermenting mass results from microbial activities, mainly yeast (Y) and acetic acid bacteria (AAB) by the production of ethanol (EtOH) and acetic acid (Ac), respectively [55]. Its importance lies in several aspects, such as its role in the seed's embryo death process [56], its possible involvement in AAB inhibition at final stages of the fermentation [57], and evaporation of volatile metabolites (i.e., EtOH and Ac) [16].

In regard of including temperature in a quantitative approach, López-Pérez *et al.* [58] stands as the only attempt among the few mathematical models of cocoa bean fermentation that has implemented it as another state variable using the cardinal temperature model as proposed by Rosso *et al.* [59]. Opposed to this approach, here we present the use of the Arrhenius equation applied to maximum specific growth and mortality rates coupled to the mathematical expression accounting for temperature produced by microbial dynamics as stated by Boulton [60].

6.2 Formulation

6.2.1 Heat transfer

According to Boulton [60], the temperature rate change in a fermentation process can be considered as an energy balance between the rates of heat generation of the fermenting mass and its surroundings. Thus, this rate is:

$$\frac{dT}{dt} = \frac{\Delta Q}{\rho\, C_{\mathrm{p}}\, V}\,\frac{dS}{dt} - \frac{UA}{\rho\, C_{\mathrm{p}}\, V}\,(T - T_e)\ , \tag{23}$$

where, for our case, ΔQ is the heat released during the fermentation per mg of substrate ($\frac{kJ}{\mathrm{mg}_{\mathrm{substrate}}}$), ρ is the density of cocoa beans ($\frac{\mathrm{kg}_{\mathrm{bean}}}{m^3}$), C_{p} is the cocoa beans heat capacity ($\frac{kJ}{\mathrm{mg}_{\mathrm{bean}} K}$), V is the volume of the fermenting mass (m^3), T_e is the environmental temperature (K), U is the overall heat transfer coefficient ($\frac{kJ}{m^2 K}$), A is the contact area of the fermenting mass with the environment (m^2), and $\frac{dS}{dt}$ is the substrate rate change. With the purpose of facilitating any optimization routine, $\frac{\Delta Q}{\rho\, C_{\mathrm{p}}\, V}$ and $\frac{UA}{\rho\, C_{\mathrm{p}}\, V}$ can be comprised into single parameters each. Then, let's define a substrate-to-heat yield coefficient ($Y_{Q|\mathrm{substrate}}$) such as:

$$Y_{Q|\mathrm{substrate}} = \frac{\Delta Q}{\rho\, C_{\mathrm{p}}\, V}\,, \tag{24}$$

and a heat-loss (Q_L) parameter as

$$Q_L = \frac{UA}{\rho\, C_{\mathrm{p}}\, V}\,. \tag{25}$$

Hence, equation (23) can be re-written as:

$$\frac{dT}{dt} = Y_{Q|\mathrm{substrate}}\,\frac{dS}{dt} - Q_L\,(T - T_e)\ . \tag{26}$$

The inclusion of Eq. (26) into the baseline model proposed in Eqs. (1.MM) to (8.MM) is then reduced to appropriately replacing $\frac{dS}{dt}$. For this, we account the total release of heat product of the growth of all microbial groups involved. With this in mind, Eq. (26) takes the form:

$$\frac{dT}{dt} = Y_{Q|\mathrm{Glc}}\,\left(Y_{\mathrm{Glc|Y}}\,v_1 + Y_{\mathrm{Glc|LAB}}\,v_3\right) + Y_{Q|\mathrm{Fru}}\,\left(Y_{\mathrm{Fru|Y}}\,v_2\right) + Y_{Q|\mathrm{EtOH}}\,\left(Y_{\mathrm{EtOH|AAB}}\,v_4\right) + Y_{Q|\mathrm{LA}}\,\left(Y_{\mathrm{LA|AAB}}\,v_5\right) - Q_L\,\left(T - T_e\right)\,, \tag{27}$$

where $\frac{dS}{dt}$ is replaced by the necessary growth rates and yield coefficients explained in Tables 1.MM and 2.MM, $Y_{Q|\mathrm{Glc}}$, $Y_{Q|\mathrm{Fru}}$, $Y_{Q|\mathrm{EtOH}}$ and $Y_{Q|\mathrm{LA}}$ are the glucose-to-heat, fructose-to-heat, ethanol-to-heat and lactic acid-to-heat yield coefficients, respectively.

6.2.2 Temperature effect on kinetic parameters

Among several ways to include temperature in bacterial growth models, the Arrhenius equation is one of the most widely utilized. Regardless its limitations in interpretation and recommended application only in sub-optimal growth conditions [61], here we introduce it as a first approximation for how temperature affects kinetic parameters of our baseline model.

Thereby, considering that Eq. (27) adds 6 parameters to baseline model, a simple manner to relate kinetic parameters to temperature would be by besetting Arrhenius terms to as few as possible and thus reducing risks of over-parametrization.

Then, the original Arrhenius equation is expressed as

$$k(T) = A\,e^{-\frac{Ea}{RT}}\,. \tag{28}$$

As in Eqs. (24) and (25), energy of activation (Ea) and gas constant (R) can be comprised into a single term:

$$k(T) = A\,e^{-\frac{b}{T}}\,. \tag{29}$$

Assuming that all growth and mortality rates share common energies of activation b_1 and b_2, respectively, these can be then rewritten in function of temperature as

$$\begin{aligned}
\mu_{\max}^{\mathrm{Y_{Glc}}} &= \mu_1\,e^{-\frac{b_1}{T}}\\
\mu_{\max}^{\mathrm{Y_{Fru}}} &= \mu_2\,e^{-\frac{b_1}{T}}\\
\mu_{\max}^{\mathrm{LAB_{Glc}}} &= \mu_3\,e^{-\frac{b_1}{T}}\\
\mu_{\max}^{\mathrm{AAB_{EtOH}}} &= \mu_4\,e^{-\frac{b_1}{T}}\\
\mu_{\max}^{\mathrm{AAB_{LA}}} &= \mu_5\,e^{-\frac{b_1}{T}}\\
k_{\mathrm{Y}} &= k_1\,e^{-\frac{b_2}{T}}\\
k_{\mathrm{LAB}} &= k_2\,e^{-\frac{b_2}{T}}\\
k_{\mathrm{AAB}} &= k_3\,e^{-\frac{b_2}{T}}\,,
\end{aligned}$$

where μ_i and k_i represent the growth rates of corresponding microorganisms and mortality rates at exponential phases, respectively.

Finally, above expressions are incorporated into baseline model (Eqs. (1.MM) to (8.MM)).

6.3 Model simulations

Posterior distributions of the parameters for a model as described above can be computed using the same Bayesian framework and priors as proposed in Eqs. (13.MM) and (14.MM), respectively. The only addition to this scheme is the extra parameter not considered before.We refer to term T_e in Eqs. (23) and (27), which represents environmental temperature. For this, we treated T_e as another free parameter with a more informative prior. Assuming that environmental temperature during the fermentation process had a mean value of 26 °C, we can add random noise by assigning a prior of the form:

$$T_e \sim \mathcal{N}(26,\, 0.01)\,, \tag{30}$$

which after time series scaling by their maximum value becomes:

$$T_e \sim \mathcal{N}\left(\frac{26}{T_{\max}},\, \frac{0.01}{T_{\max}}\right)\,, \tag{31}$$

where $T_{\max}$ is the maximum value observed in each temperature time series.

6.4 Results

As mentioned in the main manuscript, addition of temperature to the model does not improve fits in terms of their PSIS-LOO, posterior predictions and adjustment of problematic parameters' ranges to more plausible values. Here, we present each of these claims on basis of fitting this model to datasets *ghhp1* [1] (known for accounting non-identifiable parameters), *brwb1* and *brwb2* [4].

6.4.1 PSIS-LOO deviance values

As depicted in Figure S32, PSIS-LOO deviance values between baseline model (MI(0)) and baseline model added temperature as state variable (MI(0+t)), there are not significant improvements in prediction adequacy between them both, over datasets *ghhp1* [1], *brwb1* and *brwb2* [4].

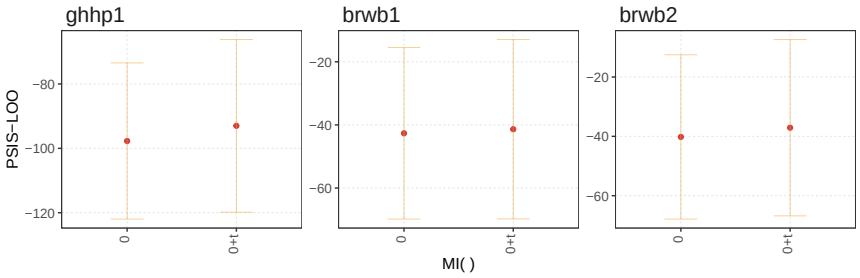


Figure S32: PSIS-LOO deviance values of baseline model (MI(0)) and baseline model added temperature (MI(0+t)) over datasets *ghhp1* [1], *brwb1* and *brwb2* [4].

6.4.2 Posterior predictions

Besides MI(0+t) being capable of fitting remarkably well temperature, no outstanding enhancement of posterior predictions for the rest of state variables is perceptible compared to its counterparts with no temperature interactions (see figures S33, S34 and S35 for posterior predictions of *ghhp1*, *brwb1* and *brwb2*, respectively).

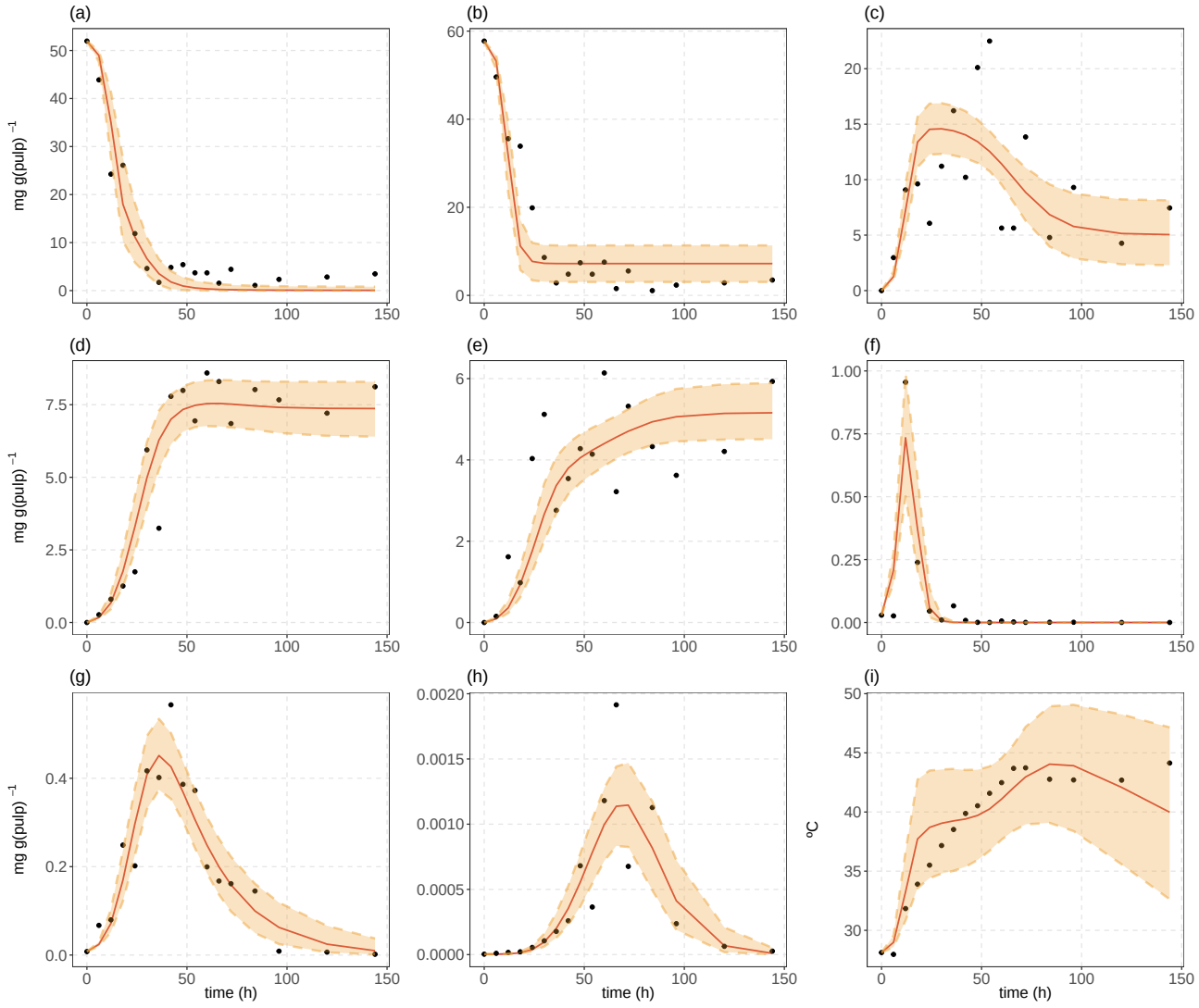


Figure S33: Posterior predictions of MI(0+t) over dataset *ghhp1* reported by Camu *et al.* [1]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. (i) Temperature. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

6.4.3 Parameter ranges

Finally, regarding ranges for parameters likely affected by non-identifiability, as it can be seen in Table S10, do not change in a great extend nor begin to fall within reported ranges in the literature.

6.5 Conclusion

From this approach of including temperature in our baseline model, no great improvements are clearly seen on the basis of statistics used for assessing the models in this work.

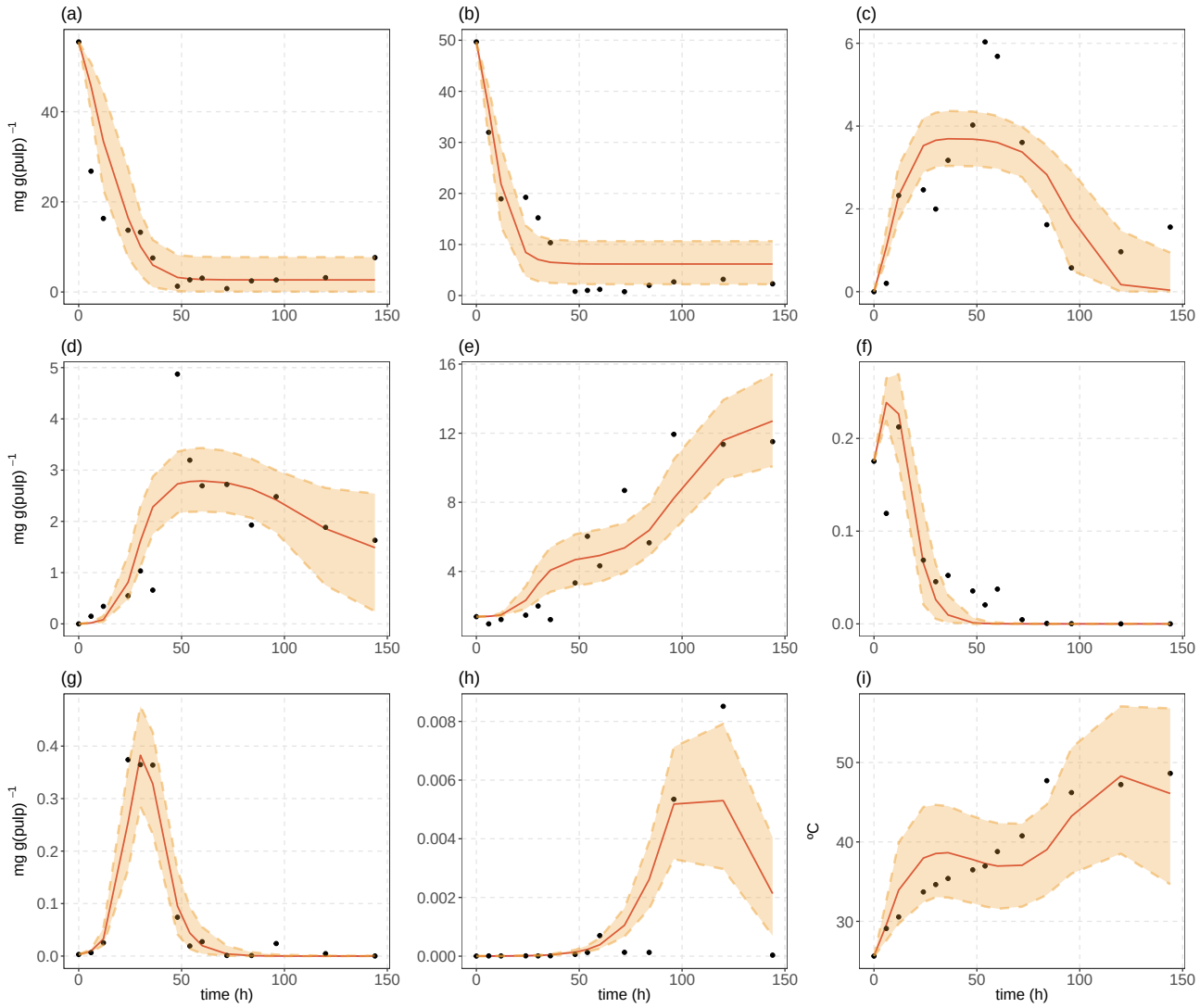


Figure S34: Posterior predictions of MI(0+t) over dataset *brwb1* reported by Papalexandratou *et al.* [4]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. (i) Temperature. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

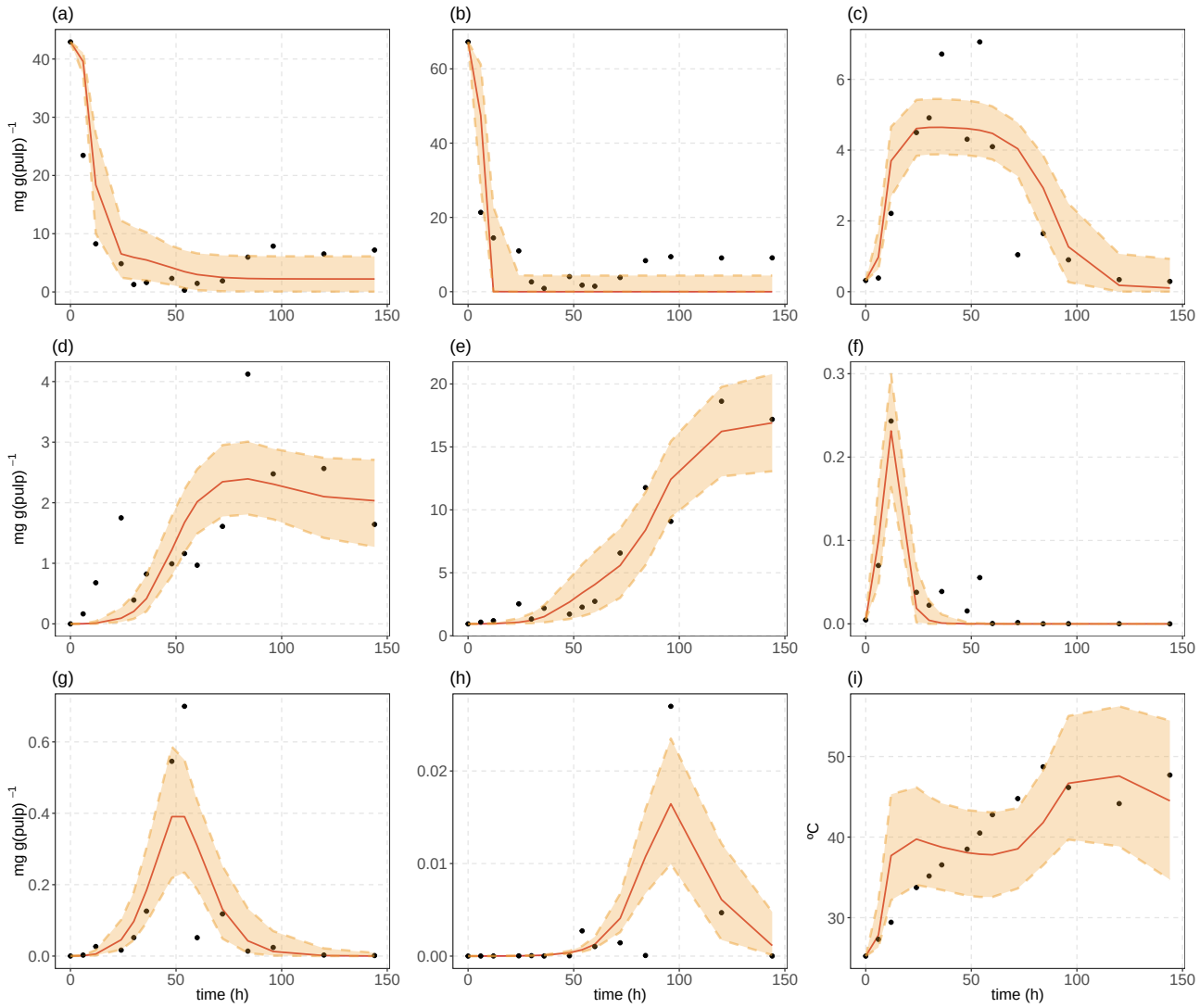


Figure S35: Posterior predictions of MI(0+t) over dataset *brwb2* reported by Papalexandratou *et al.* [4]. Metabolites: (a) glucose, (b) fructose, (c) ethanol, (d) lactic acid and (e) acetic acid. Microbial groups: (f) yeast, (g) lactic acid bacteria and (h) acetic acid bacteria. (i) Temperature. Solid red lines represent medians of the posterior predictions, solid black points denote experimental data and orange ribbons describe 95% credible intervals of posterior predictions.

Table S10: Parameter estimates of baseline model (MI(0)) and baseline model with added temperature (MI(0+t)) calibrated over datasets *ghhp1* [1], *brwb1* and *brwb2* [4]. Means and standard deviations (within parenthesis) of the posterior parameters where temperature was not explicitly included are shown. All reported values in literature have same units as depicted in Table 2.MM with the exception of substrate saturation constants ($\mathrm{mg(substrate)\,mL^{-1}}$). Green coloured cells indicate parameter estimates that lie within reported ranges. Orange colourd cells indicate parameter estimates that are far from reported ranges by less than the estimate either divided or multiplied by 10. Parameters fitted with the temperature model are indicated with (t) next to dataset coding.

Parameter	ghhp1	ghhp1 (t)	brwb1	brwb1 (t)	brwb2	brwb2 (t)	Reported values	Reference
K^{Y}_{Glc}	35.178 (13.809)	36.343 (13.275)	34.607 (14.961)	36.069 (15.006)	30.097 (10.79)	30.704 (10.528)	9.73×10^{-4} - 0.5	[26, 28]
K^{Y}_{Fru}	35.682 (15.148)	36.708 (13.806)	25.172 (13.874)	27.683 (13.632)	41.026 (17.479)	27.152 (16.461)	5.82 - 161.45	[33, 35, 36]
K^{LAB}_{Glc}	37.949 (12.291)	35.876 (11.578)	31.821 (14.1)	21.171 (10.839)	18.936 (10.595)	15.779 (9.424)	0.790 - 178.0	[40–42]
K^{AAB}_{EtOH}	16.201 (5.559)	15.063 (5.355)	3.796 (1.666)	3.832 (1.585)	4.104 (1.941)	4.197 (1.965)		
K^{AAB}_{LA}	2468.808 (1246.01)	2504.518 (1230.219)	314.451 (153.855)	330.123 (154.135)	80.936 (41.545)	81.353 (41.802)		
$Y_{Glc\|Y}$	33.833 (11.286)	35.245 (10.925)	241.469 (57.414)	247.991 (56.39)	120.043 (35.606)	128.349 (34.287)	1.56 - 66.67	[24–26, 30, 31, 50, 51]
$Y_{Glc\|LAB}$	29.094 (10.903)	26.525 (10.147)	19.96 (12.78)	18.558 (12.482)	3.289 (3.391)	3.604 (3.381)	4.50 - 200	[42, 52]
$Y_{Fru\|Y}$	40.913 (11.183)	43.431 (10.847)	243.705 (47.332)	248.639 (46.601)	231.885 (63.675)	274.426 (58.585)	43.48 - 200	[31, 32]
$Y^{Glc}_{EtOH\|Y}$	7.575 (4.505)	7.351 (4.386)	11.924 (6.043)	11.921 (5.984)	8.333 (4.507)	9.46 (4.559)	1.39 - 21.49	[26, 29, 30, 53]
$Y^{Fru}_{EtOH\|Y}$	5.87 (3.301)	6.707 (3.516)	11.102 (4.928)	11.575 (5.07)	5.886 (3.897)	6.6 (4.034)	5.679 - 5.896	[29]
$Y_{EtOH\|AAB}$	1330.24 (734.365)	1441.682 (886.113)	380.088 (151.185)	425.236 (162.987)	171.249 (55.576)	178.163 (55.176)	8.06 - 166.67	[46, 48, 54]
$Y^{Glc}_{LA\|LAB}$	10.573 (1.518)	10.651 (1.507)	2.781 (0.665)	2.853 (0.666)	2.128 (0.783)	2.169 (0.751)		
$Y_{LA\|AAB}$	1923.213 (1182.916)	1862.392 (1191.561)	286.443 (141.585)	287.649 (137.542)	62.138 (38.388)	61.628 (37.531)	7.51 - 8.34	[48]
$Y^{Glc}_{Ac\|LAB}$	5.586 (0.907)	5.65 (0.902)	3.249 (1.094)	3.344 (1.094)	2.879 (1.856)	2.973 (1.83)		
$Y^{EtOH}_{Ac\|AAB}$	107.838 (113.797)	100.5 (132.648)	579.784 (271.612)	550.655 (276.068)	321.488 (136.809)	312.075 (133.57)		
$Y^{LA}_{Ac\|AAB}$	1418.047 (868.763)	1390.92 (860.409)	665.158 (348.199)	634.929 (339.7)	385.605 (182.323)	383.586 (179.92)		
σ	0.149 (0.01)	0.142 (0.009)	0.168 (0.014)	0.161 (0.013)	0.167 (0.013)	0.162 (0.012)		

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